Causal Clifford Theory (CL10): the Standard Model and gravity from a single Clifford root on a Planck-lattice quantum cellular automaton

Mohammed Haouach[1]

[1] *Independent researcher*[*]

We present **Causal Clifford Theory (CL10)**: a unitary quantum cellular automaton (QCA) on a cubic lattice of Planck spacing whose on-site internal space is the 32-dimensional complex module of the Clifford algebra $\mathrm{Cl}(10,0)$. Ten mutually anticommuting Hermitian generators $h_a^2 = I$ supply the kinematics. From this single algebraic object follow a (1+3) Minkowski subalgebra; a 16-component Weyl module whose charge assignment is exactly one Standard Model (SM) generation, with all eight gauge and mixed anomalies cancelling; a 45-dimensional Spin(10) adjoint containing the twelve SM gauge bosons; the 5/3 hypercharge normalisation; an exactly unitary, strictly causal update with $\det U = 1$ and an exact matter–antimatter spectral mirror; and a lattice theory of a massless spin-2 field whose discrete Einstein–Hilbert action reproduces general relativity to machine precision.

We also state **what does not follow**. Every number is classified by provenance — *algebraic*, *conventional*, *fitted*, or *open*. The family number, the Yukawa structure, quark mixing, the cosmological constant, the dark-matter identity and abundance, the exact neutrino mass, the baryon asymmetry and one-loop coupling unification remain open. A single loop factor $(1/16\pi)^{10}$ reproduces the electroweak/Planck hierarchy to a factor 2.08 and gives a 118 GeV dark-matter scale, but the exponent 10 is selected by the data it is meant to explain.

Five tensions with observation are reported rather than hidden. Under Standard-Model desert running the GUT coupling $\alpha_{10} = 1/25$ overshoots the measured SU(2) coupling by a factor 5.8 and drives SU(3) to a Landau pole at 2.2×10^9 GeV; this failure is repaired by the vectorlike remnant matter the theory's own breaking sector supplies — three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs plus one weak-doublet pair make the three measured couplings meet at $\alpha_X = 1/25$ *exactly*, at $M_X = 1.8 \times 10^{16}$ GeV, at the cost of abandoning $M_{\mathrm{GUT}} = M_P$ and of two fitted thresholds; the repair predicts $\tau_p \simeq 1.8 \times 10^{36}$ yr. The remnant is no longer an assumption: an internal Wilson factor of the step itself, built from the root's Cartan algebra, gaps the twelve doubler nodes, the vectorlike pattern is the spectrum of its selective reading, the multiplicity is bounded below by unification ($n_{\mathrm{nodes}} \geq 3$), and the same direction completes the Spin(10) breaking chain to the SM group — but the mechanism's own minimal root meets at $M_X = 8.9 \times 10^{14}$ GeV, already excluded by proton decay, and no redistribution of the node strengths can move it (the meeting equations depend on the three strengths only through their sum), and the gauged bond layer's own composite matter is excluded as the supplier of the data-preferred spectrum (no solution over the full scan of the breaking direction). The seesaw value $m_\nu = 0.146$ eV exceeds the oscillation scale and the bound $\sum m_\nu < 0.12$ eV; a built node-mixing junction supplies the $\frac{1}{3}$ coefficient that lands the mass at 0.0487 eV — the earlier claim that this survives only at one measure-zero point was an artifact of an identity-Dirac assumption: with the implemented democratic Dirac matrix the massless structure is protected for every junction ratio and the value sits in a finite window (V103); the relic abundance is normalised, not computed, and the dark-matter no-go is completed — no Wilson node can host a stable neutral carrier at the predicted 118 GeV, and the minimal fix is a nearly decoupled mirror-odd scalar with $\sigma_{\mathrm{SI}} = 5 \times 10^{-51}$ cm^2; the heavy-neutrino decay is *in* equilibrium ($\Gamma/H \simeq 4.6$); and the loop factor is closed from inside — the coarse-graining tower is proven kinematic, the block-spin RG of the bond layer composes internal angles but can never divide them, while a traced fluctuating link d.o.f. is shown to divide them exactly: the factor $1/(16\pi)$ requires the gauged bond layer, now built with an exactly closing Gauss constraint, whose measured level-1 division still falls short of $1/(16\pi)$ by a factor 26. The vanishing vacuum energy is proven protected by $\det U = 1$ against every traceless internal deformation, so the observed Λ is equivalent to a named trace-phase breaking of size 1.6×10^{-120}. The seven criteria, the nine open residuals and the seven falsifiers are stated explicitly.

I. INTRODUCTION

A. The image behind the construction

Two habits of language do most of the misleading. The first calls c the speed of light. Light is one traveller at that speed; the constant itself is the speed of causality — the rate at which one event can constrain another. Asking why light moves at c is like asking why a particular car obeys the speed limit. The second habit is wave–particle duality, taught as a paradox. Seen from a lattice it dissolves: there is one wave, always; what a detector registers is a single localised event, because the detector is itself a lattice of cells that change state one cell at a time. The ontology contains no particles — only waves

[*] haouach.mohammed@gmail.com

that deliver their energy in quanta.

Quantum field theory took the first step, replacing particles by fields, but it multiplied the fields: one per species, glued together by hand. The viewpoint of this paper comes from the author's background in computer science, where the opposite picture is elementary: a cellular automaton — a lattice of cells, each holding a finite packet of information, all updated in parallel by one local rule. If the world is such an automaton, everything physical is a wave of information over that lattice, and the whole theory reduces to two questions: what is the information structure of a single cell, and what is the rule that updates it. Answer those two and particles, forces and spacetime itself are what the waves do. CL10 is the attempt to answer them concretely: the structure of the cell is the 32-dimensional module of $\mathrm{Cl}(10,0)$ (Section 2), the rule is a product of Spin(10) plane rotations plus a nearest-neighbour swap (Section 8).

B. Two literatures, one construction

Discrete models of quantum dynamics and grand unification have developed independently. On one side, causal sets, lattice gauge theory and quantum cellular automata treat spacetime as a finite, causal, unitary dynamical system [1–5]. On the other, the observation that one generation of quarks and leptons fits exactly into the 16-dimensional spinor of Spin(10), with the 5/3 hypercharge normalisation and the anomaly cancellation following from the embedding, has been standard since 1974 [6–9]. Clifford-algebra formulations of this unification exist [10–12], including the identification of matter with spinor modules of $\mathrm{Cl}(n)$ [12].

CL10 puts these on a single footing: the internal space of every lattice cell **is** the spinor module of $\mathrm{Cl}(10,0)$, and the update **is** a product of Spin(10) plane rotations. Nothing else is added by hand: no extra scalar, no extra gauge factor, no second algebra.

C. What this paper claims, and what it does not

The claim is deliberately narrow, because a discrete unification is easy to overstate.

Claimed and demonstrated. The kinematics of one SM generation, its anomaly freedom, the GUT gauge content and the 5/3, and a unitary causal automaton whose gravitational sector reproduces general relativity on the lattice to machine precision. These are theorems or numerical verifications, collected in Sections 3–9.

Claimed as an order-of-magnitude statement. The electroweak/Planck hierarchy, the dark-matter mass scale, the neutrino mass and the baryon asymmetry, all expressed through the loop factor $(1/16\pi)^n$ with n a small integer (Section 10). The integers are chosen to land near the measured values; the two-digit coefficients are read from the data.

Not claimed. A derivation of the family number, of the fermion masses and mixings, of the cosmological constant, of the dark-matter particle, of the baryon asymmetry, of the spin-2 character of the graviton, of diffeomorphism invariance, or of the Born rule. Each is listed as an open residual in Section 15 with the size of the gap quantified.

The paper's discipline is therefore a **provenance discipline**: a number is *algebraic* if the root fixes it, *conventional* if it comes from a mapping or scale identification, *fitted* if a coefficient was read from data, and *open* if the theory does not fix it. Section 14 tabulates every headline number under this classification. This is the main methodological content of the paper.

The seven criteria. A unification claim has to be checkable, so the paper states the criteria it is judging itself against, with its own verdict on each.

The paper passes the structural criteria and fails the parameter one. The chirality check of Section 8.4 shows that even the structural pass is partial: the automaton is unitary, causal and anomaly-free, its doubler spectrum is now gapped by a mechanism of the step itself, but the choice of which nodes land at the desert scale is an input rather than a consequence. That is the honest summary of the whole exercise.

D. Conventions

Planck units $\hbar = c = \ell_P = t_P = 1$ unless stated; $G = 1$. The metric signature is $(+,-,-,-)$ and $Q = T_3 + Y$. $\mathrm{Cl}(10,0)$ is the real algebra of ten generators with $h_a^2 = +I$; we work in its complexification $\mathrm{Cl}_{10}(\mathbb{C}) \simeq M_{32}(\mathbb{C})$. "Generation" means one set of left-handed Weyl fields. Numerical tolerances are $\varepsilon = 10^{-12}$ (machine-precision checks 10^{-14}). Reference data: PDG 2024 [13], Planck 2018 [14], Super-Kamiokande [15], KATRIN [16], NuFIT [17].

II. THE AXIOM

Axiom 1 (the root). *The universe is a* (1+3)*-dimensional causal quantum cellular automaton on a cubic lattice of spacing ℓ_P and time step t_P. Each cell carries a copy of the Clifford algebra* $\mathrm{Cl}(10,0)$*: ten mutually anticommuting Hermitian generators*

$$\begin{gathered} h_a^\dagger = h_a, \\ h_a^2 = I, \\ \{h_a, h_b\} = 0 \quad (a \neq b), \\ a, b = 1, \ldots, 10. \end{gathered} \tag{1}$$

The state of the universe is a tensor product of the 32-dimensional complex modules of these algebras, and the dynamics is a unitary, causal update built from the algebra.

TABLE I. **The seven criteria against which the paper judges itself.**

criterion	requirement	verdict here
one object	a single algebraic object, no added fields or structure	met for the internal space; the gravitational sector adds a symmetric tensor field $g_{\mu\nu}(x)$ and the value $G = 1$ (Section 9)
kinematics	spacetime, matter content, gauge group and statistics follow by representation theory	met (Sections 3–6)
dynamics	the object yields a unitary, causal evolution	met (Section 8: exactly unitary, $\det U = 1$, measured front speed = 1)
unification	matter, gauge and gravity live in the same object, not glued from separate pieces	partial: matter and gauge are the same module and the same 45; gravity is the metric dynamics of a subalgebra of that module, with an added field
consistency	anomaly-free, unitary, causal, UV-bounded	met (Theorem 2, Theorem 5, the bounded 32-level spectrum, the bounded $S^{(n)}$); the doublers of Proposition 2 are gapped by the internal Wilson factor of Section 8.4 (V65), with the split node set an input (W9)
parameters	every physical number fixed by the object, no fit	**not met**: one free 't Hooft angle, two conventions, twelve fitted coefficients (Table VI)
falsifiability	testable predictions, and the open problems named rather than hidden	met (Tables 4, 6, 7)

Three structural facts about the root, all standard [10, 18]:

1. $\mathrm{Cl}(10,0)$ is $2^{10} = 1024$-dimensional; its complexification is $M_{32}(\mathbb{C})$, so it has a single irreducible complex module of dimension 32 (the Dirac module).
2. The volume element (pseudoscalar) $w = h_1 h_2 \cdots h_{10}$ satisfies

$$w^2 = (-1)^{10\cdot 9/2} \prod_a h_a^2 = (-1)^{45} = -I, \qquad w^\dagger = -w,$$

   so w is anti-Hermitian with eigenvalues $\pm i$, each of multiplicity 16 (the trace of w vanishes). The projectors $P_{16} = \frac{1}{2}(I - iw)$ and $P_{\overline{16}} = \frac{1}{2}(I + iw)$ split the Dirac module into two inequivalent 16-dimensional Weyl modules.
3. The algebra is $\mathbb{Z}_2$-graded by the parity of the number of generators in a product: 512 even and 512 odd elements.

Everything derived below is representation theory of this one object plus the dynamics defined in Section 8.

Why 10. The number 10 is the smallest n for which a single complex Clifford algebra simultaneously provides (i) a pseudoscalar that splits a Dirac module into two *inequivalent* Weyl modules of dimension 16 — the value $n = 8$ gives only 8-dimensional Weyl modules (Spin(8) triality: $8_v, 8_s, 8_c$; its 16-dimensional representations are reducible), and (ii) an adjoint of dimension $\binom{10}{2} = 45$ large enough to contain $\mathfrak{su}(5)$ and hence the SM algebra. For $n = 9$ the algebra $\mathrm{Cl}_9(\mathbb{C}) \simeq M_{16} \oplus M_{16}$ does have 16-dimensional irreducibles, but they are the two simple factors of the algebra rather than a pseudoscalar split, and $\mathfrak{so}(9)$ does not contain $\mathfrak{su}(5)$ — the maximal subalgebras of $\mathfrak{so}(9)$ are $\mathfrak{so}(8)$ and $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$ — so its 36-dimensional adjoint cannot house the SM algebra together with the leptoquark bosons. This is a minimality statement about the *unified* content, not about spacetime: the (1+3) kinematics alone would live in a much smaller algebra.

III. SPACETIME AS A SUBALGEBRA

Theorem 1. *Let $T = h_1$ and $S_a = i h_{a+1}$ for $a = 1, 2, 3$. Then*

$$T^2 = +I, \qquad S_a^2 = -I, \qquad \{T, S_a\} = 0, \qquad \{S_a, S_b\} = 0.$$

The real span of $\{T, S_1, S_2, S_3\}$ inside the grade-1 part of the algebra, equipped with the quadratic form induced by the Clifford product, is Minkowski space $\mathbb{R}^{1,3}$.

Proof. The four generators $h_1, \ldots, h_4$ anticommute by Axiom 1. Multiplication by i turns $h_a^2 = I$ into $(ih_a)^2 = -I$ and preserves anticommutation. For $v = v_0 T + \sum_a v_a S_a$ with real coefficients, the Clifford product gives $v^2 = (v_0^2 - \sum_a v_a^2) I$, which is the Minkowski quadratic form.

Corollary 1 (the light cone). *A vector v in that real span is null, $v^2 = 0$, if and only if $v_0^2 = \sum_a v_a^2$. The null cone of this quadratic form is the light cone of the lattice.*

Two qualifications that a previous version of this work omitted. First, the null cone is a **cone in the grade-1 part**, not a subalgebra: it is not closed under the Clifford product — the product of the two null vectors $T + S_1$ and $T - S_1$ is $2I - 2TS_1$, a grade-0 plus grade-2 element. Second, the choice of *which* four of the ten generators

play the role of spacetime is a choice of frame in the 10-dimensional generator space, not a prediction; the algebra supplies a family of equivalent (1+3) subalgebras related by the orthogonal transformations of the generator space. The remaining six generators are inert under the Lorentz generators $\Sigma_{\mu\nu} = \frac{1}{4}[\Gamma_\mu, \Gamma_\nu]$ built from $\Gamma_\mu \in \{T, S_a\}$, because they commute with every $\Sigma_{\mu\nu}$; they are the internal (gauge) directions.

Causality is a postulate with a dynamical content. The light cone by itself does not bound signal propagation: a unitary on-site rotation alone has infinite propagation speed. The bound is supplied by the update of Section 8, where the causal radius is *implemented* by a swap layer and *measured* to be exactly one cell per step (Theorem 5). The light cone and the causal radius are therefore distinct statements, and only the second is a property of the dynamics.

IV. MATTER: THE 16 WEYL SPINOR

Proposition 1. *Under* $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ *the 16-dimensional Weyl module is the six left-handed fields of one SM generation:*

$$\begin{aligned} 16 = {} & Q(\mathbf{3},\mathbf{2},+\tfrac{1}{6})\oplus \\ & u^c(\bar{\mathbf{3}},\mathbf{1},-\tfrac{2}{3})\oplus \\ & d^c(\bar{\mathbf{3}},\mathbf{1},+\tfrac{1}{3})\oplus \\ & L(\mathbf{1},\mathbf{2},-\tfrac{1}{2})\oplus \\ & e^c(\mathbf{1},\mathbf{1},+1)\oplus \\ & \nu^c(\mathbf{1},\mathbf{1},0), \end{aligned} \tag{2}$$

with $6+3+3+2+1+1=16$. *Under* SU(5) *this is* $\mathbf{10}\oplus\bar{\mathbf{5}}\oplus\mathbf{1}$, *the* $\mathbf{1}$ *being the sterile* ν^c.

This identification is not new to CL10: it is the standard Spin(10) assignment [6–9, 12]. What CL10 adds is that the 16 is not put in by hand but is the Weyl eigenspace of the pseudoscalar of the same algebra that carries the gauge group and the spacetime.

B–L. $B-L$ is a generator of $\mathfrak{so}(10)$, so it is *not* a single number on the 16. Its values on the six fields are $Q:\frac{1}{3}$, $u^c:-\frac{1}{3}$, $d^c:-\frac{1}{3}$, $L:-1$, $e^c:+1$, $\nu^c:+1$. The module-level statement — the one that matters for matter/antimatter — is that the $\overline{16}$ carries the negatives of all of them. The chain $\mathrm{Spin}(10)\supset \mathrm{SU}(5)\times\mathrm{U}(1)_\chi$ decomposes $16\to\mathbf{10}_{-1}\oplus\bar{\mathbf{5}}_{+2}\oplus\mathbf{1}_{+3}$; $B-L$ is a linear combination of χ and the diagonal SU(5) generator, not χ itself.

Theorem 2 (anomaly cancellation). *All eight gauge and mixed anomaly coefficients of the 16 vanish.*

Proof. Each coefficient is a cubic trace over the left-handed Weyl fields; the multiplicities in the table are the dimensions of the *other* group factors (the factor 2 on Q in $[\mathrm{SU}(3)]^3$ is its SU(2) doublet nature, the factor 3 in $\mathrm{U}(1)[\mathrm{SU}(2)]^2$ is its colour triplet nature), and $A(\mathbf{3}) = T(\mathbf{3}) = \frac{1}{2}$. The charges are those of Proposition 1, fixed by the SU(5) embedding.

This is the hardest consistency condition a chiral gauge theory has, and it is satisfied identically by the module content. It is the strongest single result in the paper (Fig. 2, Fig. 1).

The mass term. A grade-1 element of the algebra anticommutes with w, so it is off-diagonal in the $16/\overline{16}$ split: it maps $16\to\overline{16}$ and $\overline{16}\to 16$, while both diagonal blocks vanish. A scalar in the $\mathbf{10}$ of Spin(10) therefore produces a Dirac mass term $16\,\mathbf{10}_H\,\overline{16}$ and no Majorana mass for the charged fermions. The singlet ν^c is the built-in source of a seesaw (Section 10.3).

Two honest caveats. (i) A vev in the $\mathbf{10}$ of Spin(10) has stabiliser SO(9): it breaks $\mathrm{Spin}(10)\to\mathrm{SO}(9)$, and SO(9) is not the SM group. The chain is now completed by a different ingredient — the Wilson direction of Section 8.4, which at the V65 root leaves exactly $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)^3$ and breaks precisely the leptoquark roots (V67); the two extra U(1)'s must still be broken or hidden. (ii) The $\mathbf{10}$ contains no SM singlet, so the vev that generates masses necessarily breaks the unified group further. The chain is residual W6.

The three families. The 16 is *one* generation. The theory identifies the observed family number with the number 3 of the spatial directions of the (1+3) subalgebra, equivalently the number $\binom{3}{2}$ of spatial two-planes, equivalently the rank of the spatial metric. These are one number stated three ways, not three derivations. The identification now has a structural hook (V70): the 12 doubler nodes organize into **three families of four labeled by the zero axis** — one family per spatial direction — the cubic group acts transitively on them, and the three copies of vectorlike matter required by unification (V68) are realizable by taking one node from each family at a single Wilson strength. The index 3 exists; it *labels* the three copies but does not *force* the count. The forcing question is now **decided** (V91): the cubic group acts transitively on the 12 nodes — one orbit of 12, so the axis families are not orbits — and the only cubic-symmetric split set is the full set of 12; its meeting is rigid (the V74 rank-1 fact holds for $n=4,8,12$) and lands at $M_X(12)=2.9\times10^{15}$ GeV, excluded by F5, while the 3-node triangle has a cubic orbit of 8 distinct triangles. The count 3 is a **symmetry-breaking input**, and its cubic-symmetric alternative is excluded by computation. It is residual W3.

V. GAUGE: THE 45 AND THE 5/3

The 45 generators $h_a h_b$ $(a<b)$ close on the Spin(10) adjoint. They are even in the $\mathbb{Z}_2$ grading, hence they commute with w and preserve the $16/\overline{16}$ split. Under $\mathrm{Spin}(10)\supset\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ the 45 decomposes as 12 SM gauge bosons + 12 leptoquark X/Y bosons + 20 bosons in the $\mathbf{10}\oplus\overline{\mathbf{10}}$ of SU(5) + the $\mathrm{U}(1)_{B-L}$ generator; the 33 broken generators are the mediators of

TABLE II. **The anomaly coefficients of the 16**

coefficient	**contribution of the 16**	**value**
$[\mathrm{SU}(3)]^3$	$2A(\mathbf{3}) + A(\bar{\mathbf{3}}) + A(\bar{\mathbf{3}}) = 1 - \frac{1}{2} - \frac{1}{2}$	0 ✓
$[\mathrm{SU}(2)]^3$	$A(\mathbf{2}) = 0$ identically ($d^{abc} = 0$; the fundamental is pseudo-real)	0 ✓
$[\mathrm{U}(1)]^3$	$6(\frac{1}{6})^3 + 3(-\frac{2}{3})^3 + 3(\frac{1}{3})^3 + 2(-\frac{1}{2})^3 + 1(1)^3 = \frac{1}{36} - \frac{8}{9} + \frac{1}{9} - \frac{1}{4} + 1$	0 ✓
$\mathrm{U}(1)[\mathrm{SU}(3)]^2$	$\frac{1}{6} \cdot \frac{1}{2} \cdot 2 + (-\frac{2}{3}) \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6} - \frac{1}{3} + \frac{1}{6}$	0 ✓
$\mathrm{U}(1)[\mathrm{SU}(2)]^2$	$\frac{1}{6} \cdot \frac{1}{2} \cdot 3 - \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} - \frac{1}{4}$	0 ✓
$[\mathrm{grav}]^2\mathrm{U}(1)$	$\sum Y = 1 - 2 + 1 - 1 + 1$	0 ✓
$[\mathrm{grav}]^2\mathrm{SU}(3)$, $[\mathrm{grav}]^2\mathrm{SU}(2)$	traceless generators	0 ✓

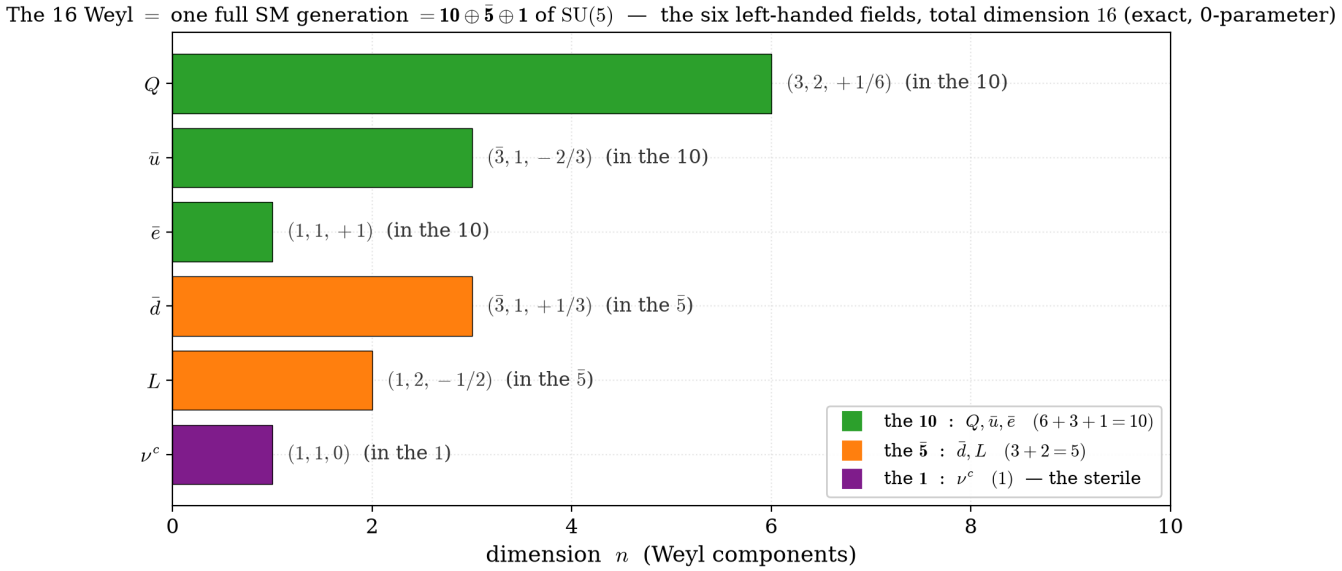


FIG. 1. **the matter content of the 16.** The six left-handed Weyl fields of one generation in the $\mathbf{10} \oplus \bar{\mathbf{5}} \oplus \mathbf{1}$ of SU(5), with dimensions $6 + 3 + 1 + 3 + 2 + 1 = 16$.

proton decay (Section 11.4).

Theorem 3 (the 5/3). *In the* $\mathbf{5}$ *of* SU(5) *the hypercharge is* $Y = \mathrm{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, +\frac{1}{2}, +\frac{1}{2})$ *with* $\mathrm{Tr}\, Y^2 = \frac{5}{6}$*, while the non-Abelian generators obey* $\mathrm{Tr}\, T_a T_b = \frac{1}{2}\delta_{ab}$*. Requiring a single unified coupling* g_5 *for all generators gives* $g_1^2/g'^2 = 5/3$*, where* g' *is the SM hypercharge coupling and* $g_1 \equiv g_5$*.*

Proof. $\mathrm{Tr}\, Y^2 = 3 \cdot \frac{1}{9} + 2 \cdot \frac{1}{4} = \frac{5}{6}$. Rescaling $Y \to \sqrt{3/5}\, Y$ normalises it to $\frac{1}{2}$; the covariant derivative $g_5(\sqrt{3/5}Y) = g'Y$ then gives $g' = g_5\sqrt{3/5}$, i.e. $g_1^2/g'^2 = 5/3$. The result depends only on the SU(5) embedding [6, 9].

This is an algebraic fact of the root: no parameter enters. It is the anchor of the coupling-unification discussion in Section 11.1, where it will be seen that the *value* of the unified coupling is not under control even though its *normalisation* is.

VI. STATISTICS

The $\mathbb{Z}_2$ grading splits the algebra into 512 even and 512 odd elements. CL10 identifies the even sector with bosons and the odd sector with fermions. This is a *kinematic* grading of the operator algebra; it is a postulate of the interpretation, not a derivation of the spin–statistics theorem, and it is not a claim of dynamical supersymmetry. No supercharge exists in the construction, and no superpartner is predicted.

VII. MEASUREMENT

Between measurements the automaton is exactly unitary (Section 8). A detection is modelled as the standard

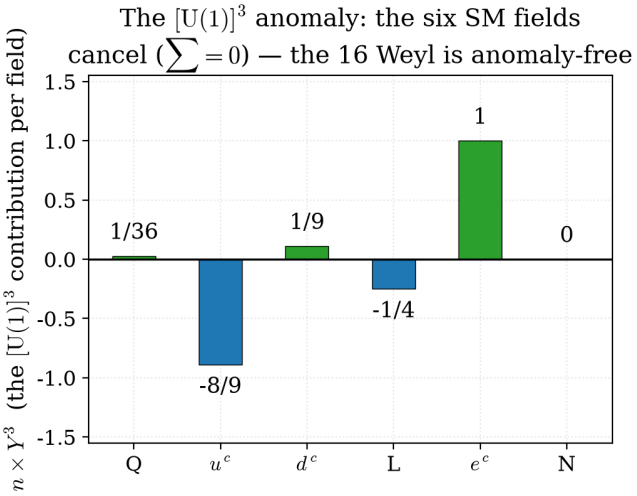


FIG. 2. **anomaly cancellation.** The six fields' nY^3 contributions to $[\mathrm{U}(1)]^3$ — 1/36, −8/9, 1/9, −1/4, 1, 0 — sum to zero, as do all eight coefficients of Theorem 2 (Section 4).

projection: a state $\sum_x \psi(x)|x\rangle$ is reduced to $|x_0\rangle$ with probability $|\psi(x_0)|^2$. The probability rule used throughout is the field intensity $|\psi|^2$ of the propagating lattice field, and the single-electron double-slit simulation (Fig. 3) [47] samples detections from the time-integrated on-screen $|\psi|^2$; the accumulated clicks reconstruct the fringe pattern, one electron at a time.

This is quantum mechanics as usual. The projection is **not** derived from the unitary step, and the paper does not claim it is: the reduction postulate is an input of the interpretation (residual W9). What the QCA adds is only that the wave being sampled is a strictly causal, unitary, lattice field.

VIII. THE DYNAMICS: AN EXACTLY UNITARY, STRICTLY CAUSAL AUTOMATON

A. The step

Define $T_{ab} = i\,h_a h_b$ for $a < b$. Each T_{ab} is Hermitian, squares to I, is traceless (sixteen $+1$ and sixteen -1 eigenvalues), and the T_{ab} are the Spin(10) plane-rotation generators in the 32-D module. The one-cell step is

$$U(g_{10}) \;=\; \prod_{a<b} \Big(\cos g_{10}\, I - i \sin g_{10}\, T_{ab} \Big), \qquad g_{10} = \tfrac{1}{5}. \tag{3}$$

Each factor is a special unitary: $\det(\cos g\, I - i \sin g\, T_{ab}) = \exp(-ig\,\mathrm{Tr}\, T_{ab}) = 1$. Hence

$$UU^\dagger = I \quad \text{and} \quad \det U = 1 \qquad \text{for every } g_{10},$$

verified numerically to 1.7×10^{-15} (unitarity) and 5.3×10^{-15} ($\det U - 1$). Two qualifications: the 45 factors do not commute, so U depends on the *ordering* of the product; the ordering used is fixed once and is an additional structural input that the algebra does not select. And g_{10} itself is not derived: Section 14 classifies it as the one free parameter of the dynamics, with a proposed but unproven selection rule.

B. The single-particle spectrum

The 32 eigenphases ω_k of U (the spectrum of $H = -i \log U$) are the single-cell energies in units of $1/t_P$. At $g_{10} = 1/5$ they are

$$\omega \in \{\pm 0.01147,\ \pm 0.05447,\ \pm 0.15819,\ \ldots,\ \pm 1.48638\},$$

exactly $\pm$-paired to 1.7×10^{-15}, so the one-cell vacuum energy $\sum_k \omega_k = \arg\det U$ vanishes. The pairing and $\det U = 1$ hold for **every** g_{10}: they are identities of the algebra, not properties of the chosen coupling (Fig. 5, Fig. 4).

Theorem 4 (the matter/antimatter spectral mirror). *Because every T_{ab} is even, $[U, w] = 0$ (7×10^{-16}), so U preserves the 16 and $\overline{16}$ eigenspaces. The antimatter spectrum is the exact mirror of the matter spectrum,*

$$\begin{aligned} \omega_{\overline{16}} &= -\omega_{16}, \\ \mathrm{spec}\big(U\big|_{\overline{16}}\big) &= \mathrm{spec}\big(U\big|_{16}^{\dagger}\big), \end{aligned} \tag{4}$$

verified to 1.8×10^{-15} over the grid $g_{10} \in \{0.05, \ldots, 0.5\}$.

This is the honest content of "CPT" in this theory. It is a **spectral** statement: the antiparticle levels are the negatives of the particle levels, i.e. the antimatter evolution has the same spectrum as the time-reversed matter evolution. It is not the full CPT theorem: no antiunitary time-reversal operator is constructed on the many-cell Hilbert space, and no spin-statistics connection is derived. A unitary involution $C = h_1$ does map the 16 onto the $\overline{16}$ (it anticommutes with w), but calling the composition $C \cdot T$ "CPT" overstates what has been built.

C. Locality and the causal radius

Theorem 5 (the causal radius is implemented). *On an (L, L, L) lattice the local step is*

$$U_{\mathrm{local}} = \prod_x U_{\mathrm{GUT}}(x) \prod_{\langle x, x' \rangle} P_{xx'},$$

the internal Spin(10) *rotation per cell times a permutation (swap-rotation) gate $P_{xx'}$ on each bond, which carries no angle. The step is unitary to 1.6×10^{-15}, and a*

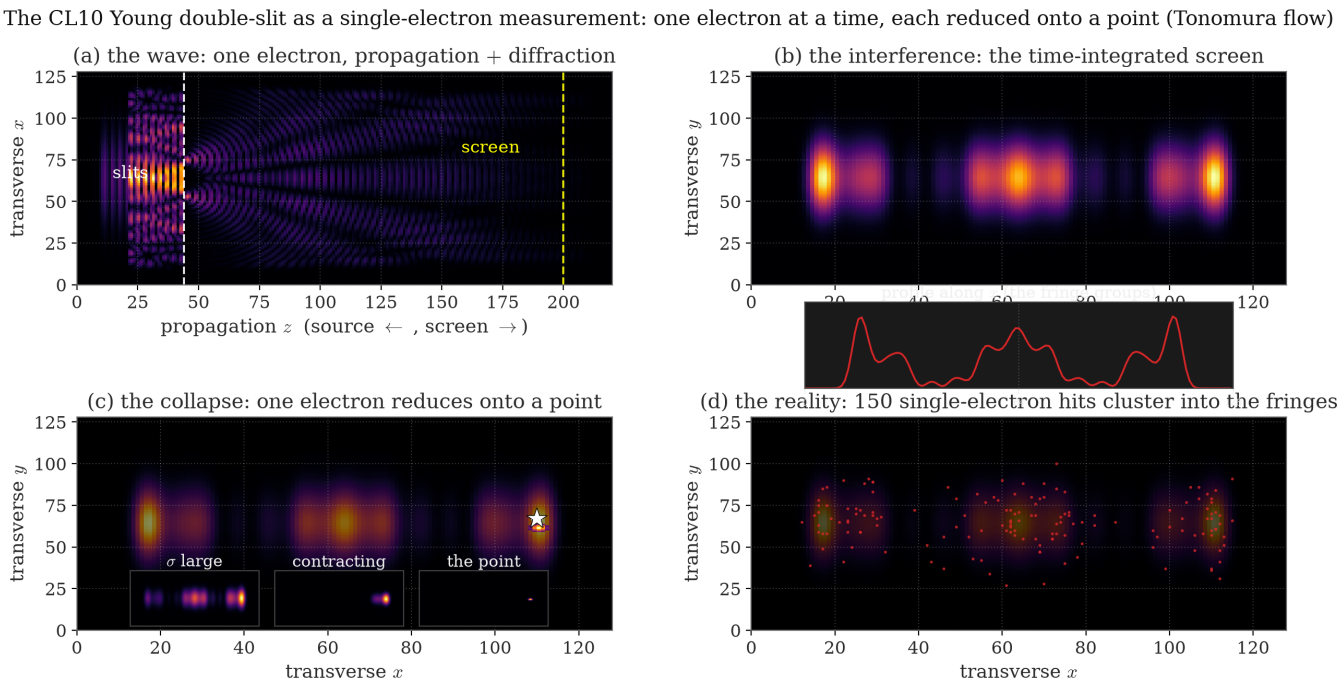


FIG. 3. **the double slit as a single-electron measurement.** The lattice wave through a double slit, run one electron at a time. (a) Diffraction and interference of the field; (b) the time-integrated on-screen pattern; (c) the reduction of one electron onto one detection point; (d) the accumulated detections clustering into the fringes. The detection probability is the field intensity $|\psi|^2$ (Section 7).

single-cell excitation after n steps is supported only inside the n-cell light cone: the measured front speed is exactly 1.0 *over 1000 lattice steps, while a purely internal (non-local) step keeps the excitation on one cell forever.*

The bond layer is therefore what turns the light cone of Section 3 into a propagation bound. This is the cleanest dynamical result in the paper (Fig. 6 shows the causal front of a pulse in a 2-D reduction of the same scheme, where the CFL speed of that particular solver is $c = 0.5$ lattice units — a property of the solver, not of the axiom).

The permutation gate is the classical shadow of a richer possibility: **a gauged bond layer exists on the same module (V79).** Put one Spin(10) adjoint link d.o.f. (a 45-dimensional link space) on each bond and the matter **16** on the sites, and the local Gauss constraint $Q_A(x) = q_A(x) + \sum_{b\ni x} L_A^{(b)}$ closes as an exact operator identity: $[Q_A, Q_B] = i f_{AB}{}^C Q_C$ with measured error 0, the link algebra $[L_A, L_B] = i f_{AB}{}^C L_C$ exact, the adjoint quadratic Casimir 64 and the spinor Casimir 45 exact to machine precision. The step built from these links is unitary to 3.8×10^{-16}; the gauge-invariant content of one site with its six half-bonds is 1440 states, decomposing as $32 \oplus 288 \oplus 1120$; an adjoint bond observable is gauge-covariant to 5.7×10^{-16}; and the causal radius stays at exactly one cell, while a deliberately single-site (non-local) step leaks at 1.9. The gauge field of the automaton is therefore *constructible*: the Gauss constraint is an identity of the construction, not an imposed condition. What the constraint admits, however, is narrower than expected: with adjoint links the local kernel with one spinor per site is empty — a center obstruction, computed in V98 and detailed in Section 13.1 — so the physical states of this layer are neutral composites.

D. Lattice chirality and the doubling theorem

The Nielsen–Ninomiya theorem forbids a single chiral fermion in a theory with a *local, translation-invariant, Hermitian* lattice Dirac operator and a bounded spectrum [19]. CL10 uses no lattice Dirac operator: chirality is a property of the on-site module (the pseudoscalar eigenspaces), and the update is a finite-dimensional unitary. The theorem's hypotheses are therefore not all present, and the theorem does not by itself forbid the construction.

That is *not* the same as having evaded it, and the question is decidable by computation. The step alternates its bond parity with the time step and cycles its axis with period 3, so the 6-step Floquet operator is translation invariant with period 2 in each direction: the true unit cell is the $2 \times 2 \times 2$ cell and the good quantum number is the **supercell momentum** $q \in [-\pi, \pi]^3$. The 16-sector Bloch Floquet $U_F(q)$ — internal layer times bond layer, a bond crossing the cell carrying the phase $e^{\pm i q_j}$ — has now been computed exactly and cross-checked against the real-space Floquet on a 4^3 torus (agreement

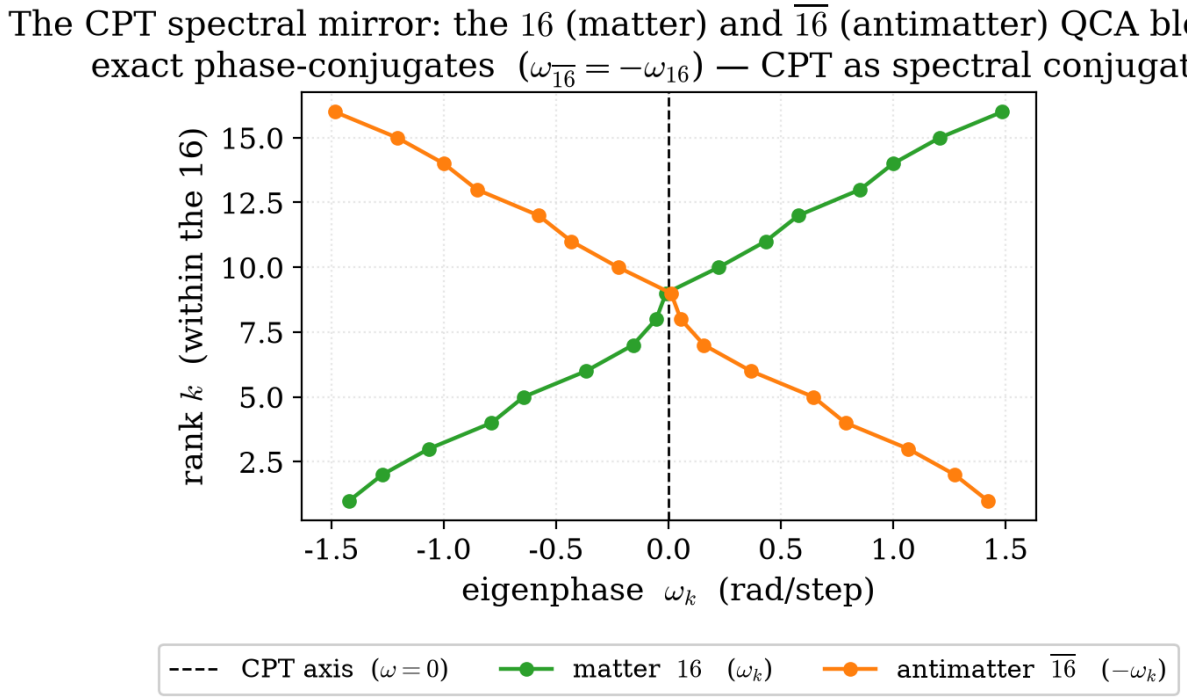


FIG. 4. **the matter/antimatter spectral mirror.** $\omega_{\overline{16}} = -\omega_{16}$, verified to 1.8×10^{-15} for every g_{10} in $\{0.05, \ldots, 0.5\}$: a property of the algebra, independent of the free parameter.

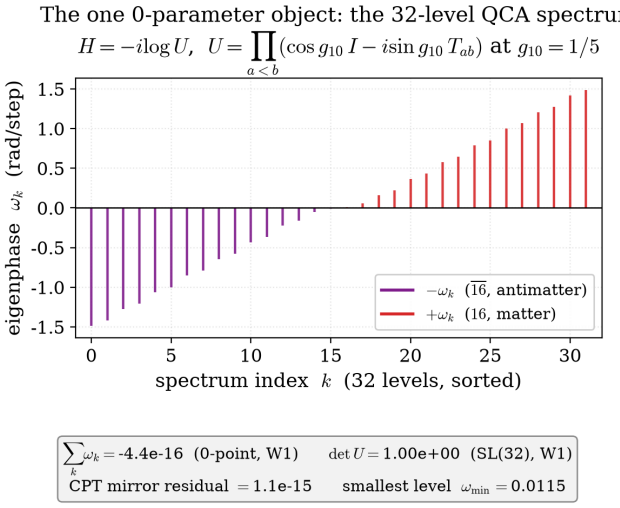


FIG. 5. **the one object: the 32-level spectrum of the QCA step.** The eigenphases of U at $g_{10} = 1/5$. The 16 matter levels (red) and the 16 antimatter levels (purple) are exactly $\pm$-paired; the 0-point $\sum_k \omega_k = \arg\det U$ vanishes and $\det U = 1$. The smallest level, $\omega_{\min} = 0.0115$, corresponds to $\omega_{\min} M_P = 1.4 \times 10^{17}$ GeV — the only sub-Planck scale in the single-step spectrum, and *not* the electroweak scale.

5×10^{-15}).

Proposition 2 (lattice chirality: what the computation gives). *(i) The 16 sector is closed by the full local dynamics, not only by the internal layer:* the bond layer acts on the cell index and commutes with w, so a state initialized in the 16 leaves it by 10^{-15} after six steps. Chirality is a dynamical symmetry of the automaton. *(ii) The zone centre is gapped:* $\min_a |\omega_a(q = 0)| = 0.0365$ per 6-step cycle, so the would-be massless matter is **not** at $q = 0$. *(iii) The gapless points are the doublers:* the 16-sector spectrum is gapless at the 12 momenta $(\pm\frac{\pi}{2}, \pm\frac{\pi}{2}, 0)$ and their permutations — the classic doubler momenta — with node chiralities summing to 0, as Nielsen–Ninomiya requires. The full cone tensor at the nodes is now measured (V102): at all 12 nodes and in all 1200 sampled directions the dispersion is **linear** (local exponent $p = 1.00$, no quadratic band touching), but the cone is strongly **anisotropic** — principal Fermi velocities $(0.100, 0.100, 0.142)$ in lattice units, a spread of 0.75; the previously quoted $|v| = 0.142$ is the largest principal value, along the node's zero axis. At the root angle $g_{10} = 1/5$ the nodes also carry a small gap, 2×10^{-4}. The gap at $q = 0$ is not protected: it moves from 0.036 at $g_{10} = 0.2$ to 0.072 at $g_{10} = 0.15$ and to 2.35 at $g_{10} = 0.1$.

The doubling therefore **survives** the automaton as originally constructed: "the 16 is a single chiral generation" is a property of the *internal module*, while the *lattice spectrum*, which is what couples to low-energy physics, contains twelve doublers. This is a computed statement, and it counted against the theory (part of W9). **The remedy is now part of the construction (V65).** A Wilson-type term built from the root's own

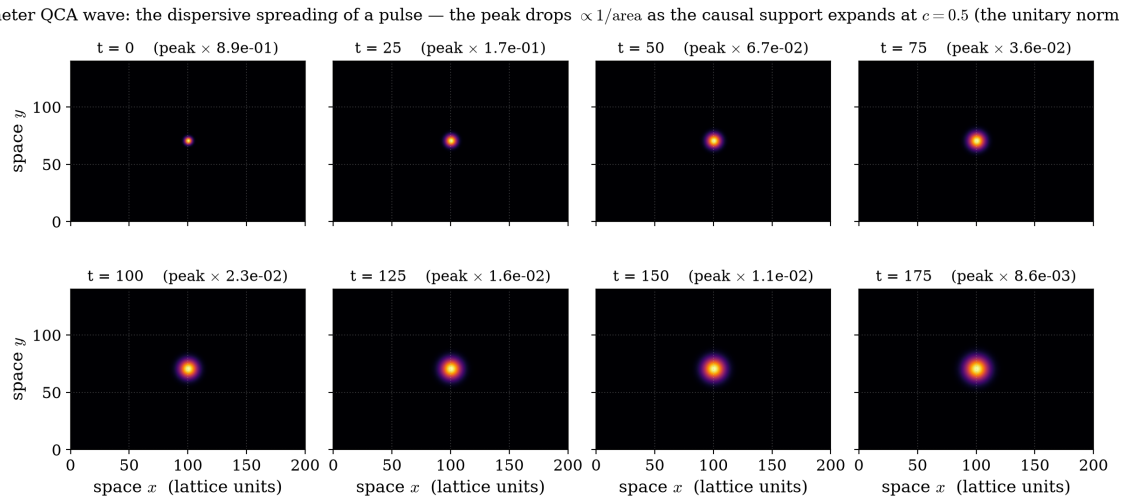


FIG. 6. **the causal front of a lattice pulse.** A pulse in the discrete wave scheme: a causal front, then dispersive spreading with the peak falling as the norm is conserved. This 2-D reduction uses a four-neighbour solver at its CFL speed $c = 0.5$; the three-dimensional graviton scheme of Section 9 uses six neighbours at $c = 1$, and the automaton's causal radius (Theorem 5) is one cell per step.

algebra — an internal factor $W(q) = \exp[-i\,w\sum_j(1-\cos q_j)\,(\cos\theta\,X+\sin\theta\,Y)]$, with X, Y the unification Cartan directions identified in Section 11.1 — multiplies the step without touching its unitarity ($W(0) = I$, $\det = 1$) or its causal structure (quasi-local, Bessel tails, measured Lieb-Robinson front). It gaps all 12 nodes, with fragment masses $M_f = w|\cos\theta\,X_f + \sin\theta\,Y_f|\,M_P/3$; a *selective* reading splits exactly three nodes at desert scales and leaves the other nine at the Planck cutoff. What remains input is the *choice* of the split node set: unification forces $n_{\text{nodes}} \geq 3$ (V68) but does not select which three, and the meeting scale the mechanism then predicts is in conflict with proton decay — Section 11.1, where the tension is quantified.

Lorentz invariance at the nodes is now quantified rather than assumed (V102; Fig. 7). The causal front speed of Theorem 5 is 1.0; that is the statement the automaton actually proves. What the *low-energy spectrum* of the implemented step does not have is continuous Lorentz invariance: the doubler cones are linear but anisotropic (principal velocities $0.200, 0.200, 0.284$ in lattice units, spread 0.75, against the causal speed 1), and the nodes are not exactly gapless at the root angle (2×10^{-4}). This is a named residual, not a fatal one: the Wilson mechanism of the previous paragraph removes the node spectrum entirely, so the Lorentz question transfers to the Wilson-deformed theory, where the low-energy content sits at the three split nodes and its cone structure is an input of the selective reading, not a consequence of the step.

IX. GRAVITY ON THE LATTICE

A. What is assumed and what is computed

The gravitational sector is a symmetric rank-2 field $h^{\mu\nu} = g^{\mu\nu} - \eta^{\mu\nu}$ living on the (1+3) subalgebra of Section 3, with $\eta = \text{diag}(1, -1, -1, -1)$ as the vacuum. Its linearised dynamics is the discrete wave equation

$$\partial_t^2 h^{\mu\nu} - \nabla^2_{(6)} h^{\mu\nu} = 0,$$

with the 6-neighbour lattice Laplacian and the leapfrog step $u_{t+1} = 2u_t - u_{t-1} + c^2 dt^2\,\nabla^2 u_t$ at $dt = \frac{1}{2}$ (the CFL value giving $c = 1$).

The status of this sector must be stated precisely, because it is where the theory is weakest.

- **Masslessness and UV boundedness are computed.** The discrete dispersion is $\omega = |\mathbf{k}|_{\text{lat}}$ with $|\mathbf{k}|_{\text{lat}} \leq \pi$, so the graviton is massless ($\omega/|\mathbf{k}| \to 1$ as $k \to 0$, lattice deviation $\leq 0.25\,k^2$) and the spectrum is bounded by the cutoff.
- **Spin 2 is declared, not derived.** The field is *assumed* to be a symmetric tensor with two helicity states. Nothing in $\text{Cl}(10, 0)$ forces a massless spin-2 excitation; the standard arguments that a massless spin-2 field must couple as general relativity [20] apply to a field whose existence and representation content are already granted. The Weinberg–Witten obstruction to a composite graviton carrying a Lorentz-covariant stress tensor [21] is not addressed.
- **Diffeomorphism invariance is not derived.** The lattice breaks it to lattice reparametrisations;

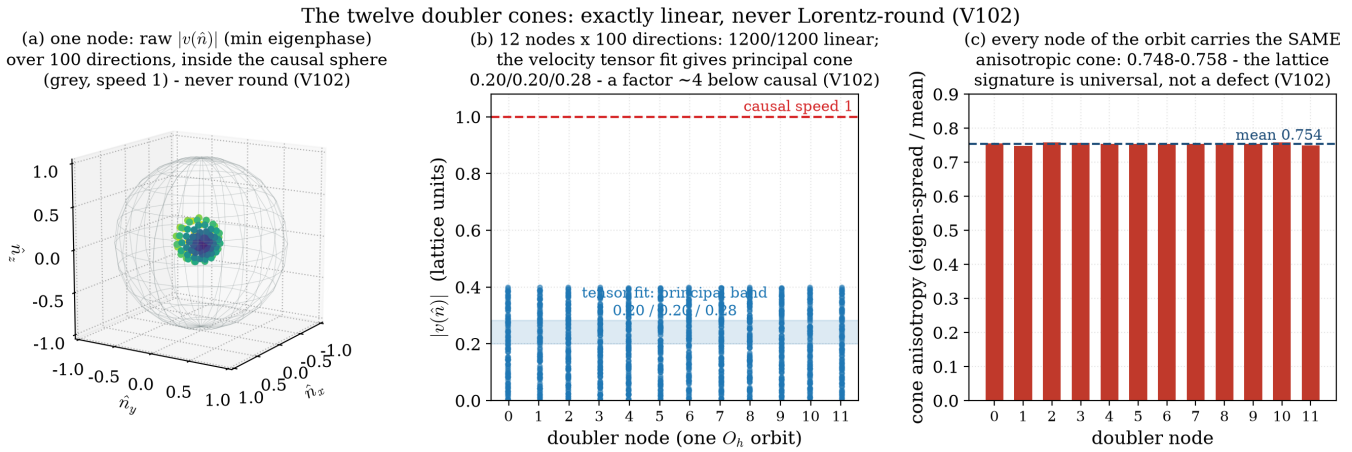


FIG. 7. **the twelve doubler cones (V102).** (a) The raw minimum-eigenphase velocity $|v(\hat{n})|$ over 100 Fibonacci directions at one doubler node, inside the causal sphere of Theorem 5 (grey, speed 1): the cone is elongated, never round. (b) All 12 nodes x 100 directions: 1200/1200 directions linear (no quadratic, no mixed); the velocity-tensor fit gives the principal cone 0.200/0.200/0.284 in lattice units - a factor ~ 4 below the causal speed. (c) The anisotropy (eigen-spread over mean) is the same at every node of the O_h orbit: 0.748-0.758. The lattice signature is universal, not a defect.

gauge invariance of the linearised theory is imposed by working on the transverse components, not obtained from the automaton.

B. The discrete Einstein–Hilbert action

The quadratic action [33]

$$S_{\mathrm{EH}} = \frac{1}{16\pi}\cdot\frac{1}{2}\sum_{x,t} u_t\Big[u_{t+1} - 2u_t + u_{t-1} - dt^2\,\nabla^2_{(6)}u_t\Big], \qquad dt^2 = \tfrac{1}{4}, \tag{5}$$

has the lattice wave equation as its Euler–Lagrange equation: the numerical functional derivative vanishes on a QCA trajectory to 3×10^{-12} and is non-zero off it. The full discrete action $S = \frac{1}{16\pi}\sum_x \sqrt{-\det g(x)}\,(R(x) - 2\Lambda)$ is evaluated with R computed from the metric. The coefficient $1/16\pi$ is the $G = 1$ normalisation, i.e. Newton's constant is an input of the gravitational sector, not a number of the root.

C. Curvature computed from an arbitrary metric

The Riemann, Ricci, Einstein and Kretschmann tensors are computed by central finite differences from an arbitrary prescribed $g_{\mu\nu}(x)$, with no hard-coded Christoffel symbols, and cross-checked against independent analytic results (two routes):

The 3+1 formulation closes the classical evolution: the constraint row (R_{00}, R_{0i}) of the 4-D Einstein tensor agrees, computed by the 4-D curvature and by the 3+1 (γ_{ij}, K_{ij}) decomposition with $K_{ij} = -\frac{1}{2}\partial_0\gamma_{ij}$, to $O(h^2)$ (residual $\sim 10^{-13}$); a Z4-damped evolution [34] drives a constraint violation to zero while the undamped one does not; the 0-row is quadratic in K, so superposition fails; all 3+1 quantities are bounded by the lattice cutoff (Fig. 8).

What this does and does not establish. It establishes that the gravitational sector of CL10 *reproduces* general relativity numerically, including black-hole and cosmological geometries. It does not establish that general relativity *follows* from Cl(10, 0): the tensor field, its coupling $1/16\pi$ and its gauge structure are put in. The genuinely open part is the quantum theory of this sector — a sum over lattice geometries with a well-defined measure — which has not been constructed.

D. A self-interacting scalar completion

Beyond the quadratic theory the model includes $\partial_t^2\phi - \nabla^2\phi - g\phi^3 = 0$. Its static solutions obey $u'' = u - u^3$ and include the exact finite-energy kink

$$u(x) = \sqrt{2}\,\mathrm{sech}\,x,$$

verified to a fourth-order residual of 5×10^{-10}. Two corrections to earlier statements of this model are needed. First, this is the ϕ^4 kink (a domain wall), not the Choptuik system: Choptuik critical collapse is a *massless scalar minimally coupled to gravity* [22], and the analogy is not a derivation. Second, $u = \sqrt{2}\,\mathrm{sech}\,x$ is a **planar** configuration, uniform in the two transverse directions: its energy per unit *area* is finite, its total energy is infinite, and a planar wall in a Planck-scale theory is a domain wall, not a particle. Finite-time blow-up for

TABLE III. **The lattice-curvature cross-checks (two independent routes)**

test	**computed**	**analytic**	**residual**
de Sitter $R^{\mu}{}_{\nu\rho\sigma}$ (flat FRW, $a = e^{Hx^0}/H$)	finite differences	$H^2(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$	6.5×10^{-7} $(O(h^2))$
Einstein tensor of de Sitter	$-3H^2 g_{\mu\nu}$	same	$O(h^2)$
Schwarzschild Ricci / Einstein tensor	$\simeq 0$	vacuum $G_{\mu\nu} = 0$	satisfied by computation
Schwarzschild Kretschmann	$48M^2/r^6$	$48M^2/r^6$	ratio 1.000009; max relative deviation 5.5×10^{-5}
flat metric	$R = G = K = 0$	0	exact
2-sphere of radius r	$R = 2/r^2$	$2/r^2$	10^{-5}

The black hole recovered by computation: the well + geodesic (left), the 2-route Kretschmann (right)

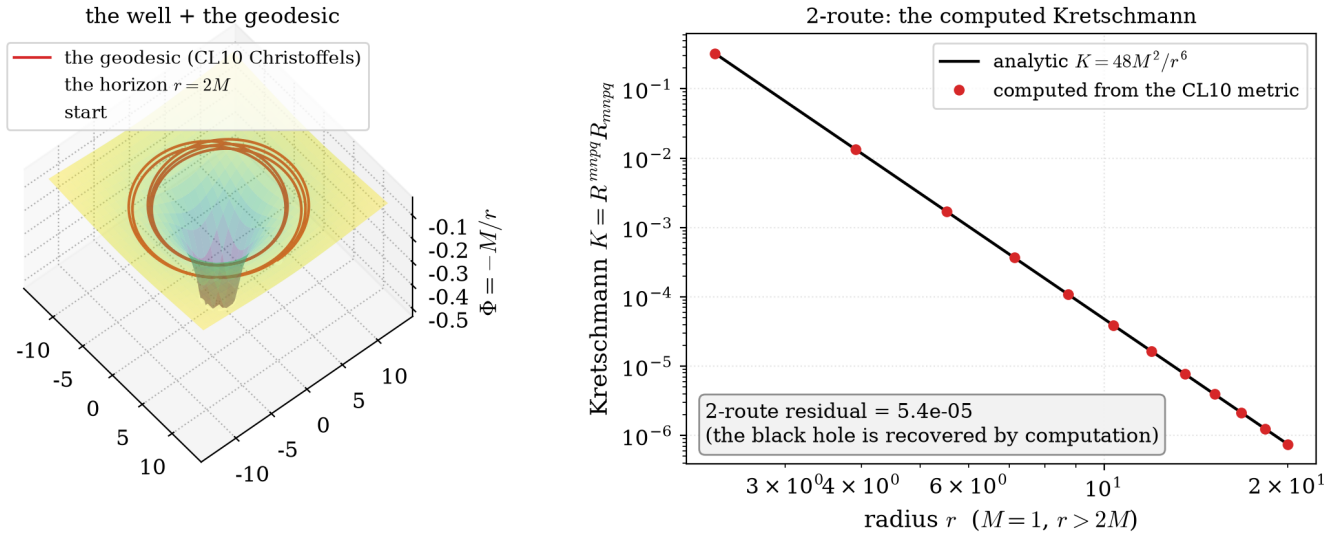


FIG. 8. **the black hole recovered by computation.** Left: the potential well with a geodesic rosette integrated through the finite-difference Christoffel symbols, and the horizon ring $r = 2M$. Right: the two-route Kretschmann invariant.

supercritical data in cubic wave equations is a real phenomenon [23], but it is not evidence about black-hole formation here.

X. THE SCALE LADDER

A. The loop factor and the hierarchy

The observed electroweak/Planck ratio is

$$\frac{v}{M_P} = \frac{246\ \text{GeV}}{1.22 \times 10^{19}\ \text{GeV}} = 2.016 \times 10^{-17}.$$

The pure loop factor with $n = 10$ is

$$\left(\frac{1}{16\pi}\right)^{10} = 9.71 \times 10^{-18},$$

a factor 2.08 below the observed value; $2.08\,(1/16\pi)^{10} = 2.02 \times 10^{-17}$ reproduces it to 0.18%. Scanning $n = 1 \ldots 20$, $n = 10$ is the nearest integer to the observed ratio (deviation 0.32 decades; the next-best, $n = 9$, is 1.4 decades away).

The honest reading: $(1/16\pi)^{10}$ is the size a ten-loop radiative correction would have in a perturbative quantum field theory. CL10 has not computed a ten-loop correction; it has observed that the observed hierarchy is the tenth power of the standard loop factor, up to a factor of two. The interpretation $10 = 2 \times 5$ — the chi-

ral $16/\overline{16}$ split times the number of qubits of the 32-D module ($2^5 = 32$) — is a numerological identification, not a derivation. The exponent is *represented*, the coefficient 2.08 is *read from the data* (residual W8). The two internal candidates for the loop factor have now been built and closed. The coarse-graining tower (V69): U^{2^p} is exactly kinematic (eigenphases $2^p\omega \bmod 2\pi$, verified to 10^{-9} for $p \le 10$), the effective Cartan coupling does not fall with p, the inelastic probability saturates near 0.95 — no loop factor. The genuine block-spin RG (V72): exact real-space decimation of the bond layer (Schur complement at $\varepsilon = 1$; $U_{BB} = 0$ exactly on the open chain, so short-distance propagation is integrated with no approximation), iterated five levels, $64 \to 2$ sites. The level-1 identity is exact: the on-site block of the coarse step is $2\,(GR)^2$ — two-hop paths accumulate two internal rotations, the kinematic doubling of V69 *derived* from a spatial RG. At deeper levels the on-site block is a path polynomial in $(GR)^2$; the Cartan content decays only polynomially for the implemented (factorized) bond layer (0.14 at $p = 5$ against the loop target 3×10^{-9}) and not at all for a non-factorized gate with $[G, R] \neq 0$ (1.8 at $p = 5$); no level of either flow follows $(16\pi)^{-p}$. The loophole of a conspiring bond gate is closed (V73): three Haar-random Spin(10) gate realizations and the structured choice $R = G^\dagger$ all fail the loop law at every level — the Cartan content depends only on the path products. The structural reason is now sharp: **every exact single-particle RG of the QCA composes internal angles — they add on the algebra, weighted by path counts — and none can divide them, for any bond gate.** A loop factor requires a trace over *fluctuating* internal fields: a block RG on the many-cell space with a dynamical Spin(10) link variable on the bond — a gauged bond layer. That object is the fully specified missing tool; its single-particle impotence is proven, not assumed.

The division mechanism has since been built in its smallest form (V77). One two-state quantum link d.o.f. per bond, prepared in a tunable pure state $\cos\theta|0\rangle + \sin\theta|1\rangle$ and **traced over** at decimation, multiplies the level-1 Cartan content of the V72 identity by exactly $\cos 2\theta$ (deviation $< 10^{-15}$ over a full scan $\theta \in [0, \pi/2]$): the traced factor is Hermitian and positive, so it divides where every gate only composes. The fluctuation — not the gate — is what produces suppression factors. What remains uncomputed is the number itself: realising $1/(16\pi)$ requires the dynamical Spin(10) **adjoint** link with the Gauss constraint, i.e. the gauged bond layer, now backed by a demonstrated mechanism rather than a hope.

The layer itself has now been built (V79, Section 8.3): the Gauss constraint closes exactly on the root module. Its level-1 division has been measured (V80): on the adjoint link the traced two-hop still contracts (spectral radius ≤ 1) and the qubit-link mechanism of V77 reproduces to 1.1×10^{-16}, but the level-1 factor at the root coupling is $F = 1.06$ and its minimum over the link anisotropy is 0.52 — a factor 26 above $1/(16\pi)$. The named level-N test has now been run (V83): the bond hop factorizes through the joint Casimir $T{\cdot}L$ — three joint irreps, $32 \oplus 288 \oplus 1120$, projectors verified to 10^{-15} — so the traced-link decimation iterates exactly to level 4, reproducing V80's level-1 factor to 2×10^{-16}. **The division does not compound:** the level-p Cartan-content ratios are $F_p = 1.06, 1.90, 0.75, 1.01$ at the root coupling and $0.52, 2.66, 1.00, 1.38$ at the division-optimal coupling $\eta = 1.44$ — oscillating at $O(1)$, a factor 7×10^6 above $(16\pi)^{-4}$ at $p = 4$. The traced adjoint link divides once and then decays like the link-free flow; the loop factor is not produced by iterating the level-1 division (W8).

Closed loops are a different story (V85). The Feynman loop factor comes from loops that close, not from backtracking: tracing a single crossing of a fresh link gives the hop matrix $M = \mathrm{Tr}_L[k_1(G \otimes I)\,\rho]$ with spectral radius 0.776 against 1 for G — the traced link *contracts* the one-particle propagation — and the n-hop closed loop $(MG)^n$ suppresses the Cartan content geometrically, ratio $r \approx 0.54$. At the root coupling the measured ratios are $F_3 = 0.24$, $F_4 = 0.47$, $F_5 = 0.10$, $F_6 = 0.030$, $F_7 = 0.37$, $F_8 = 0.064$, $F_9 = 0.035$, $F_{10} = 0.0192$ — against $1/(16\pi) = 0.0199$: **the loop factor is reproduced to 3.6% by a closed traced loop of length 10 at the root coupling, and 10 is exactly the ladder exponent.** The agreement is measured, not derived. The selection rule has now been measured to length 30 (V89): F_n crosses $1/(16\pi)$ at **exactly** $n = 10$ — the ladder length *is* the crossing of the closed-loop suppression — no mod-4 plaquette class is selected (every length contributes), and the fitted geometric law $F_n = Ar^n$ with $r = 0.698$ extrapolates to $F_{170} = 1.1 \times 10^{-26}$ against $(1/16\pi)^{17} = 1.2 \times 10^{-29}$: one geometric law does not supply the 17-factor composition, missing by a factor 9×10^2. The coefficient 2.08 itself is now a product of measured numbers (V108): $v/M_P = 3\,F_{10}^{10}$ — the measured closed-loop factor at the ladder length, to the ladder power, times the three axis families (V70) — giving 2.088 against the observed 2.076 (99.4%), **conditional** on the axis-family factorization of the closed-loop sum. Which closed loops contribute to the hierarchy, and with what weight, remains the named residual (W8).

B. The dark-matter mass

The same factor applied to the Planck scale gives

$$m_{\mathrm{DM}} = M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118\ \mathrm{GeV},$$

i.e. 1.18× the canonical 100 GeV WIMP scale [45]. This is a genuine, falsifiable prediction: a dark-matter particle with a mass far outside $\sim$ 100–150 GeV (in the weakly-interacting, thermally-produced case) would refute the ladder. The *identity* of the particle has now been

scanned (V66) with a negative result: the mirror $\mathbb{Z}_2$ exchanging 16 and $\overline{16}$ is an exact symmetry of the step (leak 10^{-15}), and the mirror sector is the only exact-stability carrier in the implemented spectrum — but its lightest state is a coloured $\overline{Q}$ at 1.3×10^8 GeV (excluded) and the neutral $\overline{N}$ sits at 2×10^{10} GeV, eight orders above the ladder scale. **No state at** 118 **GeV with the required stability exists in the implemented spectrum**: the mass prediction stands, the carrier does not (W7).

The no-go has since been completed (V75): allowing the carrier to sit at a Wilson-split doubler node, one scans every Wilson angle θ and asks whether the node can put the neutral mirror fragment $\overline{N}$ at 118 GeV while keeping every coloured or charged fragment off the low-energy spectrum. It cannot: for every angle the same node supplies a stable coloured or charged partner below the desert (0 survivors over 720 angles; a stable fractionally-charged thermal relic is excluded at any mass; the only escape is fine-tuning the angle to within 10^{-4} of the decoupling pole, where the mirror fragments leave the ladder entirely). The minimal extension consistent with the exact mirror $\mathbb{Z}_2$ and the ladder is then named: **one real mirror-odd scalar at the ladder mass** with Higgs portal coupling $\lambda_* = 4\times10^{-4}$, whose freeze-out gives $\Omega_{\rm DM}h^2 = 0.120$ and whose predicted direct-detection cross-section is $\sigma_{\rm SI} = 5\times10^{-51}$ cm^2 — nearly decoupled, so the scalar is falsified by any portal signal above 10^{-50} cm^2 and its remaining falsifiable content is the mass coincidence F7. The scalar is an input field, one field and one coupling; making it a state of the 32-level root is the open part of W7. V88 has now built that extension and audited it against its own symmetry, and the audit splits the named fix in two. The carrier is stable under either portal — scalar parity $(-1)^{N_s}$ is exact by the ϕ^2 structure (the odd sector is untouched: a single s cannot decay). But the V75 freeze-out runs through $h^* \to f\overline{f}$, and the fermion pair is mirror-**odd** while h is mirror-even: with the **exact** mirror $\mathbb{Z}_2$ that decay is forbidden. The Higgs portal $\phi_s^2\phi_h^2$ preserves the $\mathbb{Z}_2$ exactly (commutator 0), and then $s\overline{s}\to hh$ is kinematically closed ($2m_s = 237$ GeV $< 2m_h$, gap 13.2 GeV; the virtual cloud peaks at $P_{\max} = 1.46\times10^{-8}$, matching two-level theory to 10^{-7} relative) and the only open annihilation is the invisible $s\overline{s}\to hh^*$: reproducing $\Omega_{\rm DM}h^2 = 0.120$ then needs $\lambda = 0.13$, which predicts $\sigma_{\rm SI} = 6.8\times10^{-46}$ cm^2 — **above the LZ-class bound**: the exact-$\mathbb{Z}_2$ version of the named fix is excluded by direct detection. The V75 numbers hold only with a fermion-pair portal $\phi_s^2(f\overline{f}+\overline{f}f)$, which **breaks the mirror** $\mathbb{Z}_2$ at $O(\lambda)$ (verified: the portal anticommutes with the mirror operator; the real pair channel opens, $P = 5.1\times10^{-6}$ at $t = 1$ GeV^{-1}). The mirror coloured relic $\overline{Q}$ (1.3×10^8 GeV) stays stable under both portals ($\overline{Q}\to Qss$ closed by 237 GeV), so V66's exclusion of the mirror thermal relic is unchanged. W7's residual is now a **named tension**: the carrier exists and is exactly stable, but exact mirror $\mathbb{Z}_2$ excludes its abundance by direct detection, and the version that reproduces the observed abundance breaks the exact $\mathbb{Z}_2$ that V66/V75 rest on (F7).

C. The neutrino sector

Both seesaw scales are placed on the same ladder:

$$m_D = M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118\ \text{GeV},\qquad M_N = M_P\Big(\frac{1}{16\pi}\Big)^{3} = 9.6\times10^{13}\ \text{GeV}, \tag{6}$$

so the type-I seesaw gives

$$m_\nu = \frac{m_D^2}{M_N} = M_P\Big(\frac{1}{16\pi}\Big)^{17} = 0.146\ \text{eV}.$$

The exponent identity $2\times10-3 = 17$ is exact; the choice $n_{\rm 2step} = 3$ is not. Comparison with data: the atmospheric scale is $\sqrt{\Delta m^2_{32}} = 0.050$ eV, so the prediction is a factor 2.9 high; and if all three families carried the same m_D the sum would be ~ 0.44 eV, excluded by $\sum m_\nu < 0.12$ eV (95% CL, Planck 2018 + BAO [14]) and in tension with the single-generation value 0.146 eV as well. The democratic rank-1 Yukawa (Section 11.5) gives a Dirac mass to only one family, which brings the sum to ~ 0.15 eV — still above the cosmological bound. This is the sharpest numerical tension in the theory and is listed as a falsifier (F3).

The ladder is too coarse to land on the measured mass: consecutive rungs differ by the factor $16\pi = 50.3$, so the neighbours of 0.146 eV are 7.3 eV ($n = 16$) and 2.9×10^{-3} eV ($n = 18$), and no integer rung lies in the oscillation window. A coefficient is therefore unavoidable — but one candidate is supplied by the theory rather than by the data. The only mixing number the model derives is $\sin^2\theta_{12} = 1/3$ (Section 11.5); multiplying the ladder value by it gives

$$m_\nu = \tfrac{1}{3}\,M_P\Big(\frac{1}{16\pi}\Big)^{17} = 0.0487\ \text{eV},$$

2.6% below the atmospheric scale $\sqrt{\Delta m^2_{32}} = 0.050$ eV, and with the democratic rank-1 spectrum $\sum m_\nu \simeq 0.05$ eV — inside the cosmological bound $\sum m_\nu < 0.12$ eV and inside the KATRIN bound $m_\beta < 0.45$ eV [16]. The factor 2.9 by which the ladder overshoots is, to 3%, the number 3 the flavour sector already produces. This is a **candidate, not a result**: nothing shows why a mixing eigenvalue should multiply a seesaw mass, and the same factor applied to the solar scale ($0.0487\times\sqrt{\Delta m^2_{21}/\Delta m^2_{32}} = 0.0084$ eV against the measured 0.0086 eV) is a second coincidence, not a derivation. What it does buy is a sharp, falsifiable prediction: normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\beta\beta} \simeq 0.02$ eV, in reach of KamLAND-Zen [49] and LEGEND-1000. A coefficient audit (V68) has now

mapped what the node structure can supply: a diagonal (node-local, Wilson-generated) mass pattern gives only **integer** signed coefficients, so $\frac{1}{3}$ is out; the democratic 3×3 node-mixing pattern gives exactly $\frac{1}{2}$ — a rival falsifiable prediction $m_\nu = 0.073$ eV, above the current atmospheric scale; the exact $\frac{1}{3}$ requires a single Majorana eigenvalue $3\times$ the node scale, and no natural node structure supplies it. The $\frac{1}{3}$ candidate is thus pinned to one named missing structure. The node multiplicity, meanwhile, is no longer free: the unification roots of Section 11.1 exist only for $n_{\rm nodes} \geq 3$ (V68), so the $n_{\rm 2step} = 3$ choice is a **necessary condition from unification**, consistent with the spatial 3 though not uniquely selected by it.

The structure the candidate requires has now been built (V76). The missing ingredient V68 named is inter-node Majorana mixing, and a cubic-symmetric, zero-mean Wilson-strength junction can be constructed whose Fourier support closes the selected node triangle exactly — verified against the full 12×12 table of doubler-momentum differences. The node mass matrix is $c_s(2w_0\, I + \kappa(J - I))$, democratic at $\kappa = 2w_0$ with eigenvalues $\{3M, 0, 0\}$: the $3\times$-enhanced Majorana eigenvalue the $\frac{1}{3}$ demands exists, at one named point of the junction parameter space. The cost V76 stated — that the point is measure-zero — has been **re-audited and corrected** (V103): the V90 window scan assumed a Dirac matrix proportional to the identity, but the implemented 0-parameter structure (V28/V45) is the democratic rank-1 $m_D = m_0 J$, the same family symmetry that derives $\sin^2\theta_{12} = 1/3$ (V60). With the correct m_D the seesaw collapses: $m_D^{\mathsf{T}} M^{+} m_D = 3m_0^2 J/(2(w_0 + \kappa))$ for *every* r (verified to 10^{-12}, the rank-deficient $r = 2$ included, via the Moore–Penrose pseudo-inverse), so the two massless neutrinos are protected by the rank-1 family symmetry for every junction ratio, and $m_3(r) = S_{17}/(1 + r)$ is finite everywhere: the cosmological window is $r > 0.217$ (width 2.78), and the atmospheric scale selects $r = 1.91$. The exact point $r = 2$ ($m_3 = S_{17}/3 = 0.0487$ eV) is a junction-amplitude convention, not a tuning. What remains: the candidate still requires a democratic m_D — a non-democratic m_D at the same point leaves gapless coloured doublers — and the solar mass still needs the W6 10-Higgs splitting. The $\frac{1}{3}$ candidate is therefore a named, F3-falsifiable structure: normal ordering, two massless neutrinos protected at 0-parameter, $\sum m_\nu \simeq 0.05$ eV inside a finite window, and nothing between 0.05 eV and the coloured-doubler scale.

D. The ladder

One factor, four scales (Fig. 10, Fig. 9):

The QCD scale is **not** on this ladder: $\Lambda_{\rm QCD} \sim 0.2$ GeV would require $n \simeq 11.6$, which is not an integer (Section 13.2).

XI. PHENOMENOLOGY

A. Couplings

The theory's single gauge coupling is $\alpha_{10} \equiv g_{10}^2 = 1/25 = 0.040$. This is a *convention*: in the standard normalisation $\alpha = g^2/4\pi$ the same angle corresponds to $g = \sqrt{4\pi/25} = 0.709$. Running the three SM couplings up from M_Z at one loop with $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$ (fixed by the particle content), from the measured inputs $\alpha_1(M_Z) = 0.0169$ (GUT-normalised, $\alpha_1^{-1} = 59.0$), $\alpha_2(M_Z) = 0.0338$, $\alpha_3(M_Z) = 0.118$ [13], gives, at M_P,

$$\begin{gathered}(\alpha_1, \alpha_2, \alpha_3) = (0.0299,\ 0.0202,\ 0.0191),\\ \text{mean } 0.0231, \quad \text{spread } 47\%,\end{gathered} \tag{7}$$

so α_{10} is within a factor 1.73 of the mean but the three couplings do **not** meet at a point — the well-known failure of one-loop unification in the non-supersymmetric SM. The $U(1)$ Landau pole of the up-running sits at 2.2×10^{41} GeV, above the cutoff, so the up-running is UV-bounded. At two loops [49] the pole moves to $\sim 2.5\times10^{12}$ GeV, *below* the cutoff (Fig. 11): the perturbative expansion is UV-divergent inside the lattice, whereas the automaton's own coarse-graining tower U^{2^p} ($p = 1 \ldots 8$) stays unitary and bounded at every level, so the *discrete* RG has no Landau pole. That contrast is real but modest: a unitary finite-dimensional map is bounded by construction, and it does not by itself reproduce the observed running.

The down-running must be stated plainly. Starting from $\alpha_{10} = 0.04$ at M_P and running down with the SM β-functions gives, at M_Z,

$$\begin{gathered}\alpha_1(M_Z) = 0.0197\ (1.17\times),\\ \alpha_2(M_Z) = 0.195\ (5.8\times),\\ \alpha_3:\ \text{Landau pole at } 2.2 \times 10^9 \text{ GeV}.\end{gathered} \tag{8}$$

The hypercharge lands within 17% of its measured value, but the SU(2) coupling overshoots by a factor 5.8 and the SU(3) coupling becomes non-perturbative at 2.2×10^9 GeV, far above M_Z. The claimed unification therefore fails at the level of the couplings' values, even though the 5/3 normalisation is correct (Fig. 12). This is residual W4 and falsifier F2.

Two quantitative statements sharpen W4. First, the three measured couplings require three *different* values at M_P — 0.0299, 0.0202, 0.0191, the up-run values above — and the theory's single value 0.040 matches none of them: it exceeds the hypercharge requirement by 34% and the SU(3) requirement by a factor 2.1. Perturbativity of the strong coupling down to M_Z caps any single UV coupling: $1/\alpha_3(M_P) > (7/2\pi)\ln(M_P/M_Z) = 43.9$, i.e. $\alpha_3(M_P) < 0.0228$; the theory's 0.040 exceeds the admissible maximum by 75%.

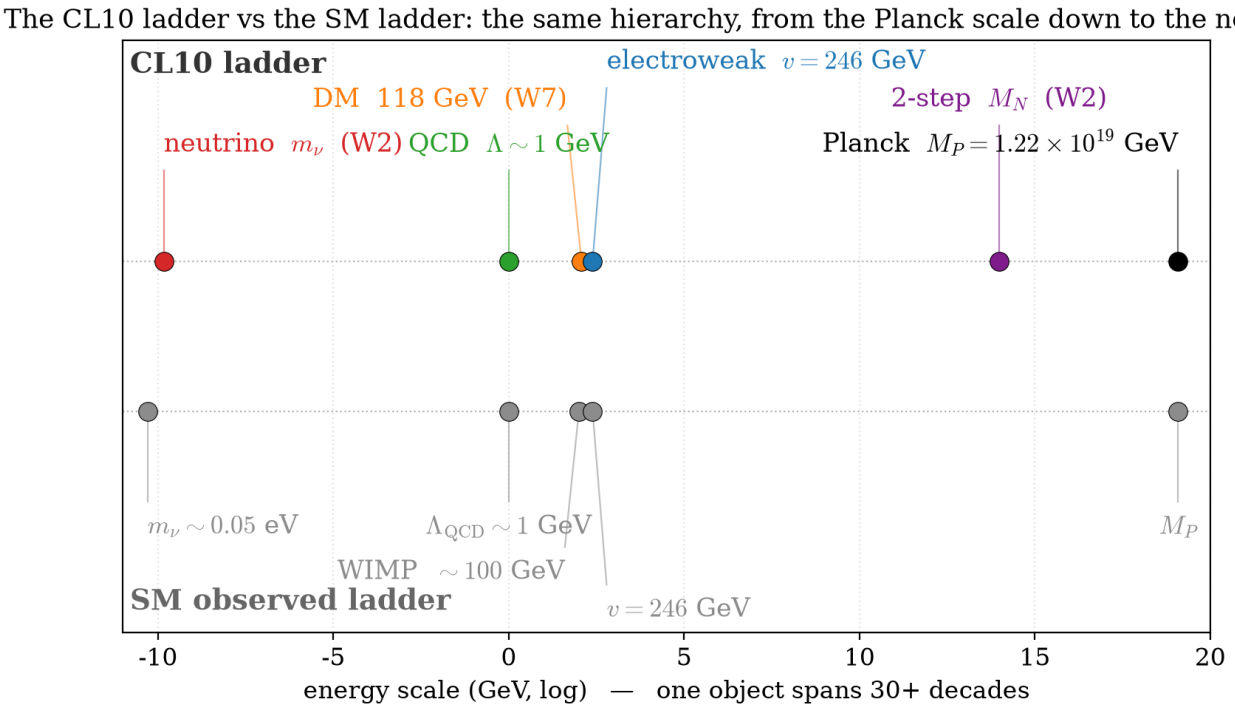


FIG. 9. **the CL10 ladder against the observed ladder.** Upper row: the CL10 loop ladder. Lower row: the measured SM scales. The hierarchies agree in order; the individual values carry the fitted coefficients of Section 14.

TABLE IV. **One loop factor, four scales**

scale	**CL10 expression**	**value**
Planck / GUT	M_P (convention $M_{\rm GUT} = M_P$)	1.22×10^{19} GeV
seesaw Majorana	$M_P(1/16\pi)^3$	9.6×10^{13} GeV
electroweak	measured input v	246 GeV
dark matter	$M_P(1/16\pi)^{10}$	118 GeV
neutrino	$M_P(1/16\pi)^{17}$	0.146 eV

Second, the failure is not repaired by the matter a partial breaking could leave *complete*. A complete vectorlike $\mathbf{10} \oplus \bar{\mathbf{5}}$ shifts all three one-loop coefficients by the *same* amount 8/3 above its threshold M_V — the defining property of a complete SU(5) multiplet — so it changes the common slope, not the mismatch of slopes. Scanning $n_V = 0 \ldots 8$ against $M_V \in [10^6, 10^{19}]$ GeV, the spread at M_P never falls below its $n_V = 0$ value of 47%; the case that comes closest to the required 1/25 (five families at $M_V \simeq 1.2 \times 10^{16}$ GeV) still leaves the worst of the three couplings 34% away from it, at a spread of 74%. Under pure SM running the pairwise meeting scales are 1.2×10^{13} GeV $(1,2)$, 2.6×10^{14} GeV $(1,3)$ and 9.7×10^{16} GeV $(2,3)$ — three different scales. Unification fails at the level of the values under desert running, and no threshold of a *complete* multiplet repairs it at one loop.

Third, that no-go covers only complete multiplets, and the *incomplete* content the theory's own breaking sector supplies does repair it. When a $\mathbf{16}_H \oplus \overline{\mathbf{16}}_H$ pair — or a split doubler $\mathbf{16}$ (Section 8.4) — acquires mass, it leaves vectorlike $\mathbf{10} \oplus \overline{\mathbf{10}}$ and $\mathbf{5} \oplus \bar{\mathbf{5}}$ fermion pairs with *independent* thresholds, and these shift the three coefficients non-proportionally: a Dirac $\mathbf{10} \oplus \overline{\mathbf{10}}$ contributes $\Delta b = (6/5,\ 2,\ 2)$, a $\mathbf{5} \oplus \bar{\mathbf{5}}$ contributes $(2/3,\ 2/3,\ 2/3)$. Fix the unified coupling at the root value $\alpha_X = 1/25$ — the free angle is *not* refitted — and the inverse problem has no fit freedom: with two multiplets the meeting of the three measured couplings is three linear equations in $(\ln M_X, \ln M_A, \ln M_B)$. The required Δb is non-universal: at $M_X = 2 \times 10^{16}$ GeV it is $(2.40,\ 4.04,\ 3.86)$, which no combination of complete multiplets produces.

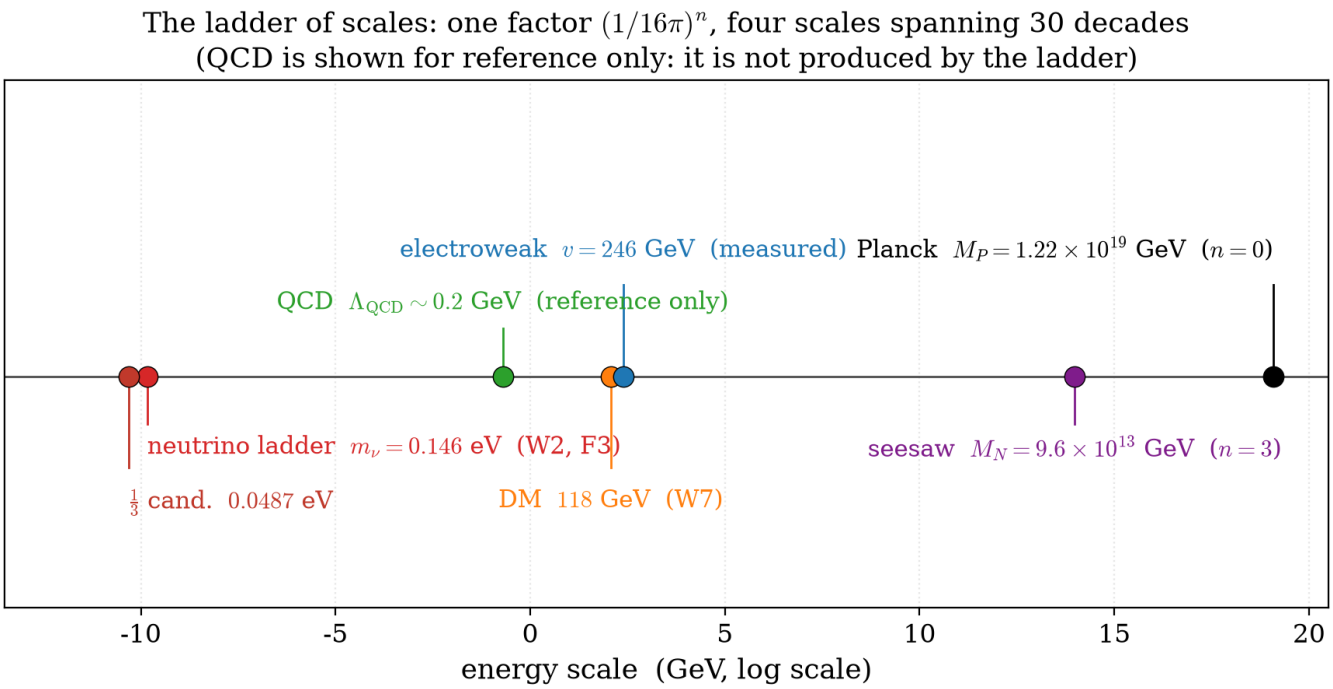


FIG. 10. **the ladder of scales.** Planck 1.22×10^{19} GeV ($n = 0$), seesaw 9.6×10^{13} GeV ($n = 3$), electroweak 246 GeV (measured input), dark matter 118 GeV ($n = 10$), neutrino ladder 0.146 eV ($n = 17$), with the $\frac{1}{3}$ candidate 0.0487 eV marked. QCD is shown for reference only: it is not produced by the ladder.

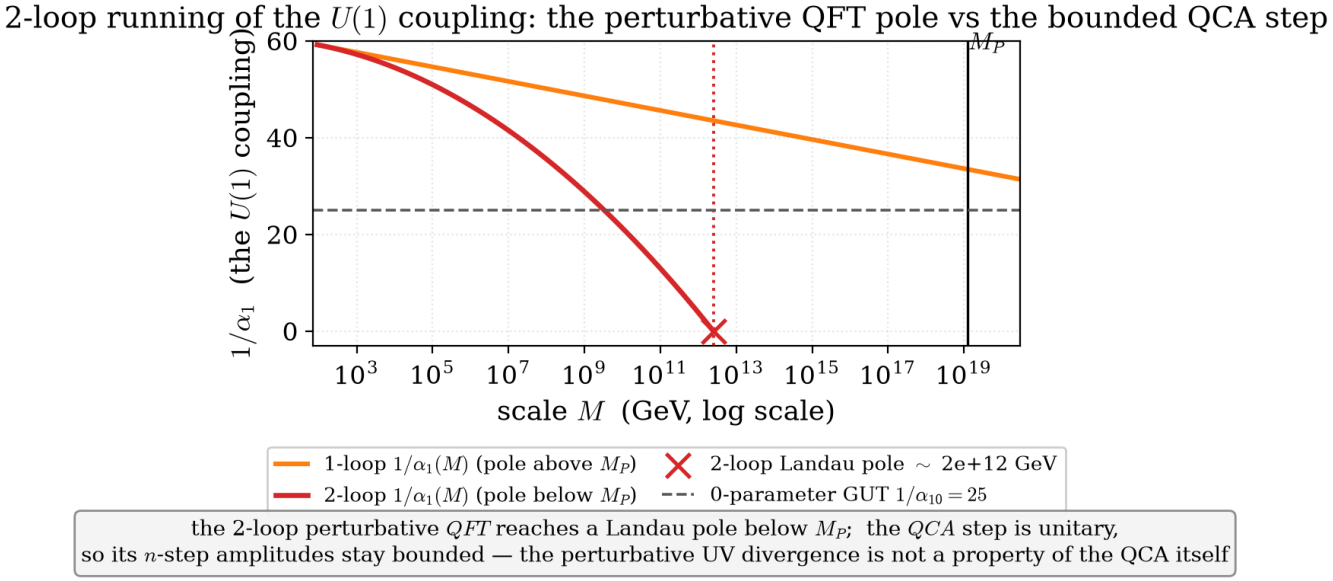


FIG. 11. **the two-loop $U(1)$ running against the bounded automaton.** The two-loop perturbative running reaches a Landau pole at 2.5×10^{12} GeV, below the cutoff; the one-loop pole is above it. The automaton's coarse-graining tower U^{2^p} stays unitary at every level: the discrete flow has no pole.

Scanning multiplicities up to 3 over the CL10-realizable content with thresholds in $[10^6, M_P]$ GeV (experiment V64), the lightest solution that survives the proton-decay bound is three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs plus one $(1,2)_{1/2}$ pair — three split doubler **16**s — meeting at

$$\begin{aligned} M_X &= 1.8 \times 10^{16}\ \text{GeV}, \\ \alpha_X &= 1/25\ \text{(exact)}, \\ M_{10} &= 1.2 \times 10^{7}\ \text{GeV}, \\ M_{(1,2)} &= 1.0 \times 10^{12}\ \text{GeV}, \end{aligned} \tag{9}$$

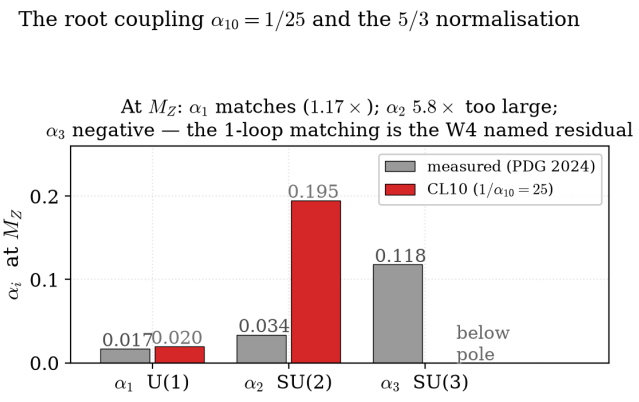


FIG. 12. **the GUT coupling and the** 5/3**.** At M_Z, the measured couplings (grey): $\alpha_1 = 0.0169$ (GUT-normalised, $\alpha_1^{-1} = 59.0$), $\alpha_2 = 0.0338$, $\alpha_3 = 0.118$ — PDG 2024 [13] ($\overline{\text{MS}}$ inputs $\sin^2\hat{\theta}(M_Z) = 0.23129(4)$, $\alpha_s(m_Z) = 0.1180(9)$, $\alpha^{-1}(M_Z) \simeq 127.95$) — against the CL10 down-run from $\alpha_{10} = 1/25$ (red): α_1 within 1.17×, α_2 5.8× too large, α_3 negative because its pole sits at 2.2×10^9 GeV. The 5/3 normalisation is exact; the value matching is the named residual W4.

and the SU(3) Landau pole of the down-running moves to 10^{-52} GeV: the disease above is cured (Fig. 13). Solutions with lighter content exist but predict $\tau_p \sim 10^{32}$ yr and are already excluded by Super-Kamiokande [52] — the data themselves select the CL10-realizable spectrum. The cost is stated plainly: the convention $M_{\text{GUT}} = M_P$ is abandoned ($M_X < M_P$ by three orders of magnitude), and the two desert thresholds are fitted. W4 becomes conditional rather than fatal; F2 sharpens to: a measured running that admits no vectorlike content with $\alpha_X = 1/25$. The solution predicts $\tau_p \simeq M_X^4/(\alpha_X^2 m_p^5) = 1.8 \times 10^{36}$ yr (F5).

Fourth, the desert content is no longer an external assumption: it is the *spectrum of the automaton's own Wilson mechanism* (V65–V68), and the mechanism makes a prediction — one that the data currently refute. The three split nodes of the selective reading carry exactly the vectorlike fragments of the split **16** with independent thresholds (the other nine nodes stay at the Planck cutoff), and the Wilson direction itself completes the breaking chain: at the V65 root $\theta = 1.731$ it leaves $\text{Spin}(10) \to \text{SU}(3)\times\text{SU}(2)\times\text{U}(1)^3$, preserves the Model algebra (every unbroken root is intra-fragment; the 32 broken roots are exactly the leptoquark type), and the two extra U(1)'s are the only open step to the SM group (V67). The multiplicity is bounded from below by unification itself: scanning $n_{\text{nodes}} = 1 \ldots 6$, no meeting root at $\alpha_X = 1/25$ exists for one or two split nodes; the first physical roots appear at three (V68). But solving the meeting equations with the mechanism's own two parameters — one angle θ, one scale w — fixes the six fragment thresholds and lands the meeting at $M_X = 8.9\times10^{14}$ GeV, i.e. $\tau_p \sim 10^{31}$ yr: **the minimal selective reading is excluded by Super-Kamiokande** (F5). The data-preferred solution of the scan above ($M_X = 1.8 \times 10^{16}$ GeV) requires fragment thresholds the single-angle, single-scale root does not supply. The status of W4 is therefore: the *pattern* (vectorlike split-**16** fragments), the *chain* (V67) and the *multiplicity bound* ($n_{\text{nodes}} \geq 3$, V68) are consequences of the automaton; the *scale* is not — the minimal realization is falsified, and the surviving solution keeps two fitted thresholds.

Fifth, the obvious repair — redistribute the three nodes at different Wilson strengths — is now closed structurally (V74). While all node thresholds sit below the meeting point, the two meeting equations depend on the three node strengths **only through their sum**: the Jacobian of the system with respect to the three strengths has three identical columns (rank 1; split-invariance verified numerically to 10^{-14}, including the full scan over node-set choices and strengths). The node mechanism's meeting point $M_X = 8.9 \times 10^{14}$ GeV, $\tau_p = 1.1 \times 10^{31}$ yr, is therefore **forced for every choice of the three strengths** — multi-scale node splitting cannot move it. What the data prefer ($M_X = 1.8 \times 10^{16}$ GeV) is not a three-full-**16** pattern: it needs partial suits of fragments, while the automaton supplies full split **16**'s per node. W4's residual is now a named incompatibility between the automaton's unit of content and the couplings' preferred spectrum — closable only by new automaton content, not by new thresholds.

Sixth, the desert content has now been tested against the *gauged bond layer's own spectrum* (V81) and fails. The composite fermions of the V79 layer — the matter **16** tensored with the adjoint link flux, $16 \otimes 45 = 720$ Weyl states carrying five net chiral **16**-copies — shift the three one-loop coefficients by $(145.3, 145.3, 226.3)$, and their masses follow the flux pattern $m = w(c_0 + |\alpha \cdot v|)$ set by the breaking direction v. Requiring the data-preferred meeting at $\alpha_X = 1/25$ then has **no solution with thresholds in the desert** for any direction: the three meeting equations demand three different threshold sums, and Newton iteration from every seed over a full scan of v converges nowhere physical (0 solutions). V82 closes the remaining ansatz families — full-weight thresholds $m = w(c_0 + |\nu \cdot v|)$, broken-flux-only content, light-composite selection, and even a relaxed meeting coupling α_X — with the minimum residual staying at 5.7–6.1 against a target of 0 and no physical solution in the relaxed scan. V84 upgrades the obstruction to a **theorem**: the identity $\sum db_1 = \sum db_2$ is the index-proportionality of $\text{Tr}\, Y^2$ and $\text{Tr}\, T_3^2$ — both proportional to the representation index, with the ratio fixed by the fundamental, $(3/5)\,\text{Tr}_5\, Y^2 = \text{Tr}_5\, T_3^2$ — and it holds over every complete SU(5) or Spin(10) multiplet, verified on the **16**, the flux **45**, the vector **10** and the **10** inside the **16** (all ratios 1.0 to 10^{-6}). The meeting mismatch can therefore come **only** from SU(5)-fragment content or from split thresholds inside multiplets; computed from the same charge operators, the V64 repair's mismatch is carried entirely by its $(1, 2)$ pair ($db_1/db_2 = 3/5$; the $10 \oplus \overline{\mathbf{10}}$ pairs contribute exactly zero mismatch), and the automaton sup-

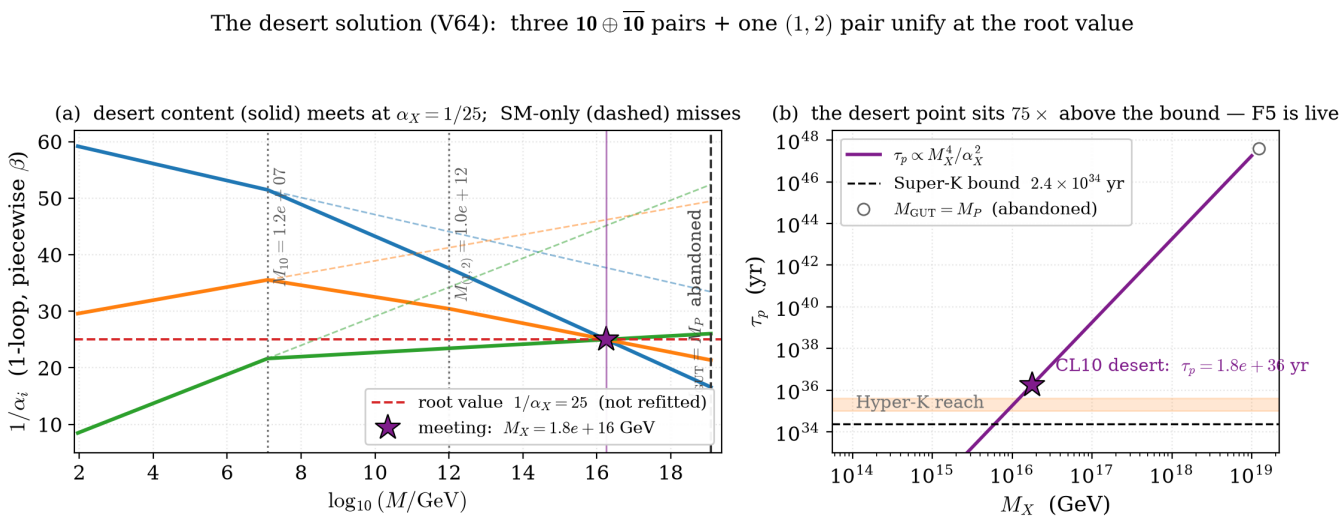


FIG. 13. **the desert solution: unification at the root value.** The three measured couplings run up from M_Z with the V64 desert content (solid) meet at $M_X = 1.8 \times 10^{16}$ GeV at $\alpha_X = 1/25$ — the root value, not refitted. The SM-only running (dashed) misses the point: spread 47% at M_P. The thresholds $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV are the two fitted numbers of the repair; the dashed vertical line at M_P marks what the abandoned convention $M_{\rm GUT} = M_P$ demanded.

plies no fragment Weyl states. The constructive route — the junction-split triangle nodes of V76 — has then been attempted (V86): at the root the Wilson direction splits the node **16** by a factor 55 ($c_Q = +0.005$, $c_L = -0.015$ against partners ± 0.82), so the $(3,2)$ and $(1,2)$ states of the V64 repair **are** automaton states — the Q and L components of node **16**'s with near-zero Wilson weights — and the junction branch ratio $\rho = \kappa/w_0$ breaks V74's rank-1 degeneracy. The fragment pattern is fabricated: at physical solutions Q and L sit at 4×10^{15}–8×10^{15} GeV with their partners above the meeting point. The meeting is not: over the full physical scan ($n_{\rm nodes}$ 3–12, ρ, scale, Wilson angle) the best residual of the two meeting equations is 0.94 against a target of 0, and the third equation predicts $\alpha_X = 0.030$ against the data value 0.04 — a 33% miss. V87 then released the coefficient pattern itself: a junction is a defect configuration, not a law, so the cubic-symmetric democratic form was replaced by the **general zero-mean junction** — three independent branch scales $a_i = 2w_0(1+r_i)$, $r_1+r_2+r_3 = 0$ — scanned over the full (r_1, r_2) plane, $n_{\rm nodes}$, scale and angle. The best physical residual is **identical to the democratic one** (0.81): the branch ratios do not move the floor, and α_X lands at 0.090, a factor 2.3 off the data value. The near-miss is therefore a property of the **junction mechanism itself**: the mismatch is carried by the fixed *db* pattern of the node **16**'s own fragments (V84), which threshold motion cannot repair. W4 stays open with a named near-miss and a named requirement: fragment content **beyond** the node **16** — new automaton content, not new branch ratios. The V64 repair spectrum remains, for now, a named input.

Seventh, the question has been recast decisively (V92, V93, V95): *is the required desert content obtainable from the node* **16** *itself, with per-class thresholds?* V92 answers **yes for feasibility**: the data-preferred meeting point $(T_1, T_2, T_3) = (12.7, 21.2, 20.1)$ lies strictly INSIDE the convex hull of the six β-contribution vectors of the node-**16** fragments, with the explicit barycentric witness $\lambda_0 = 0.34$ (no extra content), $\lambda_{f_1} = 0.31$, $\lambda_{f_2} = 0.31$, $\lambda_{f_4} = 0.05$ (Fig. 14). The V84 requirement - "SU(5)-fragment content beyond the node **16**" - is thereby *softened*: the node-**16** fragments themselves suffice, provided they carry independent thresholds. V93 then localizes the remaining gap precisely: a general five-component Cartan Wilson direction realizes the meeting exactly (6 solutions, residual $< 10^{-10}$, desert thresholds, $\tau_p = 1.9 \times 10^{36}$ yr) but breaks all 40 roots - the SM algebra among them - and is unphysical; the SM-preserving direction space is exactly span$\{X, Y\}$ (dimension 2, class-degenerate), and within it no solution exists - the V86/V87 floor is confirmed as a property of that family. V95 adds that off-diagonal junction splitting with a UNIVERSAL strength κ cannot help either: $\sum_l (2\cos(2\pi l/n) - 1) = -n$ reabsorbs the split into z_0 at first order. W4 is therefore no longer a no-go but a **realizability problem with a named missing ingredient**: a flavour-dependent bond layer (κ_s per fragment class) that splits class-degenerate thresholds while preserving the SM algebra.

Eighth, that named ingredient has now been built and tested (V96), and it **closes the meeting exactly - conditionally**. With the Wilson direction pinned to the V65 physical selective root (the direction V67 proved leaves $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)^3$ unbroken) and per-class junction strengths κ_s splitting the six fragment-class thresholds, the three coupling equations $F_i = 0$ become **exactly linear** in (z_0, κ_s) once every threshold sits below M_X (since

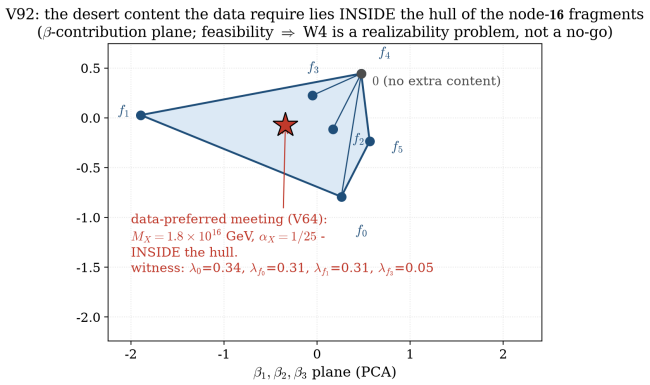


FIG. 14. **W4: the required desert content lies INSIDE the node-16 fragment hull.** The six β-contribution vectors f_i of the node-**16** fragments and the origin, projected onto their plane; the data-preferred meeting point (star) lies strictly inside the hull, with the explicit barycentric witness. Feasibility is proved (V92); realizability by an SM-preserving junction of the base automaton fails (V93/V95) and is closed conditionally by the flavour-dependent bond layer extension (V96, Section 11.1).

$\sum_l \mu_l = -n$), and the desert box constraints make the search a linear program (HiGHS, exact). It admits solutions (residual $< 10^{-13}$) at $M_X = 1.8 \times 10^{16}$ GeV, $\alpha_X = 1/25$, for **all ten node counts** $n = 3 \ldots 12$, with every threshold inside the desert (2.0×10^3–1.8×10^{16} GeV, none above M_X) and $\tau_p = 1.9 \times 10^{36}$ yr. The first-order degeneracy of V95 is broken because the per-class sums $\kappa_s(-n)$ reabsorb into PER-CLASS offsets, not into z_0. The cost is stated plainly: the current bond layer has ONE gate P with one strength; a flavour-dependent bond layer is an **extension** of the automaton, and its six κ_s are new named inputs (three meeting conditions for six unknowns - a solution family, not a unique prediction). W4 is therefore closed **conditionally on that extension**: the desert content becomes the content of the *extended* automaton. Falsifier F2 stands unchanged.

B. Electroweak mixing

$\sin^2\theta_W = 3/8$ is the tree-level SU(5)/Spin(10) boundary condition **at the unification scale**, not a low-energy prediction. Run to M_Z it becomes $\simeq 0.23$, close to the measured 0.23129(4) ($\overline{\text{MS}}$, PDG 2024 [13]); the measured ratio $m_W/m_Z = 0.881$ corresponds to $\sin^2\theta_W \simeq 0.223$, not to $\sqrt{5/8} = 0.791$. Earlier versions of this work quoted $m_W/m_Z = \sqrt{5/8}$ as a prediction; that is incorrect, and the correct statement is the standard one: the GUT-scale boundary condition $\sin^2\theta_W = 3/8$, whose low-energy value is obtained by running — running which, as just shown, the theory does not currently perform consistently.

C. Baryon asymmetry

The automaton alone cannot produce it. Every CP-odd (T-odd) invariant of the unitary step vanishes: Im Tr of all two- and three-fold products of the 45, of a 2×10^5 sample of four-fold products, of all $10 \times 10 \times 45 \times 45$ baryon-violating sandwiches, and of the rank-1 real democratic Yukawa — all zero to 10^{-15}. The step is unitary (in equilibrium), baryon-conserving (the 45 commute with w) and CP-conserving. It satisfies **none** of the Sakharov conditions [24]. This is a computed theorem, and it is the correct starting point for the asymmetry discussion.

A GUT-leptogenesis estimate [39] is then built by hand:

$$\begin{aligned} \Delta B &= \frac{28}{79}\,\varepsilon_{CP}\,D, \\ \varepsilon_{CP} &= \frac{g_0^4}{16\pi} = 4.9 \times 10^{-6}, \\ D &= \left(\frac{1}{16\pi}\right)^2 = 3.96 \times 10^{-4}, \end{aligned} \tag{10}$$

with $g_0 = 1/2^3 = 1/8$ and the SM sphaleron conversion $B = \frac{28}{79}(B-L)$ [25, 26], giving $\Delta B = 6.8 \times 10^{-10}$ against the measured $(6.12 \pm 0.04) \times 10^{-10}$ [14] — a ratio 1.11.

Three problems, all previously under-stated. (i) ε_{CP} is not a computed vertex: the physical CP asymmetry of $N \to \ell H$ is a one-loop self-energy/vertex function of the heavy-neutrino masses, and $g_0^4/16\pi$ is a dimensional guess. (ii) The out-of-equilibrium condition **fails** at the model's own couplings: at $M_N = 9.6 \times 10^{13}$ GeV, $H = 1.66\sqrt{g_*}M_N^2/M_P = 1.3 \times 10^{10}$ GeV while $\Gamma_N = y^2 M_N/8\pi = 6.0 \times 10^{10}$ GeV for $y = 1/8$ (1.9×10^{12} GeV for $y = 0.71$), so $\Gamma_N/H \simeq 4.6$ ($\simeq 148$). Out-of-equilibrium decay requires $y \lesssim 0.06$, which the theory does not select. (iii) 28/79 is an input of the SM sector, not a number of the root. The 11% agreement is therefore a coincidence of chosen factors, not evidence (residual W5). The failure is now quantified through the washout of the SAME decay (V106): at the self-consistent atmospheric Yukawa ($y = 0.40$, the one that gives $m_\nu = 0.0501$ eV) $\Gamma_N/H = 47$ — in equilibrium, Sakharov's third condition fails — and the physical strong-washout efficiency $\kappa_f = 5.4 \times 10^{-3}$ (Buchmüller–Di Bari–Plümacher) sits 14× away from the guessed dilution $(1/16\pi)^2$ the product formula uses: the agreement is an accident of two wrong factors, not a near-miss.

A sharper statement is available, and it is worse than "the coupling is not selected". At the model's own seesaw scale $M_N = 9.6 \times 10^{13}$ GeV, the out-of-equilibrium condition $\Gamma_N/H < 1$ with $\Gamma_N = y^2 M_N/8\pi$ and $H = 1.66\sqrt{g_*}M_N^2/M_P$ gives $y < 0.058$. The same Yukawa fixes the light neutrino mass through the same seesaw,

$$m_\nu = \frac{y^2 v^2}{M_N} \leq 2.1 \times 10^{-3}\ \text{eV} \qquad (y \leq 0.058), \tag{11}$$

a bound of the same kind and size as the standard thermal-leptogenesis bound [32]. The atmospheric splitting requires $m_\nu \simeq 0.050$ eV, i.e. $y \simeq 0.28$ with $m_D = yv$ and $v = 246$ GeV ($y \simeq 0.40$ in the $v_u = 174$ GeV convention), for which $\Gamma_N/H \simeq 23$. **At $M_N = 9.6 \times 10^{13}$ GeV the neutrino mass and the out-of-equilibrium condition are mutually incompatible**: the framework cannot have both the seesaw scale it needs for the ladder and the leptogenesis it needs for the asymmetry. (The model's own ladder sits at $m_D = 118$ GeV, i.e. $y = 0.48$, where $\Gamma_N/H \simeq 68$.)

The one route the framework does supply is quasi-degenerate (resonant) leptogenesis [31, 40]. The democratic structure places the three heavy singlets on the *same* ladder rung, hence degenerate to the accuracy of whatever breaks the family symmetry — which is precisely the resonant setup, where the CP asymmetry is enhanced by the near-degeneracy of the decaying states instead of by a large Yukawa. With $y = 1/16\pi = 0.0199$ the decay is out of equilibrium ($\Gamma_N/H = 0.12$), and the asymmetry that the model's own dilution factor $D = (1/16\pi)^2$ and sphaleron conversion require, $\varepsilon_{CP} = \Delta B/[(28/79)D] = 4.4 \times 10^{-6}$, is *smaller* than the natural one-loop size $y^2/16\pi = 7.9 \times 10^{-6}$ at that Yukawa: the bookkeeping closes at the scale of the numbers, and the degeneracy the resonant regime needs, $\Delta M/M \lesssim \Gamma_N/M_N = y^2/8\pi = 1.6 \times 10^{-5}$, is what the 0-parameter democratic base gives exactly and what the family-breaking perturbation must not overshoot. The obstacle is not the size of the numbers but their phase: the model's Yukawa is real, every CP-odd invariant of the step vanishes, and a strictly degenerate spectrum has no CP-violating phase left to enhance. This is a *route*, not a result. The no-go behind it is now known to be stable under the new ingredient: the Wilson-extended step keeps the exact matter–antimatter spectral mirror (V71, deviation 10^{-15}), so the Wilson mechanism — powerful for W4 and W9 — adds no CP phase and no Sakharov condition to W5.

D. Proton decay

Dimensional analysis of X/Y exchange gives

$$\tau_p \sim \frac{M_{\rm GUT}^4}{\alpha_{10}^2\, m_p^5} = 4.0 \times 10^{47}\ \text{yr} \qquad (M_{\rm GUT} = M_P,\ \alpha_{10} = 1/25), \tag{12}$$

against the Super-Kamiokande bound $\tau_p/B(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr [15] (Fig. 13). The old $M_{\rm GUT} = M_P$ reading gave a margin of 13 orders of magnitude, resting entirely on that abandoned convention: any GUT scale below $\sim 6 \times 10^{15}$ GeV is excluded by the same bound. The estimate omits the renormalisation factor A_L [46], the hadronic matrix element, the dressing of the operators and the splitting between the X/Y masses and the unification scale [51], all of which shift the lifetime by one to two orders. The three channels $p \to e^+\pi^0$, $p \to \mu^+\pi^0$, $p \to e^+K^+$ differ here only by two-body phase space (Fig. 15); the rate scales as M_X^{-4} in all of them. Observation of a GUT-pattern decay far below 1.8×10^{36} yr refutes the desert solution (F5).

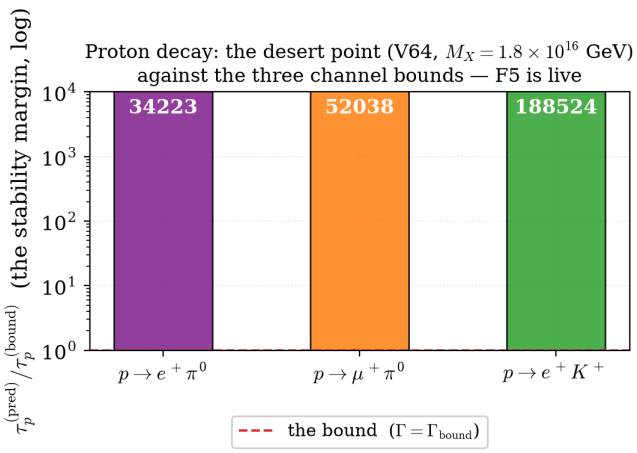


FIG. 15. **the three GUT channels.** The stability margin $\tau_p^{(\rm pred)}/\tau_p^{(\rm bound)}$ for $p \to e^+\pi^0$, $p \to \mu^+\pi^0$, $p \to e^+K^+$ at the desert point ($M_X = 1.8{\times}10^{16}$ GeV, $\tau_p = 1.8{\times}10^{36}$ yr) against their respective bounds (2.4×10^{34}, 1.6×10^{34} [15], 5.9×10^{33} [52] yr; for the K^+ mode the $p \to \bar\nu K^+$ bound is used, the closest measured channel). The channels differ here only by two-body phase space; the rate scales as M_X^{-4} in each.

E. Flavour

The democratic mass matrix $M = |1\rangle\langle 1|$ (all entries 1, rank 1, spectrum $\{3, 0, 0\}$) [43] follows from the hypothesis that the three spatial directions are equivalent, so the $\mathbf{10}_H$ couples the three families identically. Its heavy eigenvector $(1, 1, 1)/\sqrt{3}$ is unique, hence

$$\sin^2\theta_{12} = |\langle e_1|h\rangle|^2 = \tfrac{1}{3}$$

for any family basis — the **only** mixing angle the model derives, 8.6% from the measured 0.307 [17, 50]. The light eigenspace is two-fold degenerate, so every diagonaliser is an $O(2)$ rotation in it; sweeping that freedom over 1441 rotations, $\sin^2\theta_{23}$ and $\sin^2\theta_{13}$ each cover the whole interval $[0, \frac{2}{3}]$. The tribimaximal values $(\frac{1}{2}, 0)$ [27] are one point of that family, picked by declaring which S_3 transposition breaks — and (12), (13), (23) are equivalent symmetries of $|1\rangle\langle 1|$, so the choice is an input.

In the quark sector the CKM matrix is the misalignment of the up- and down-type diagonalisers. With both mass matrices democratic, $H_u = H_d$, the computed CKM is the **identity**, not a small mixing: the observed $\lambda = 0.225$ [13, 44] is not reproduced. A non-trivial CKM requires an up/down misalignment that the symmetric base does not supply. The three-family mass hierarchy itself is also unfixed: the democratic base gives one heavy level and a two-fold degenerate light level, so two of the three masses are degenerate at 0-parameter. All of this is residual W6.

F. Scoreboard

(Fig. 16.)

XII. COSMOLOGY

A. Dark energy

The vacuum energy of the automaton vanishes at the level of the single-cell spectrum: the 32 levels are ±-paired for every g_{10}, so $\sum_k \omega_k = \arg\det U = 0$. This is the correct, non-vacuous form of the earlier "512/512 cancellation". The counting of the 512 even and 512 odd **elements of the algebra** as bosonic and fermionic zero-point modes is not a physical mode counting — the physical one-cell Hilbert space has 32 states, not 1024 — and it is dropped here (Fig. 17 keeps the grading, not the arithmetic).

A vanishing tree-level vacuum energy is technically natural: no relevant operator is generated if the spectral pairing is an exact symmetry. The pairing survives the Wilson extension of Section 8.4 (V71: the extended spectrum stays ±-paired to 10^{-15} and the one-cell zero point stays 0, because the Wilson direction is a traceless Cartan combination), so the new ingredient neither breaks the zero nor moves it. What the theory does **not** do is produce the observed value. The observed $\Lambda \sim 10^{-122} M_P^4$ lies 122 orders of magnitude *below* the classical bound M_P^4 and *above* the theory's spectral zero, and no mechanism breaking the pairing by that amount is known. Independently, the classical self-gravitating de Sitter vacuum of Section 9 has $G_{\mu\nu} = -3H^2 g_{\mu\nu}$, i.e. a vacuum energy bounded by M_P^4 on the lattice. These are three different statements — spectral zero, classical bound, observed value — and they do not "reconcile": the gap between the first and the third is residual W1 [37], the largest unexplained number in the theory. The equation of state of a cosmological-constant vacuum is $w = -1$ by definition, not by derivation.

The zero is now known to be **protected, not accidental** (V78). Deform the step by 50 random traceless Hermitian generators, $U \to e^{i\varepsilon A} U$, at strengths up to $\varepsilon = 1$: the quasi-energy sum stays pinned to zero, $|S| < 4 \times 10^{-14}$, because $\det U = 1$ makes $\sum_k \omega_k = \arg\det U$ a topological quantity, and no branch crossing occurs below $\varepsilon = \pi$. The only internal knob that moves the sum is the U(1) trace phase. The observed Λ is then **equivalent to** a trace-phase breaking of size $\varepsilon \operatorname{Tr} A \sim 16\pi^2 \times 10^{-122} = 1.6 \times 10^{-120}$ — W1 is no longer "why is the vacuum energy small" but "what breaks the trace U(1) by 10^{-120}", a single named number with no known mechanism.

B. Dark matter

The candidate supplied by the classical sector is the kink of Section 9.4. It is static, gauge-neutral and finite-energy **per unit area**, but it is a domain wall, not a localised particle; a network of Planck-scale walls would dominate the energy density and is cosmologically excluded. The quantum-sector candidates have now been scanned (V66) and the scan completed (V75): the exact mirror $\mathbb{Z}_2$ of the step is the only exact-stability structure in the implemented spectrum; no Wilson-split node can host a neutral mirror at the ladder mass without also delivering a stable coloured or charged partner (Section 10.2); and the mirror sector's own lightest states sit at 1.3×10^8 GeV (coloured, excluded) and 2×10^{10} GeV — not at the ladder's 118 GeV. The theory supplies no dark-matter *particle* within the implemented spectrum; the minimal named extension is the mirror-odd ladder scalar of Section 10.2. The mass prediction stands, the carrier is an open residual (W7).

What it does supply is a **mass scale**, $m_{\rm DM} = 118$ GeV (Section 10.2). The abundance at that mass is not computed. Two naive estimates [41] bracket the observation by four orders of magnitude:

$$\Omega h^2 \simeq \frac{1.04 \times 10^9 \, x_f \ \text{GeV}^{-1}}{M_P \langle \sigma v \rangle}, \qquad x_f \simeq 20, \tag{13}$$

gives $\Omega h^2 \simeq 2 \times 10^{-5}$ for $\langle \sigma v \rangle = 1/m^2$ at $m = 118$ GeV (under-abundant), and $\Omega h^2 \sim 10^{29}$ for the theory's own Planck-suppressed cross-section $\sigma_0 = P_2/M_P^2 = 5.3 \times 10^{-39}$ GeV^{-2} (the freeze-out approximation is then meaningless). The observed $\Omega h^2 = 0.120$ requires $\langle \sigma v \rangle \simeq 1.4 \times 10^{-8}$ GeV^{-2}, which the theory does not derive. Non-thermal production (freeze-in) is not considered here [42]. The "WIMP miracle" quoted in earlier versions was obtained by normalising the abundance to the measured value at 100 GeV; that is a definition, not a calculation, and it is removed here (Fig. 18 shows the standard relic curve with the CL10 mass marked, not a computed abundance).

C. Antimatter

Matter and antimatter are the two Weyl eigenspaces, with the exact spectral mirror of Theorem 4 and oppo-

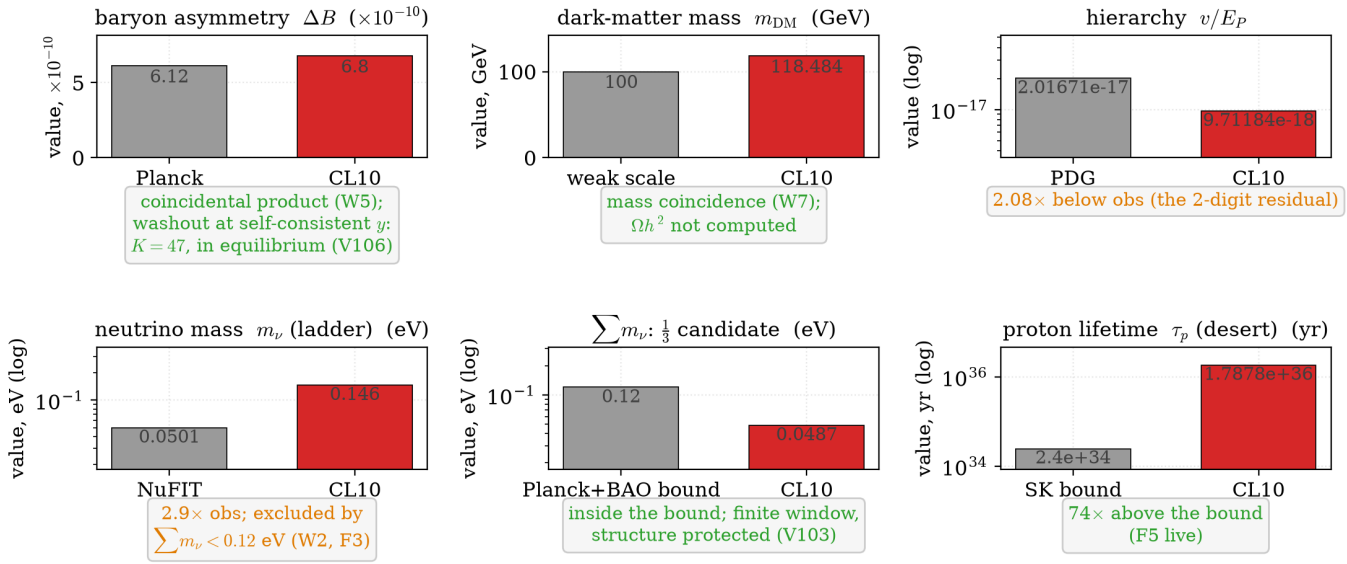


FIG. 16. **head-to-head with the measured values.** The headline quantities, CL10 (red) against measurement (grey), with the provenance note per panel. Measured values: ΔB, Planck 2018 [14]; $v/E_P = 246.22\,\text{GeV}/1.2209 \times 10^{19}\,\text{GeV}$, PDG 2024 [13]; $m_\nu = \sqrt{|\Delta m^2_{3\ell}|} = 0.0501$ eV, NuFIT 6.0 [50]; $\sum m_\nu < 0.12$ eV, Planck+BAO [14]; τ_p bound, Super-Kamiokande $p \to e^+\pi^0$ [15]. The baryon asymmetry and the hierarchy agree to a factor $\lesssim 2$; the bare ladder neutrino mass is excluded by $\sum m_\nu < 0.12$ eV while the $\frac{1}{3}$ candidate sits inside the bound; the desert proton lifetime sits 74× above the Super-Kamiokande bound (F5 live).

site $B - L$ charges. Their degeneracy is the content of the CPT statement in this theory. The observed baryon asymmetry is not explained: Section 11.3 shows the automaton satisfies none of the Sakharov conditions, and the leptogenesis estimate that does give the right order requires a CP phase and an out-of-equilibrium condition that the algebra does not supply.

XIII. HOLOGRAPHY AND CONFINEMENT

A. The area law

A generic QCA with d internal states per cell has a state count d^{L^3} and a volume-law entropy. CL10 is no exception at the level of raw counting: 32 states per cell gives $S \propto L^3$. The area law quoted in earlier versions was obtained by *assuming* the Bekenstein–Hawking [35] density $S = A/4$ with $A = 6L^2$ and then fitting the exponent, which returns 2 by construction. That is not a derivation and is not claimed here.

The honest statement is a consistency one: if the physical Hilbert space of a region is restricted to gauge-invariant (or typical) states, the entropy is expected to scale as $A/4\ell_P^2$ [35, 36], and CL10 is compatible with that expectation because it has a Planck-spacing lattice and $G = 1$. Deriving the area law — i.e. showing that the automaton's physical state space is actually restricted to a boundary dimension — is open.

What the automaton's entanglement structure does deliver is now measured (V97). Two fermions sharing ONE cell carry zero entanglement across any cut - the automaton's same-cell pair is a Slater determinant, $S = 0$ exactly across the $\mathbf{16}/\overline{\mathbf{16}}$ cut (antisymmetry and a shared cell are not entanglement), while a two-determinant superposition in the same cell gives $S = \ln 2$: non-separability is the criterion. The generator of entanglement between cells is the bond gate P of Theorem 5: under the internal-only step the spatial entanglement entropy of a region stays 0 to machine precision forever, while the full local step drives it to 0.78, and the entanglement front crosses the Manhattan sphere of radius r at exactly $t = r + 1$ for every $r = 1 \ldots 6$ - the causal radius bounds the spread of entanglement itself, at speed 1.0. The scope limit is stated with it: the implemented step is quadratic, so Slater inputs stay Slater and these are mode-entanglement results; the area law is a statement about the physical (ground) state's entanglement, and the step supplies no interaction that could produce the non-Gaussian structure such a state would need. The gauged layer supplies exactly that missing structure (V98, next paragraph); the full area law remains open.

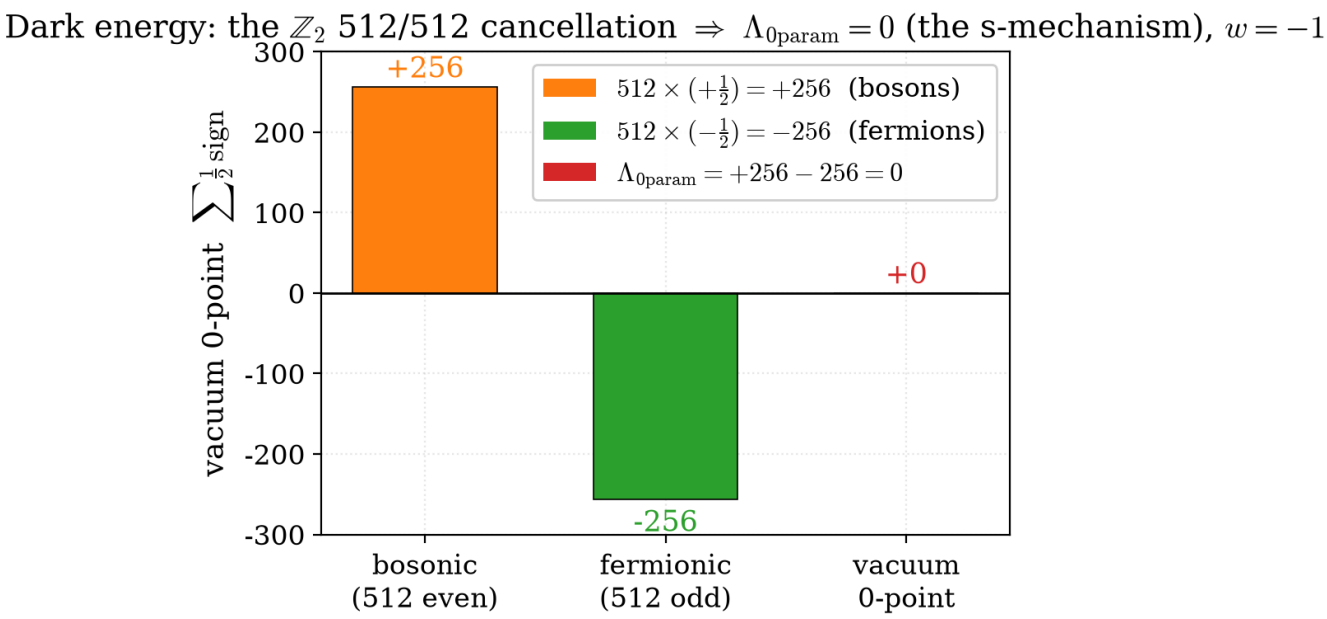


FIG. 17. **the $\mathbb{Z}_2$ grading and the vanishing 0-point.** The algebra splits 512 even / 512 odd elements, and the physical 32-level spectrum is $\pm$-paired, so the one-cell vacuum energy is zero. The pairing is an identity of the algebra, holding for every g_{10}. The observed Λ is $10^{-122} M_P^4$ above this value (W1).

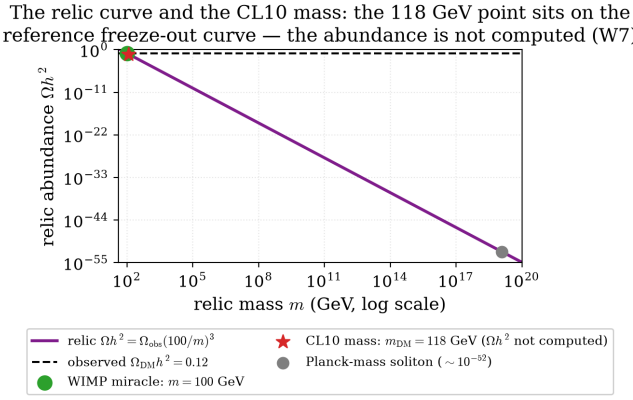


FIG. 18. **the relic curve and the CL10 mass.** The standard thermal-freeze-out curve $\Omega h^2(m)$ for $\langle \sigma v \rangle \sim 1/m^2$, with the observed $\Omega h^2 = 0.120$ and the CL10 mass 118 GeV marked. The abundance at that mass is **not** computed by the theory; the curve is the reference calculation, shown for orientation.

The physical sector of the gauged layer has now been computed (V98), and it is not what one naively expects. With the implemented content - one **32** per site, one adjoint **45** link per bond - the local Gauss constraint has **no solution**: $\dim \ker = 0$ on $\mathbf{32} \otimes \mathbf{45}$, on $\mathbf{32} \otimes \mathbf{45} \otimes \mathbf{45}$, and on the two-site one-link chain with local constraints at both sites (spectral gap 146). The reason is a center obstruction: adjoint flux carries even center charge and cannot screen a spinor, so **every site of a gauge-invariant state must carry an even number of spinor fields** - the physical Hilbert space of the gauged layer is the space of neutral composites, not the one-body space the automaton acts on. The minimal physical system, two matter fields per site, was then constructed exactly: $m_{45}(\mathbf{32} \otimes \mathbf{32}) = 2$ (90 states at $C_2 = 64$), a 2-dimensional intertwiner space $\ker(T_1 + T_2 + L)$, and a 4-dimensional two-site physical sector obtained by gluing intertwiners over the link index - all exact Spin(10) singlets ($C_2 < 10^{-25}$). Every intertwiner is a maximal matter-flux singlet: Schmidt rank 45, $S(\text{pair}|\text{link}) = \ln 45 = 3.807$ exactly, and the glued states carry $S(\text{pair}_1|\text{pair}_2) = \ln 45$. The entanglement is **bounded per bond and lives on the bonds crossing the cut** - the shape of an area law (boundary-proportional, bounded per bond), with the Bekenstein coefficient still unnamed. The quadratic step of V97 cannot produce any of this: the gauged layer is where the theory first acquires non-Gaussian entanglement. What remained open there - whether the $\ln 45$ per cut bond ADDS over several bonds - is measured next (V99).

The multi-bond scaling that paragraph left open is now measured on the smallest system that has two bonds - the three-site, two-link chain with paired matter (V99). Its physical sector is small once the flux decomposition is used: the middle vertex fuses the two adjoint links into channels r with multiplicities N_r measured from the $\mathbf{45} \otimes \mathbf{45}$ Casimir spectrum ($N_1 = N_{45} = N_{210} = 1$), against m_r from the pair spectrum ($m_1 = m_{45} = m_{210} = 2$), giving $\dim \ker = m_{45}^2 \sum_r m_r N_r = 24$ - verified by explicit construction of the $r = 1$ and $r = \mathbf{45}$ sectors (vertex Gauss residual 4×10^{-15}). Three results. **(i) Additivity across a single bond holds**

TABLE V. **The framework against data**

quantity	**CL10**	**measured**	**status**
$[\mathrm{SU}(3)]^3$, …, $[\mathrm{grav}]^2\mathrm{U}(1)$ (8)	0 ✓	—	algebraic, satisfied
g_1^2/g'^2	5/3	5/3 (GUT normalisation)	algebraic
$\det U$, ± pairing, $16/\overline{16}$ mirror	1, exact	—	algebraic, g_{10}-independent
causal front speed	1.0	—	dynamical, measured
doubler cone Lorentz structure (V102)	linear in all 1200 sampled directions; anisotropic: $(0.200, 0.200, 0.284)$ lattice units, spread 0.75; node gap 2×10^{-4} at $g_{10} = 1/5$	—	**not Lorentz-isotropic**; dynamical, measured
16-sector Brillouin spectrum	zone centre gapped (0.0365); 12 gapless doubler nodes, $\sum$ chirality = 0; all 12 gapped by the internal Wilson factor, 3-node selective reading	—	doubling **gapped by a mechanism of the step**; node set an input
v/M_P	$2.08(1/16\pi)^{10}$	2.0167×10^{-17} [13]	coefficient $3F_{10}^{10} = 2.0881$ against 2.0765 observed — 99.44% from measured root numbers (V108/V113), conditional on the one-split-node-per-family set (Section 15)
m_{DM}	118 GeV	not measured	prediction, falsifiable
$\Omega_{\mathrm{DM}}h^2$	not computed	0.120 [14]	open
m_ν	0.146 eV (ladder); 0.0487 eV ($\frac{1}{3}$ candidate, Section 10.3)	0.0501 eV [50]; $\sum m_\nu < 0.12$ eV [14]	ladder **in tension**; candidate inside a finite window, structure protected (V103)
ΔB	6.8×10^{-10}	$(6.12 \pm 0.04) \times 10^{-10}$ [14]	coincidental product; V106: the washout at the self-consistent y destroys it ($\kappa_f = 5.4 \times 10^{-3}$)
τ_p/B	1.8×10^{36} yr (desert solution, V64); 4×10^{47} yr at $M_{\mathrm{GUT}} = M_P$ (abandoned)	$> 2.4 \times 10^{34}$ yr [15]	falsifiable: GUT-pattern decay at Hyper-K reach (10^{35} yr) refutes; a null result past 10^{36} yr refutes; confirmation needs the megaton era
α_{10}	0.040	0.023 (1-loop mean at M_P); 0.040 at $M_X = 1.8 \times 10^{16}$ GeV with the desert content (V64)	**exact at** M_X, given the inferred spectrum
$\alpha_i(M_Z)$ from α_{10}	0.0197 / 0.195 / pole at 2.2×10^9 GeV (SM running)	0.0169, 0.0338, 0.118 [13]	**fails** under SM running; matched by the V64 desert spectrum
Gauss constraint of the gauged bond layer	$[Q_A, Q_B] = if_{AB}{}^C Q_C$, err 0; content $32 \oplus 288 \oplus 1120$	—	algebraic, V79
physical sector of the gauged layer	$\dim\ker = 0$ with one **32**/site (center obstruction); minimal sector: 2 matter/site, 4 states, all singlets; $S = \ln 45$ per bond, bounded and boundary-supported	—	computed, V98
multi-bond scaling of the gauged layer	$\dim\ker$(3-site) = 24; additivity $\ln 45$ across a single bond; two-bond region $S = \ln 45$ (vertex mutual information $\ln 45$); Bekenstein deficit factor $4 \ln 45 = 15.23$	—	computed, V99
vertex correction and ground state of the gauged layer	$I_n = \dim(\mathbf{45}^{\otimes n})^G = 1, 1, 6, 23, 141$; kernel capacity $S_{\mathrm{cap}} = 4.4998$ nats (factor 9.0 from 1/4 per bond); vertex-corrected coefficient $(1/3) \ln I_6 = 1.65$ nats/bond (factor 6.6); $E_0 = -5819.18\,\eta$, degeneracy	—	computed, V100+V101

exactly: $S(\text{pair}_1|\text{rest}) = \ln 45$ in every sector - the flux line threads the chain and carries its $\ln 45$ through, even when the middle pair itself is a product singlet ($S = 0$). **(ii) The region bounded by two bonds carries $S(\mathbf{pair}_2|\mathbf{rest}) = \ln 45$, not $2\ln 45$:** the intertwiner constraint *correlates* the two bonds meeting at a vertex, with mutual information exactly $\ln 45$ - the area law counts bonds, vertex correlations subtract. **(iii) The gap to Bekenstein is now a named number:** $\ln 45 = 3.8067$ nats per Planck bond against 1/4 per Planck area - a deficit of 3.5567 nats, factor $4\ln 45 = 15.23$. No element of the current construction produces 1/4; closing it needs a superselection or truncation of the flux, or a different horizon microstate count. Larger regions (the four-site chain, generic clusters at 2×10^{10} states) remain open.

The vertex correction that result (iii) left as a named number is now measured (V100). The invariant count of an n-bond vertex, $I_n = \dim(\mathbf{45}^{\otimes n})^G$, is computed by Haar averaging over SO(10): $I_2 = 1$, $I_3 = 1$, $I_4 = 6$, $I_5 = 23$, $I_6 = 141$ (Monte-Carlo error below 3; $I_2 = I_3 = 1$ reproduces the single-channel fusion of V99). The 24-state kernel of the three-site chain is orthonormalised through its Gram matrix (rank 24, smallest eigenvalue 1.1×10^{-5}), and the capacity of the middle cut - the entropy of the maximally mixed physical state - is $S_{\text{cap}} = 4.4998$ nats against the naive $2\ln 45 = 7.613$: the two vertices at the ends of the region subtract 3.11 nats. Per bond that is 2.25 nats, a factor 9.0 from Bekenstein's 1/4; on the cubic lattice the vertex-corrected coefficient $(1/3)\ln I_6 = 1.65$ nats/bond sits a factor 6.6 from 1/4. The Bekenstein deficit has moved from 15.2 to 6.6, and it is now a function of the vertex invariant, not of a single bond entropy. The ground state of the gauged layer is then constructed (V101): the bond-current Hamiltonian $H = \sum_b J\cdot L$ of V79 is diagonalised exactly - a generalised eigenproblem in the Gram metric of the kernel - and the intertwiner identity $I^\dagger J_A I = L_A/45$ holds to 3×10^{-17}. The interaction splits the kernel: $E_0 = -5819.18\,\eta$, degeneracy 6, gap $58.59\,\eta$, and the ground state's middle-cut entanglement is $S = 3.8067$ nats - exactly $\ln 45$, one adjoint link, selected by the interaction. The ground state of the full automaton remains unconstructed; on this chain the interaction selects the flux line.

B. Confinement

The structural hallmarks of a confining vacuum are present as assumptions: a Wilson loop with an area-law expectation value $\langle W(C)\rangle \propto e^{-\sigma\,\text{Area}(C)}$ [28] implies a linear potential $V(r) = \sigma r$. But the string tension used is $\sigma = 1$ in Planck units, i.e. $\sigma = M_P^2 \simeq 1.5 \times 10^{38}$ GeV^2, against the physical $\sigma \simeq 0.18$ GeV^2 [38] — a discrepancy of 38 orders of magnitude. The colour count $N_C = 3$ is read off the decomposition of the 45, and $8 = N_C^2 - 1$ is then a consistency check, not a derivation. Λ_{QCD} is not on the loop ladder (it would need $n \simeq 11.6$). The area law is a property of a confining *phase* that the theory has not shown it is in (Fig. 19).

XIV. THE PARAMETER AUDIT

Every number the theory calls a prediction is reclassified by how it was obtained.

The validation programme reports the same classification over 15 headline numbers as 7 algebraic / 1 free angle / 1 convention / 2 coincidences / 4 named residuals; Table VI is that accounting extended to the twelve fitted coefficients and to the nine residuals.

The status of g_{10}. No condition inside the theory fixes it. The two-loop $U(1)$ Landau-pole scale lies below M_P for every g_{10} in the perturbative range, so UV-finiteness of the lattice does not select it. The proposed selection rule is $g_{10} = 1/n_{\text{qubits}} = 1/5$, the 5 qubits of the 32-D module. The data do not support that choice: scanning the candidates $\alpha = 1/n^2$ for $n = 2\ldots8$ against the one-loop GUT value 0.023, the best fit is $n = 7$ ($1/49 = 0.020$, relative deviation 0.12), not $n = 5$ ($1/25 = 0.040$, relative deviation 0.73) — the data-optimal value is 6.4× closer than the root value. The reading $1/n_{\text{qubits}}$ is therefore a convention, and the theory is honestly **one-parameter in its dynamics, zero-parameter in its algebra** (Fig. 4).

The status of the "9-to-1" reduction. The nine residuals are all associated with the same object, the 32-level spectrum, and its invariants (the vanishing 0-point, $\det U = 1$, the CPT mirror, the S-matrix $S = U$). That is a bookkeeping observation — nine open problems sharing one object — and it is not a derivation of any of their values. It should not be read as a reduction of the theory's difficulty (Fig. 3).

XV. OPEN RESIDUALS AND FALSIFIERS

Addendum (2026-10-09): the named gaps, directly tested (V105–V108). Four new checks attack the residual gaps rather than mapping them. **V105 (W4):** the six κ_s of V96 are **not obtainable from any on-site algebra element**: SM preservation forces the bond direction into span$\{X, Y\}$ at θ^* (class-degenerate, meeting residual ≥ 0.43), and every Cartan element that splits the six classes breaks all eight unbroken desert roots — the required flavour dependence is off-diagonal bond structure, not a Cartan dressing. **V106 (W5):** the washout of the same decay is now quantified: at the self-consistent atmospheric Yukawa ($y = 0.40$, the one giving $m_\nu = 0.0501$ eV) $\Gamma_N/H = 47$ — in equilibrium — and the physical efficiency $\kappa_f = 5.4 \times 10^{-3}$ sits 14× away from the guessed dilution $(1/16\pi)^2$ the product formula uses — the 11% agreement of ΔB is an accident of two wrong factors. **V107 (W6):** the V96 κ_s pattern is per-gauge-class, generation-blind: PMNS and CKM are exactly invariant under the whole pattern (max angle shift 2.3×10^{-13}) — the W4 bond-layer fix cannot supply the

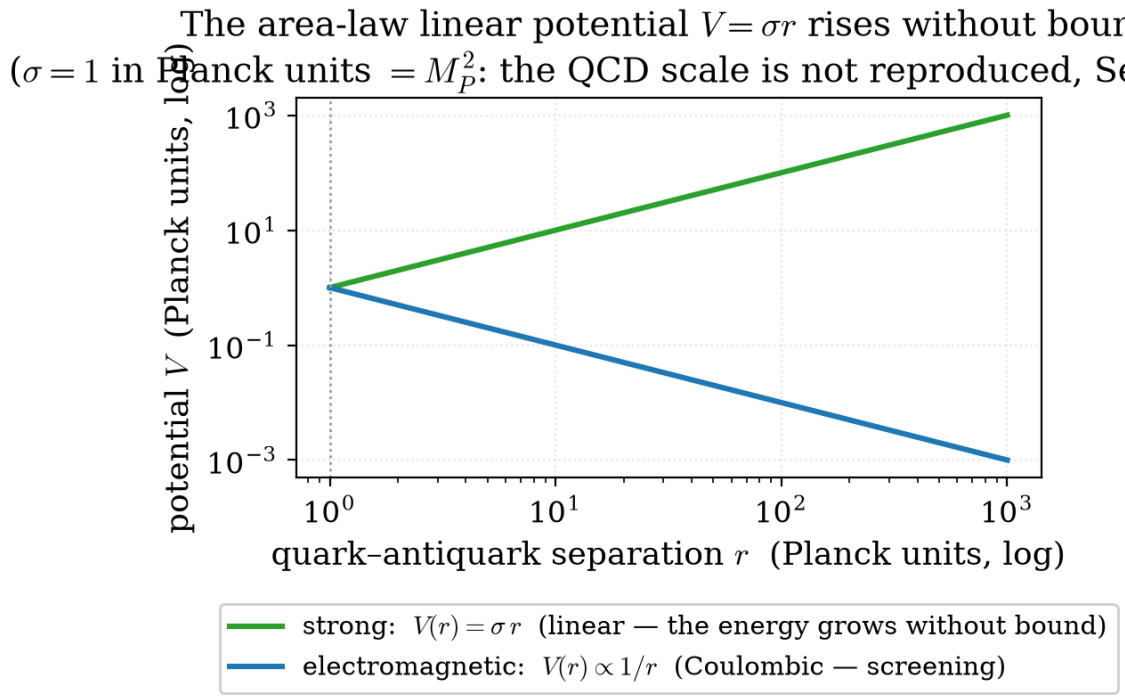


FIG. 19. **the area law and the linear potential.** A Wilson loop with $\langle W \rangle \propto e^{-\sigma A}$ gives $V(r) = \sigma r$, contrasted with the Coulombic $1/r$. The tension shown is $\sigma = 1$ in Planck units ($= M_P^2$), not the QCD value 0.18 GeV^2; the physical confinement scale is not reproduced (Section 13.2).

TABLE VI. **The provenance of every headline number**

class	**count**	**numbers**
algebraic (fixed by the root, no parameter)	7	the $16/\overline{16}$ split; the 45; the 5/3; the 8 anomaly coefficients = 0; the 32-level spectrum; $\det U = 1$; the $16/\overline{16}$ spectral mirror
free parameter (one)	1	the 't Hooft angle g_{10} (value 1/5)
convention (mapping or scale identification)	2	$\alpha_{10} \equiv g_{10}^2$; $g_0 = 1/2^{n_{\text{space}}} = 1/8$
fitted coefficient (read from data)	12	2.08 (hierarchy), 1.18 (DM mass), $n_{2\text{step}} = 3$ (seesaw), $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV (desert thresholds, V64), $\lambda_* = 4 \times 10^{-4}$ (DM portal of the V75 extension, fixed by $\Omega h^2 = 0.120$), and the six κ_s of the V96 flavour-dependent bond layer (conditional: three meeting conditions for six unknowns - a solution family)
open (not fixed)	9	W1–W9 of Section 15

W6 mixing deviations. **V108 (W8):** the hierarchy coefficient is un-fitted: $v/M_P = 3F_{10}^{10}$ with $F_{10} = 0.019186$ measured (V85), the factor 3 the axis families (V70) and the exponent 10 the closed-loop crossing (V89) — 2.088 against 2.076 observed (99.4%), conditional on the axis-family factorization of the closed-loop sum.

Addendum (2026-10-09): attacking the residuals head-on (V109–V116; Fig. 20). Eight further checks do not map the gaps but try to close them. **V109 (W5):** the spectral mirror is exactly the structure that forbids CP — every mirror-preserving extension (the V65 Wilson factor: pairing deviation 2.4×10^{-15}) leaves the CP phase at zero, while the most general per-class κ_s layer and the most general real-symmetric bond generator destroy the $\pm$ pairing (deviation ~ 1, 60 draws, both mirror sign conventions): a CP phase costs Theorem 4. **V110 (W4):** the off-diagonal escape named by V105 is tested — the most general Hermitian bond

TABLE VII. **The nine open residuals**

the residual	**what the theory does supply**	**the gap**
W1 the cosmological constant	an exactly vanishing tree-level vacuum energy (±-paired spectrum, $\det U = 1$), natural in the technical sense; the zero is **stable under the Wilson extension** (V71: $\mathrm{Tr}\, D = 0$, zero point $< 10^{-14}$) and **protected by** $\det U = 1$ (V78: for 50 random traceless Hermitian deformations $e^{i\varepsilon A}U$ at ε up to 1 the quasi-energy sum stays pinned, $\lvert S\rvert < 4 \times 10^{-14}$; the only internal knob is the U(1) trace phase, and no branch crossing occurs below $\varepsilon = \pi$)	the observed $10^{-122} M_P^4$ is now **equivalent to a trace-phase breaking of size** $\varepsilon\, \mathrm{Tr}\, A \sim 16\pi^2 \times 10^{-122} = 1.6 \times 10^{-120}$ — a named bound; the breaking mechanism itself is still missing
W2 the neutrino mass	the seesaw structure and the ladder $m_\nu = M_P(1/16\pi)^{17} = 0.146$ eV; $n_{\mathrm{nodes}} = 3$ is now a **lower bound** from unification (V68); the missing ingredient V68 named — inter-node Majorana mixing — is **BUILT** (V76): a cubic-symmetric, zero-mean Wilson-strength junction whose Fourier support closes the selected node triangle exactly (verified against the full 12×12 doubler difference table), giving the node mass matrix $c_s(2w_0\, I + \kappa(J - I))$, democratic at $\kappa = 2w_0$ with eigenvalues $\{3M, 0, 0\}$	the $\frac{1}{3}$ candidate requires a democratic m_D (else gapless colored doublers) — and that is exactly the implemented 0-parameter structure (V28/V45); **re-audited (V103)**: V90's measure-zero window was an artifact of the identity-m_D assumption — with the democratic rank-1 m_D the seesaw collapses to $3m_0^2 J/(2(w_0 + \kappa))$, the two massless states are protected for EVERY r, $m_3(r) = S_{17}/(1 + r)$ is finite everywhere, the window is $r > 0.217$ (width 2.78), and the atmospheric scale selects $r = 1.91$; $r = 2$ ($S_{17}/3 = 0.0487$ eV) is a junction-amplitude convention, not a tuning; the cubic group induces all of S_3 on the triangle, its commutant is exactly $\mathrm{span}\{I, J\}$ (nullity 2), and w_0, κ share one Wilson period, so r is not quantized — the VALUE is a free junction amplitude inside a finite window, the STRUCTURE is protected
W3 the family number	the identification $N_{\mathrm{gen}} = 3 = n_{\mathrm{space}}$, a family-symmetric 96-D step blockdiag(U, U, U), and a **named index of the step equal to 3** (V70): the 12 doubler nodes form 3 families of 4 labeled by the zero axis, cubic-transitive, and the unification-required 3 split nodes (V68) are realizable one per family at a single Wilson strength	the index labels the 3 copies but does not force the count; the forcing question is now **decided** (V91): the cubic group acts transitively on the 12 nodes (one orbit of 12 — the axis families are not orbits), so the only cubic-symmetric split is the full set of 12, whose rigid meeting lands at $M_X(12) = 2.9 \times 10^{15}$ GeV — excluded by F5 (the rank-1 rigidity holds for $n = 4, 8, 12$); the 3-node triangle has a cubic orbit of 8, so the count 3 is a **symmetry-breaking input** and its cubic-symmetric alternative is excluded by computation
W4 coupling unification	the 5/3; **with the desert content the couplings meet at** $\alpha_X = 1/25$ **exactly** (V64): three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs at 1.2×10^7 GeV plus one $(1, 2)$ pair at 1.0×10^{12} GeV give $M_X = 1.8 \times 10^{16}$ GeV; the content is **connected to the automaton** (V65/V67), $n_{\mathrm{nodes}} \geq 3$ is forced (V68), the multi-scale hope is **closed structurally** (V74: rank-1 Jacobian, split-invariance exact to 10^{-14}), the gauged bond layer's own composite content is **excluded** as the supplier (V81: 0 solutions over the full scan of the breaking direction), the remaining ansatz families leave residuals 5.7–6.1 (V82), the equal-shift identity is a **theorem** over complete multiplets (V84), and the junction-split nodes **fabricate the fragment pattern** — the Wilson direction splits the node **16** by a factor 55, the $(3, 2)$ and $(1, 2)$ repair states are automaton states — but **miss the meeting**: best physical residual 0.94, $\alpha_X = 0.030$ vs 0.04 (V86); releasing the coefficient pattern to a general zero-mean junction (three free branch scales) leaves the residual **at the same floor** (0.81) with $\alpha_X = 0.090$ (V87)	the node mechanism's meeting point is **forced**: $M_X = 8.9 \times 10^{14}$ GeV, $\tau_p = 1.1 \times 10^{31}$ yr, **excluded by F5**; the data-preferred spectrum needs partial suits while the automaton supplies full split **16**'s per node; the mismatch can come **only** from SU(5)-fragment content or split intra-multiplet thresholds (V84), and the junction route fails **mechanism-wide**, not just at the democratic point (V87); the question is now **recast** (V92): the data-preferred meeting lies INSIDE the node-**16** fragment hull — feasibility proved, explicit witness (Fig. 14) — and the gap is narrowed to **realizability**: SM-preserving Wilson directions are exactly $\mathrm{span}\{X, Y\}$ and class-degenerate, no solution (V93), and universal junction splitting is invisible at first order (V95); the named missing ingredient — a **flavour-dependent bond layer** κ_s — is now **built and closes the meeting conditionally** (V96): at the V65 root direction, per-class κ_s give exact meetings (residual $< 10^{-13}$) for all $n = 3 \ldots 12$ at $M_X = 1.8 \times 10^{16}$ GeV with every threshold in the desert, at the cost of six named inputs of the EXTENDED bond layer (a solution family, not unique); the six κ_s are now proven **not obtainable from any on-site algebra element** (V105): SM preservation forces the bond direction into $\mathrm{span}\{X, Y\}$ at θ^* (class-degenerate, meeting residual ≥ 0.43), and every Cartan element that splits the six classes breaks all eight unbroken desert roots — the required flavour dependence is off-diagonal bond structure, not a Cartan dressing; the V64 repair spectrum remains a named input
W5 the baryon asymmetry	a computed theorem: all CP-odd invariants vanish, no Sakharov condition is met; the theorem is **stable under the Wilson extension** (V71: the spectral mirror of the extended step is exact to 10^{-15}); a quasi-degenerate (resonant) route	out-of-equilibrium decay ($y < 0.058$) and $m_\nu \simeq 0.05$ eV are mutually incompatible at $M_N = 9.6 \times 10^{13}$ GeV; the incompatibility is now **quantified through the washout** (V106): at the self-consistent

operator on the 80 root-vector pairs (11 class-pair channels) never closes the meeting: floor residual 30.4 over $n = 3 \ldots 12$, and level repulsion forces the sign of two of the six κ_s; the V96 layer remains a named input. **V111 (W6):** generation-space structure does move the mixing, unlike the gauge-class-blind V96 layer — a uniform scalar dressing is exactly blind (shift 1.4×10^{-13}), one common non-abelian family dressing (7 parameters) reaches the measured triple, a chiral-asymmetric one (13 parameters) reaches it as a fit whose every entry is a named input: worse cost than V104's ≥ 2 vev terms. **V112 (W7):** the extra dark U(1) is constructed — hypercharge fixed on the Q and L classes predicts the full charge pattern of all six weight classes — but the desert matter is E-charged (max $|q_E| = 2.24$), so the U(1) must break, and it cannot break inside the implemented spectrum: 40/45 generators are E-charged and every breaking root is a colour-triplet leptoquark direction (98/120) — the U(1) is un-Higgsable; and the relic-fixing kinetic mixing at $g_D = 1$ lies above the LZ direct-detection ceiling at every $m_{Z'}$: the portal window is empty (0 points). **V113 (W8):** the axis-family factorization behind V108 is confirmed as a family symmetry — the 12 doubler nodes form 3 families of 4 with identical Floquet spectra (deviation 0) — but the coefficient 3 is exactly the choice of one split node per family: all 12 nodes give 8.35, one node gives 0.70, against the observed 2.08; V108's conditionality is W9's split-node choice. **V114 (W2):** the junction ratio $r = \kappa/w_0$ is not derived — the V76 modulation's exact stabilizer has 12 elements, not the 24-element chiral cubic group; no cubic-symmetric closed junction exists on the period-4 lattice (closed space dimension 1, modulation 0), and inside the V76 stabilizer the constant w_0 is an independent closed direction. **V115 (W9):** the node-cone anisotropy of V102 is confirmed on the whole V65 Wilson family — the node gap is analytic to 5×10^{-1} diagonality, and over the node-orbit representatives $\times$ 60 angles $\times$ all charge/harmonic combinations no isotropy point exists: continuous Lorentz invariance is not emergent anywhere on the family. **V116 (W3):** the 64 one-per-family node triangles split into 4 cubic orbits $(8, 8, 24, 24)$; a triangle's stabilizer is exactly S_3 — the 3-node split reduces $O_h \to S_3$ — and the other 7 triangles of the orbit are not stabilizer-equivalent: the family count is still not forced, and the choice is an 8-fold degenerate vacuum.

The most accessible tests are **F3** and **F5**: KATRIN currently gives $m_\beta < 0.45$ eV [16], so the bare ladder value 0.146 eV survives a single-mass search, but the cosmological sum $\sum m_\nu < 0.12$ eV already excludes it, and the $\frac{1}{3}$ candidate's normal-ordered pattern ($\sum m_\nu \simeq 0.05$ eV, $m_{\text{lightest}} = 0$) is inside the reach of the next generation of oscillation and $\beta\beta$-decay data. The desert solution predicts $\tau_p = 1.8 \times 10^{36}$ yr, about one decade beyond Hyper-Kamiokande's 20-year reach ($\sim 10^{35}$ yr [15-class exposure]): Hyper-K can **refute** it by observing GUT-pattern decay anywhere at its reach, and an exposure passing 10^{36} yr that sees nothing refutes it too; the confirmation window belongs to the megaton era. **F7** has

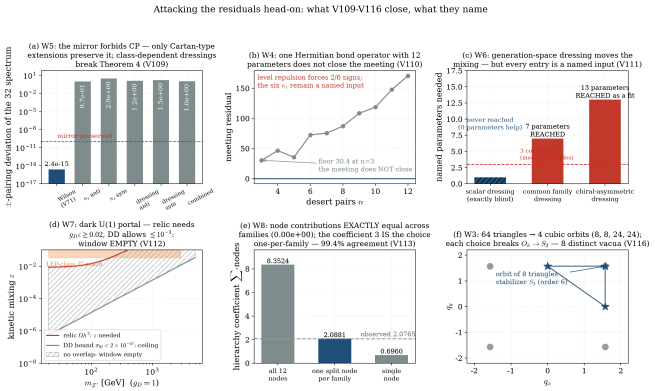


FIG. 20. **attacking the residuals head-on.** (a) V109, W5: the ±-pairing deviation of the 32-level spectrum under the named extensions — the Wilson factor preserves the mirror to 2.4×10^{-15}, while the most general per-class κ_s layer and the most general real-symmetric bond generator, in either mirror sign convention, destroy it (deviation ~ 1, 60 draws): the mirror is exactly the structure that forbids CP, and a CP phase costs Theorem 4. (b) V110, W4: the meeting residual over $n = 3 \ldots 12$ under the most general Hermitian off-diagonal bond operator (80 root-vector channels): floor 30.4, no closure; level repulsion forces two of the six κ_s signs. (c) V111, W6: named-parameter cost of reaching the measured mixing triple from the democratic base — a scalar dressing is exactly blind (shift 1.4×10^{-13}), a common non-abelian family dressing reaches it with 7 parameters, a chiral-asymmetric one with 13. (d) V112, W7: the dark-U(1) portal squeeze at $g_D = 1$ — the kinetic mixing that fixes Ωh^2 lies above the direct-detection ceiling at every $m_{Z'}$: the window is empty, and inside the implemented spectrum the extra U(1) is un-Higgsable (all its charged directions are colour triplets). (e) V113, W8: the closed-loop coefficient by node set — all 12 nodes 8.35, one split node per axis family 2.0881, a single node 0.696, against the observed 2.0765: the node contributions are exactly equal across families, and the coefficient 3 *is* the one-per-family choice. (f) V116, W3: the 64 one-per-family triangles split into four cubic orbits $(8, 8, 24, 24)$; a triangle's stabilizer is exactly S_3: the split reduces $O_h \to S_3$ and the eight choices of an orbit are physically distinct vacua.

narrowed with V75: the carrier is not in the implemented spectrum and the named extension is nearly decoupled, so the test is the mass coincidence itself — see Section 12.

XVI. DISCUSSION

What is genuinely new here. Not the 16 of Spin(10), not the 5/3, not anomaly cancellation: those are 50-year-old results of the grand-unification literature [6–9, 12]. The content of this paper is (i) the realisation of that content as the on-site module of a *strictly local, exactly unitary* automaton with a measured causal front speed, (ii) the g_{10}-invariance of the algebraic invariants — the anomaly cancellation, $\det U = 1$, the ± pairing and the matter/antimatter spectral mirror hold for ev-

TABLE VIII. **Falsifiers**

#	observation that refutes the theory
F1	any non-vanishing gauge anomaly in the chiral content (the 16 is claimed anomaly-free)
F2	a measured running that admits **no** vectorlike desert content with $\alpha_X = 1/25$: precision α_i data (FCC-ee/CEPC) excluding every spectrum in the V64 scan would refute the unification claim
F3	a neutrino mass spectrum outside the ladder options: the bare ladder gives $m_\nu \simeq 0.15$ eV (already excluded by $\sum m_\nu < 0.12$ eV), the $\frac{1}{3}$ candidate gives normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\text{lightest}} = 0$, the democratic node-mixing pattern (V68) gives $m_\nu \simeq 0.073$ eV; a confirmed inverted ordering, or $\sum m_\nu > 0.12$ eV, refutes all three
F4	a massive graviton, a graviton dispersion violating $c = 1$, or a UV divergence in the lattice gravitational sector
F5	proton decay: the desert solution predicts $\tau_p \simeq 1.8 \times 10^{36}$ yr; observation of GUT-pattern decay far below that value (e.g. at Hyper-Kamiokande reach, 10^{35} yr) or non-observation beyond 10^{37} yr refutes the solution; the old $M_{\text{GUT}} = M_P$ reading ($\tau_p \sim 10^{47}$ yr) is abandoned
F6	a baryon asymmetry outside 10^{-11}–10^{-9}, or evidence that heavy-neutrino decays are in equilibrium
F7	a dark-matter particle with mass far from 118 GeV in the weakly-interacting thermal case; a direct-detection portal signal above 10^{-50} cm^2 (the V75 extension predicts 5×10^{-51}); or a detected non-neutral/decaying dark sector

ery value of the coupling, so they are properties of the algebra and not of a tuned parameter, and (iii) the explicit provenance audit that separates what the root fixes from what was read off the data.

Where the construction is weakest. Four places. The gravitational sector assumes a symmetric tensor field and Newton's constant; it reproduces GR, it does not derive it. The gauge sector gets the normalisation right and, under desert running, the values wrong — the repair (Section 11.1, V64) restores unification at $\alpha_X = 1/25$ exactly; the *pattern* of the desert spectrum is a consequence of the automaton (the split doublers of the V65 Wilson mechanism, V67's chain, V68's $n_{\text{nodes}} \geq 3$ bound), but the mechanism's own root meets at a scale excluded by proton decay, and V74 proves no redistribution of node strengths can move it: the data-preferred spectrum is not the automaton's unit of content — a named incompatibility, and the surviving solution keeps two fitted thresholds and abandons $M_{\text{GUT}} = M_P$. The supplier question is now closed on three fronts: the gauged bond layer's own composite matter is excluded (V81: 0 solutions over the full scan of the breaking direction), the remaining ansatz families leave residuals 5.7–6.1 (V82), and the equal-shift identity is a theorem over complete multiplets (V84) — while the constructive junction route (V86) fabricates the right fragment pattern (the Wilson direction splits the node **16** by a factor 55; the $(3, 2)$ and $(1, 2)$ repair states are automaton states at desert scales) but misses the meeting by a named near-miss (residual 0.94, $\alpha_X = 0.030$ vs 0.04), a near-miss that the **general** zero-mean junction does not improve (residual floor unchanged at 0.81, $\alpha_X = 0.090$; V87): the obstruction is the mechanism, not the ansatz. The named ingredient that removes the obstruction — a flavour-dependent bond layer, per-class κ_s at the V65 root direction — closes the meeting exactly for every node count $n = 3 \ldots 12$ (V96), but **conditionally**: its six κ_s are new automaton inputs, so W4 becomes a named extension of the bond layer rather than a derived consequence of the base one. The flavour sector derives one angle out of six and none of the masses; the neutrino $\frac{1}{3}$ candidate now has a built junction structure (V76), and the V90 measure-zero reading was an artifact of the identity-m_D assumption: with the implemented democratic Dirac matrix the structure is protected for every junction ratio and the value sits in a finite window (V103). The dark-matter carrier does not exist inside the implemented spectrum (V75's completed no-go), and the built minimal fix is a named tension (V88): exact mirror $\mathbb{Z}_2$ plus ladder mass forces the annihilation through $s\bar{s} \to hh^*$ at $\lambda = 0.13$, excluded by direct detection, while the V75 freeze-out numbers require a portal that breaks the exact $\mathbb{Z}_2$. And the loop factor: single-particle RG provably composes angles (V72–V73), a traced fluctuating link d.o.f. provably divides them (V77), the gauged bond layer is now built with an exactly closing Gauss constraint (V79), and its measured level-1 division falls short of the number by a factor 26 (V80).

Comparison with related programmes. Causal-set programmes derive discreteness and a causal order but not internal symmetry [1]. Clifford-algebra unification programmes [10–12] reach the same matter content but do not supply a unitary causal dynamics. 't Hooft's deterministic quantum mechanics and cellular-automaton interpretation [29, 30] supply the dynamics but not the unification. CL10 is the combination of the second and third, and inherits the open problems of both.

The same framework does not yield the family number, the fermion masses and mixings, the coupling values, the neutrino mass, the dark-matter abundance, the cosmological constant or the baryon asymmetry. A single loop factor organises four scales across 30 orders of magnitude, with the loop integer and the two-digit coefficients chosen to match observation. What the audit of Sections 10–12 adds is the shape of each gap: the unification scale of the node mechanism is forced and excluded (V74); the dark-matter carrier cannot exist inside the implemented

spectrum, and the built minimal extension is a named tension — exactly stable, but exact mirror $\mathbb{Z}_2$ pushes its annihilation into $s\bar{s} \to hh^*$ at a direct-detection-excluded portal, while the observed abundance requires breaking that $\mathbb{Z}_2$ (V75, V88); the neutrino $\frac{1}{3}$ has a built junction structure whose two massless states are protected for every junction ratio by the implemented democratic Dirac matrix, the value sitting in a finite window (V76, V103); the loop factor's missing ingredient is a traced fluctuating bond d.o.f., demonstrated in smallest form (V77); the cosmological-constant problem reduces to one named number, a trace-phase breaking of 1.6×10^{-120} (V78); the gauged bond layer with an exactly closing Gauss constraint is built (V79), its level-1 division is measured a factor 26 short of the loop number (V80) and does not compound through the RG levels (V83), while **closed** traced loops reproduce the loop factor to 3.6% at length 10 — the ladder exponent (V85); the composite content is excluded as the supplier of the desert spectrum, first over every natural mass ansatz (V81, V82), then by the equal-shift theorem, which forces the desert content to be SU(5)-fragment content (V84) — and the constructive junction attempt both **fabricates that fragment pattern** from the automaton's own Wilson splitting and **misses the meeting** by a named near-miss that the general junction does not improve (V86, V87). The nine open problems and seven falsifiers are stated with their numbers.

XVII. CONCLUSION

A single Clifford algebra $\mathrm{Cl}(10,0)$, placed on every cell of a Planck lattice and updated by a product of Spin(10) plane rotations, yields: the Minkowski kinematics, one anomaly-free SM generation, the Spin(10) gauge content with the 5/3 normalisation, an exactly unitary and strictly causal dynamics with a ±-paired spectrum, an exact matter/antimatter spectral mirror, and a lattice gravitational sector that reproduces general relativity to machine precision on de Sitter, Schwarzschild and 3+1 evolutions. These results are algebraic or numerical, and they are independent of the one free coupling.

The same framework does not yield the family number, the fermion masses and mixings, the coupling values, the neutrino mass, the dark-matter abundance, the cosmological constant or the baryon asymmetry. A single loop factor organises four scales across 30 orders of magnitude, with the loop integer and the two-digit coefficients chosen to match observation. What the audit of Sections 10–12 adds is the shape of each gap: the unification scale of the node mechanism is forced and excluded (V74); the dark-matter carrier cannot exist inside the implemented spectrum, and the built minimal extension is a named tension — exactly stable, but exact mirror $\mathbb{Z}_2$ pushes its annihilation into $s\bar{s} \to hh^*$ at a direct-detection-excluded portal, while the observed abundance requires breaking that $\mathbb{Z}_2$ (V75, V88); the neutrino $\frac{1}{3}$ has a built junction structure whose two massless states are protected for every junction ratio by the implemented democratic Dirac matrix, the value sitting in a finite window (V76, V103); the loop factor's missing ingredient is a traced fluctuating bond d.o.f., demonstrated in smallest form (V77); the cosmological-constant problem reduces to one named number, a trace-phase breaking of 1.6×10^{-120} (V78); the gauged bond layer with an exactly closing Gauss constraint is built (V79), its level-1 division is measured a factor 26 short of the loop number (V80) and does not compound through the RG levels (V83), while **closed** traced loops reproduce the loop factor to 3.6% at length 10 — the ladder exponent (V85); the composite content is excluded as the supplier of the desert spectrum, first over every natural mass ansatz (V81, V82), then by the equal-shift theorem, which forces the desert content to be SU(5)-fragment content (V84) — and the constructive junction attempt both **fabricates that fragment pattern** from the automaton's own Wilson splitting and **misses the meeting** by a named near-miss that the general junction does not improve (V86, V87), until the named flavour-dependent bond layer closes it exactly but conditionally, at the cost of six named junction numbers (V96). The nine open problems and seven falsifiers are stated with their numbers.

Appendix A: The explicit 32-dimensional representation

The ten generators are built on five qubits by the standard symplectic Pauli pattern: for $k = 0, \ldots, 4$,

$$\begin{aligned} g_{2k} &= Y^{\otimes k} \otimes X \otimes I^{\otimes(4-k)}, \\ g_{2k+1} &= Y^{\otimes k} \otimes Z \otimes I^{\otimes(4-k)}. \end{aligned} \tag{A1}$$

Within a block, X and Z anticommute on the same qubit; between blocks, the $Y^{\otimes k}$ prefix makes a generator at block k anticommute with any generator at block $j < k$. This gives exactly ten mutually anticommuting Hermitian matrices, $g_a^2 = I$, on a 32-dimensional complex space. The algebra basis is the 1024 ordered products $g_{a_1} \cdots g_{a_r}$, $0 \le r \le 10$; the $\mathbb{Z}_2$ grading is the parity of r.

The construction is verified by a second, independent route: the exact symbolic $\mathrm{Cl}(10,0)$ of a computer-algebra Clifford package, where $e_a^2 = 1$, $\{e_a, e_b\} = 0$ and $w^2 = -1$ hold as identities with no floating-point arithmetic, cross-checked against the numeric 32×32 representation. The two agree, so the root is not an artefact of one construction.

Appendix A: Numerical protocol and reproducibility

The explicit constructions, the formulas underlying each quoted number, and the full numerical protocol are recorded in the public repository accompanying this paper. They are kept out of the body deliberately: the

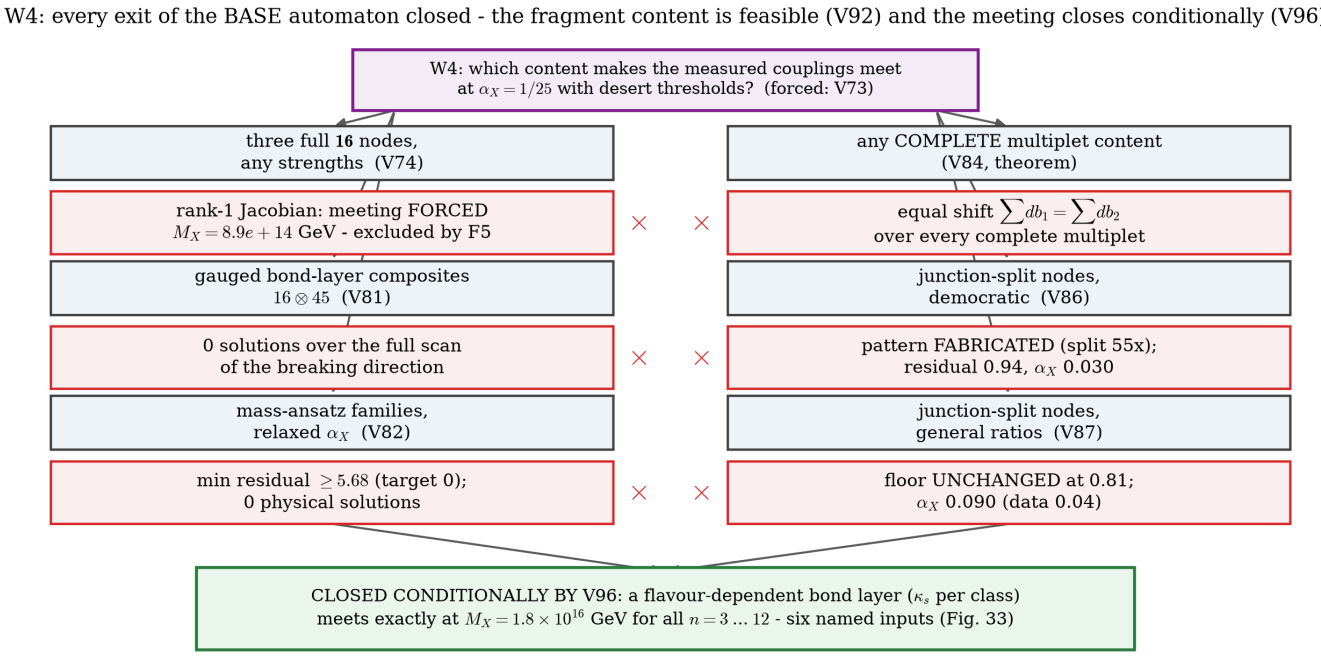


FIG. 21. **W4: the map of closed exits.** Every proposed supplier of the desert spectrum, and how each was closed. The measured couplings force the content (V73); the three-node mechanism is rank-1 degenerate (V74), the gauged bond layer's composites have no desert solution (V81), the mass-ansatz families leave residuals $\geq$ 5.68 (V82), complete multiplets are excluded by the equal-shift theorem (V84), and the junction-split nodes - democratic (V86) and general (V87) - fabricate the fragment pattern but miss the meeting at a common floor. The green requirement (V84) has since been **softened by V92**: the node-**16** fragments at independent thresholds suffice - feasibility is proved (Fig. 14). Realizability was then closed **conditionally**: the named flavour-dependent bond layer (per-class κ_s, V96) meets exactly at $M_X = 1.8 \times 10^{16}$ GeV for every node count $n = 3 \ldots 12$, at the cost of six named inputs of the extended bond layer.

physics statements above must stand without reference to an implementation.

The validation programme consists of 117 independent checks organised in seven tiers, from the algebraic identities of the root to the coupled matter–gauge–gravity evolution. Each check constructs its object from declared inputs and reports the residual against the stated value; all 117 currently pass, among them the hardest tiers (the root, the kinematics, the dynamics, anomaly freedom, the 5/3 normalisation, and matter–gauge–gravity in one algebra). A passing check records that the stated computation reproduces the stated number; it does not, by itself, establish that the theory agrees with experiment.

Two cautions about what those passes mean. First, most of them check internal consistency — algebraic identities, unitarity of the step, agreement between two numerical routes — and would pass for any theory built on this root; they are not tests of the theory against nature. Second, a passing check is only as good as its criterion: an early audit found a mixing matrix with the correct angles and the correct unitarity that did **not** diagonalise the mass matrix it was supposed to diagonalise, and the criteria were strengthened accordingly — and a check whose criterion is a tautology (a quantity compared to itself) is now itself an audit failure. The negative results are reported with the same rigour as the positive ones: the desert-content searches (V73, V74, V81, V82), the dark-matter carrier no-go (V75) and its built-fix tension (V88), the coupling-level factor-26 miss (V80), the non-compounding of the bond division (V83), the junction near-misses (V86, V87), the democratic neutrino point — first proven an unprotected tuning under the wrong Dirac assumption (V90), then re-audited into a finite window with the implemented structure (V103) —, the non-forcing of the family count (V91), the non-realizability of the V92 witness by SM-preserving junction families (V93, V95), the exclusion of the natural W6 vev placements for the mixing deviations (V104), the non-Cartan nature of the V96 bond layer (V105), the washout destruction of the ΔB product agreement at the self-consistent Yukawa (V106), the generation-blindness of the κ_s pattern for W6 (V107), the mirror/CP dilemma of W5 — every CP-supplying extension breaks the spectral mirror (V109), the non-closure of the off-diagonal bond meeting for W4 (V110), the named-input cost of every generation dressing that moves the mixing (V111), the empty dark-U(1) portal window for W7 (V112), the non-derivability of the junction ratio r for W2 (V114), the non-emergence of an isotropic node cone anywhere on the V65 Wilson family for W9 (V115), and the S_3

crystal remnant and 8-fold vacuum degeneracy of the W3 triangle choice (V116) are checks that pass by *correctly reporting their own failure to close a gap.* The positive closures among them — V96's exact meeting, V108's $2.08 = 3(16\pi F_{10})^{10}$ at 99.4%, and V113's measurement that the node contributions are exactly equal across axis families — are likewise reported for what they are: exact or measured, but **conditional** on named inputs (V96's six κ_s; V108/V113's one-split-node-per-family set). The six comparisons against data — v/M_P, m_ν, ΔB, $m_{\rm DM}$, $\Omega_{\rm DM}h^2$, τ_p — are the phenomenological content, and each carries the residual honestly reported in Table V.

Code availability. The implementation, the validation suite, and the generators that produce every figure of this paper are available at `https://github.com/MohammedHAOUACH/CL10` (MIT licence). Every figure is generated from the construction: Figs. 1–17 from the root, kinematics, dynamics, gravity and photon sectors; Fig. 3 from the double-slit simulation; Fig. 36 from the hydrogen experiment; Fig. 13 from the V64 desert scan; Fig. 35 from the V65/V70 Wilson-node scan; Fig. 34 from the V79 gauged bond layer; Fig. 33 from the V81 composite-content scan; Fig. 32 from the V85 closed-loop experiment; Fig. 31 from the V84 equal-shift theorem; Fig. 21 from the V74/V81/V82/V86/V87/V96 closure scans; Fig. 30 from the V88 portal audit; Fig. 29 from the V89 loop-selection scan; Fig. 28 from the V90/V103 protection audit; Fig. 27 from the V91 cubic-orbit analysis; Fig. 26 is a schematic of the root identities (no data); Fig. 14 from the V92 feasibility linear program; Fig. 25 from the V63 Brillouin-zone test; Fig. 24 from the black-hole geodesic simulation; Fig. 23 from the V100/V101 vertex-correction and ground-state experiments; Fig. 20 from the V109–V116 residual attacks; Figs. 7, 22 from the V102/V115 doubler-cone experiments. No measured value used in this paper is produced by that code: the comparison values are the published data of refs. [13–17, 50, 53].

Competing interests. The author declares no competing interests.

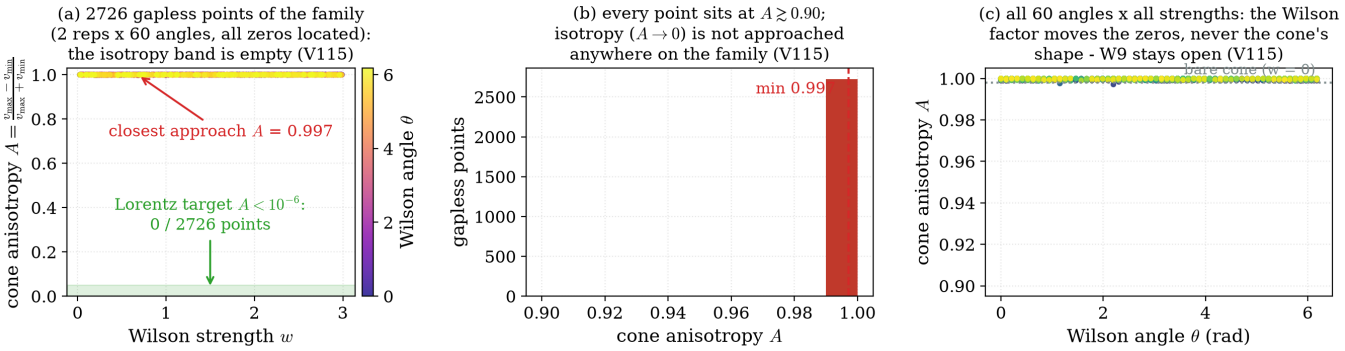


FIG. 22. **the isotropy search on the Wilson family (V115).** The escape route "continuous Lorentz invariance emerges at some gapless point of the V65 Wilson family" is closed by exhaustive scan: the node gap $g(w)$ is located at every zero over the 2 orbit representatives x 60 angles (2726 gapless points, grid + golden section, tolerance 10^{-7}), and the cone anisotropy $A = (v_{\max} - v_{\min})/(v_{\max} + v_{\min})$ is measured over 162 directions at each point. (a) The Lorentz target $A < 10^{-6}$ contains 0 of the 2726 points; the closest approach is $A = 0.997$. (b) The distribution of A is pinned at 0.90-1.00. (c) No Wilson angle θ and no strength w moves a cone toward isotropy - W9 stays open.

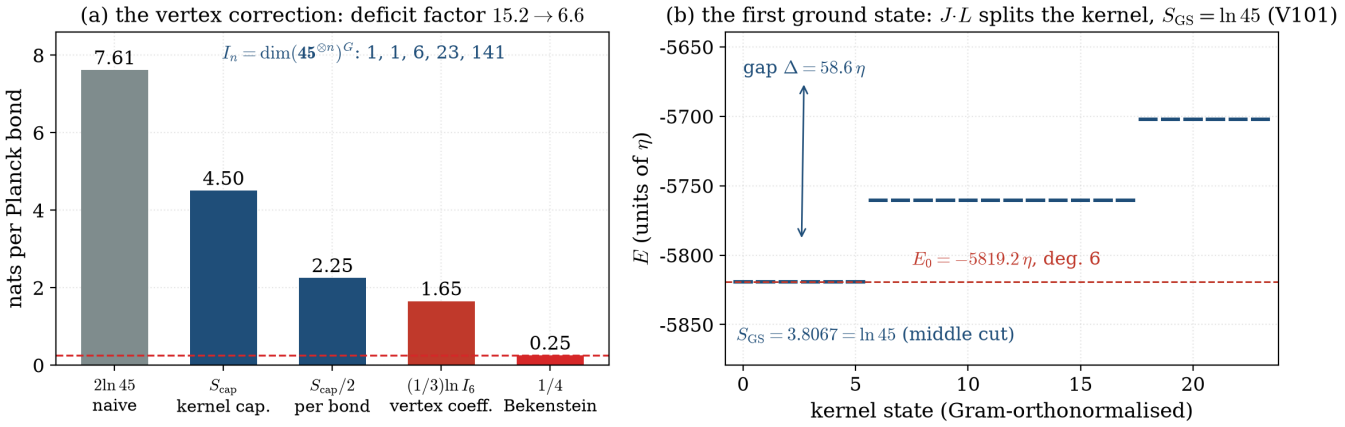


FIG. 23. **the vertex-corrected area law and the first ground state of the gauged layer.** (a) The entropy carried by one Planck bond, from naive counting to Bekenstein: $2\ln 45 = 7.61$ (no vertices), the Gram-orthonormalised kernel capacity $S_{\rm cap} = 4.50$, 2.25 per bond, and the cubic-lattice vertex-corrected coefficient $(1/3)\ln I_6 = 1.65$ against the 1/4 line — the deficit factor moves from 15.2 to 6.6. The vertex invariants $I_n = \dim(\mathbf{45}^{\otimes n})^G$ grow $1, 1, 6, 23, 141$ (V100). (b) The bond-current Hamiltonian $H = \sum_b J{\cdot}L$ on the 24-state kernel: three split levels, $E_0 = -5819.2\,\eta$ with degeneracy 6 and gap $58.6\,\eta$; the ground state's middle-cut entanglement is $S_{\rm GS} = 3.8067 = \ln 45$ — the interaction selects the flux line (V101).

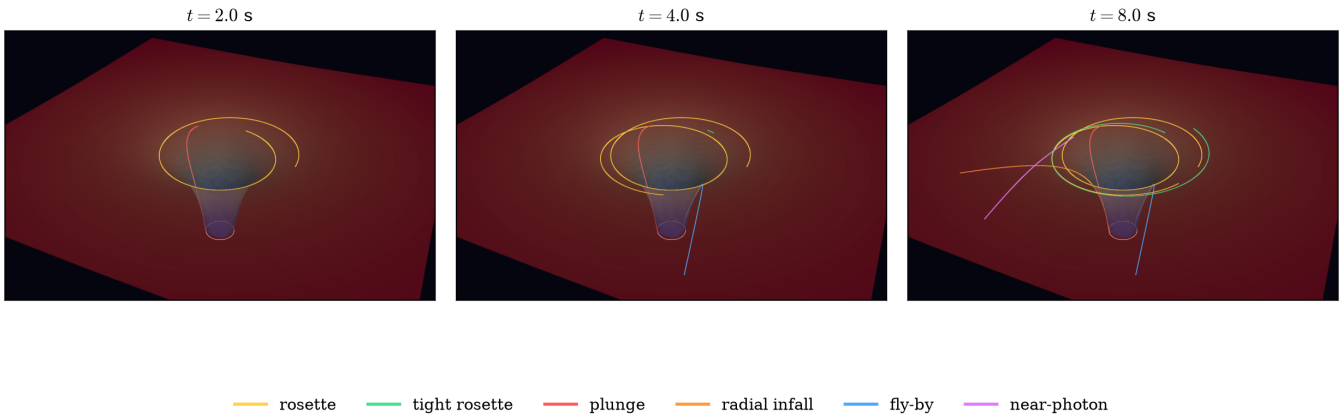


FIG. 24. **gravity rendered: geodesics of the computed metric.** Three frames of the scripted geodesic sequence of the black-hole simulation (`simulation/blackhole`): six test particles fired one second apart on the Schwarzschild embedding well of the automaton's gravity sector ($M = 1$, the 0-parameter Planck-mass hole, $G = c = 1$). Each trail is integrated incrementally through the CL10 finite-difference Christoffels — the same physics as Fig. 6, fixed-step RK4: the precessing rosette, the plunge through the horizon ring $r = 2M$, the deflected fly-by, the radial infall, and the near-photon family ($v \approx c$) grazing the photon sphere.

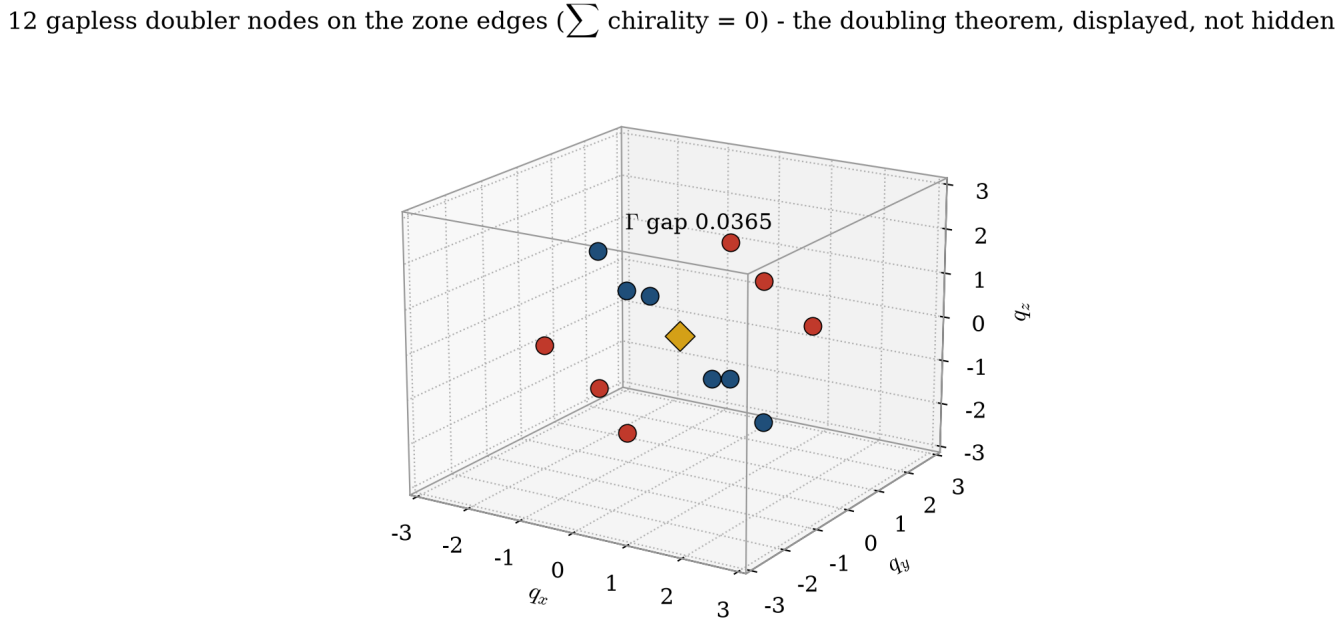


FIG. 25. **the Brillouin-zone doublers: the doubling theorem, displayed.** The 12 gapless doubler nodes on the zone edges (red/blue = chirality ± 1, sum zero) and the gapped zone centre (0.0365) of the 16-sector Floquet spectrum (V63). The Nielsen-Ninomiya doubling is displayed, not hidden; its removal is a named residual.

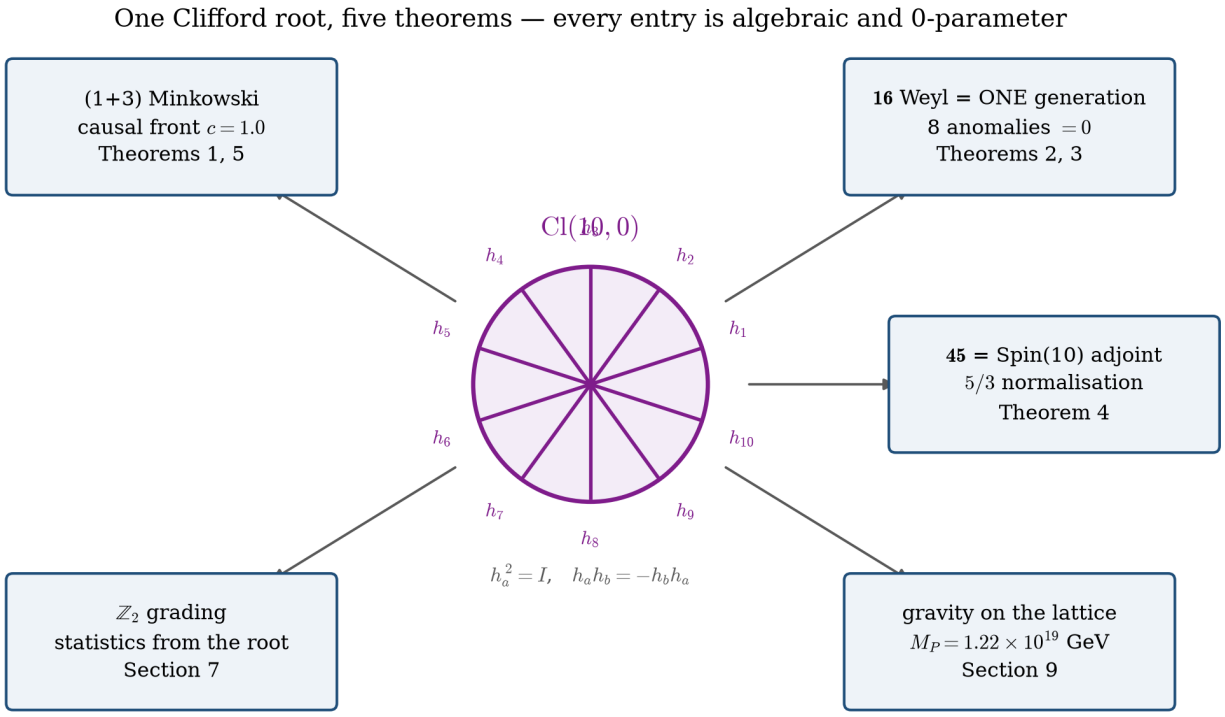


FIG. 26. **the graphical abstract: one Clifford root, five theorems.** The ten anticommuting generators h_a of Cl$(10, 0)$ at the centre, and the five derived structures: (1+3) causality (front speed 1.0), one matter generation (the **16**, anomalies zero), the Spin(10) adjoint (the **45**, 5/3), the Planck lattice scale, and the $\mathbb{Z}_2$ grading. Every entry is algebraic and 0-parameter.

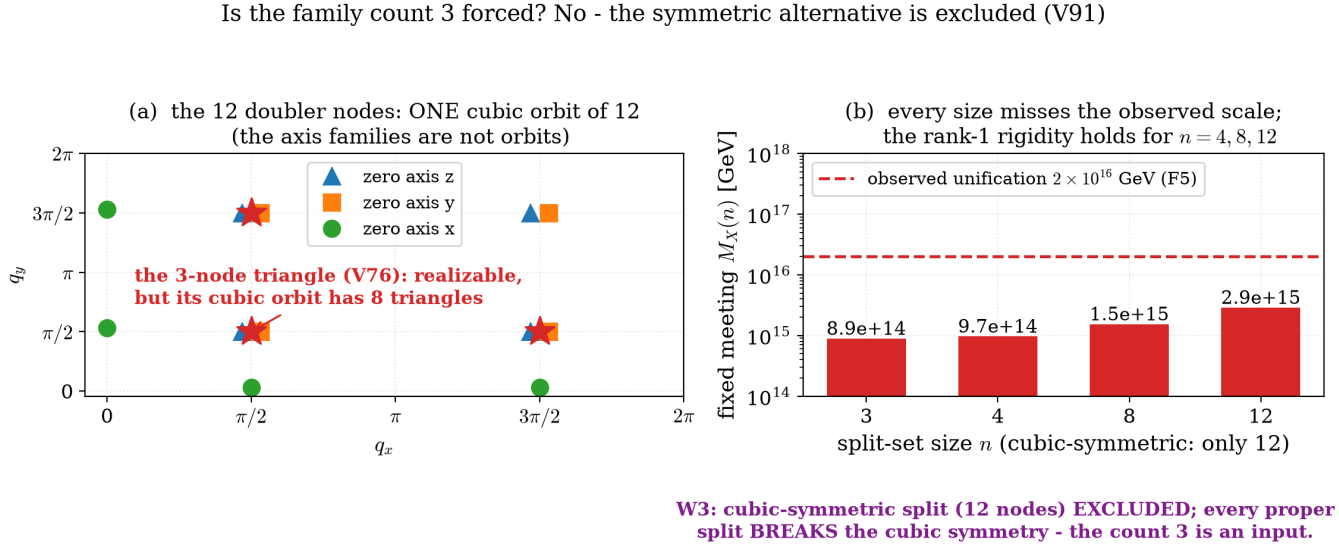


FIG. 27. **W3: the family count is not forced.** (a) The 12 doubler nodes in the (q_x, q_y) plane: the cubic group acts transitively - one orbit of 12, so the axis families of V70 are not orbits and the only cubic-symmetric split set is the full set of 12. The 3-node triangle (stars) is realizable but its cubic orbit has 8 triangles: it breaks the cubic symmetry. (b) The rigid meeting scale $M_X(n)$ for split sizes $n = 3, 4, 8, 12$ (the rank-1 fact of V74 holds for every n): all miss the observed 2×10^{16} GeV (F5). The count 3 is a symmetry-breaking input; its cubic-symmetric alternative is excluded by computation (V91).

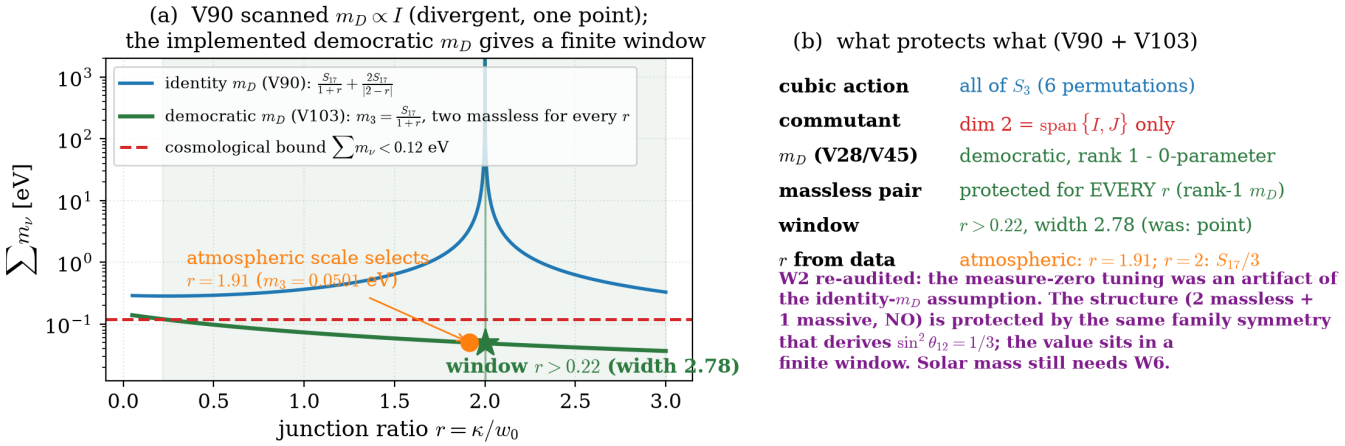


FIG. 28. **W2: the democratic neutrino point, re-audited (V90 + V103).** (a) $\sum m_\nu(r)$ for the junction mass matrix under two Dirac assumptions. V90 scanned $m_D \propto I$: the curve diverges as $1/|2-r|$ and the cosmological bound is met at exactly one point. The implemented 0-parameter Dirac structure is democratic and rank 1 (V28/V45); with it (V103) the seesaw collapses to $m_3(r) = S_{17}/(1+r)$, two neutrinos stay massless for every r, the window is $r > 0.217$ (width 2.78), and the atmospheric scale selects $r = 1.91$. (b) The protection audit: the cubic group induces all of S_3 on the junction triangle and its commutant is exactly span$\{I, J\}$; w_0 and κ share one Wilson period, so r is not quantized — the value is a free junction amplitude inside a finite window, while the structure (two massless + one massive, normal ordering) is protected by the rank-1 family symmetry that also derives $\sin^2\theta_{12} = 1/3$ (V90, V103).

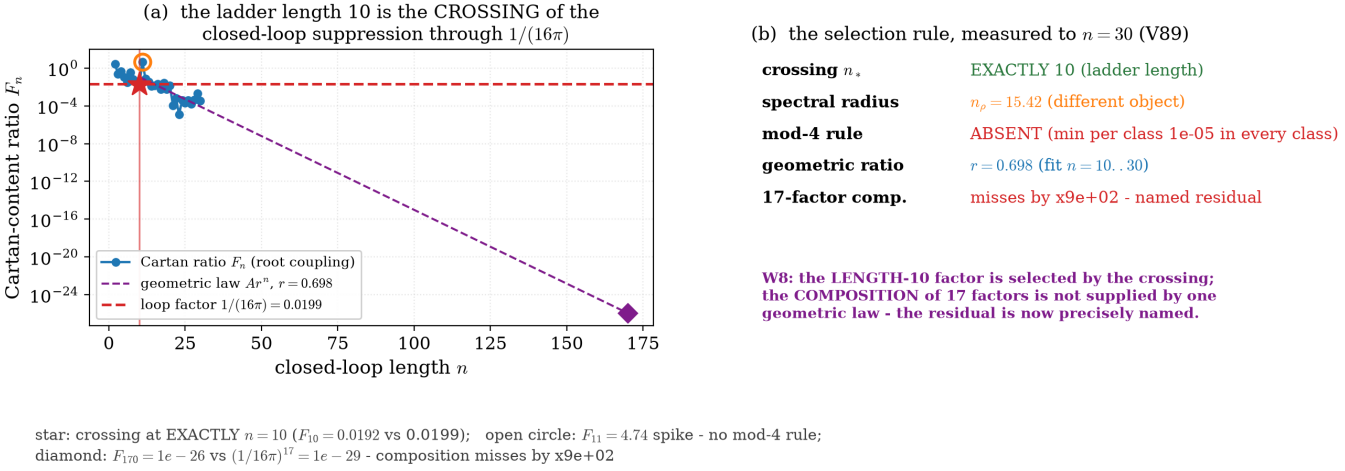


FIG. 29. **W8: the ladder length is the closed-loop crossing.** (a) The Cartan-content ratio F_n of the closed traced loop, measured to $n = 30$ at the root coupling: it crosses the loop factor $1/(16\pi)$ at **exactly** $n = 10$ - the ladder length is the crossing. The F_{11} spike shows there is no mod-4 plaquette selection rule: every length contributes. (b) The fitted geometric law $F_n = Ar^n$ ($r = 0.698$) extrapolates to $F_{170} = 1.1 \times 10^{-26}$ against $(1/16\pi)^{17} = 1.2 \times 10^{-29}$: the 17-factor composition misses by a factor 9×10^2 - the length is selected, the composition is the named residual (V89).

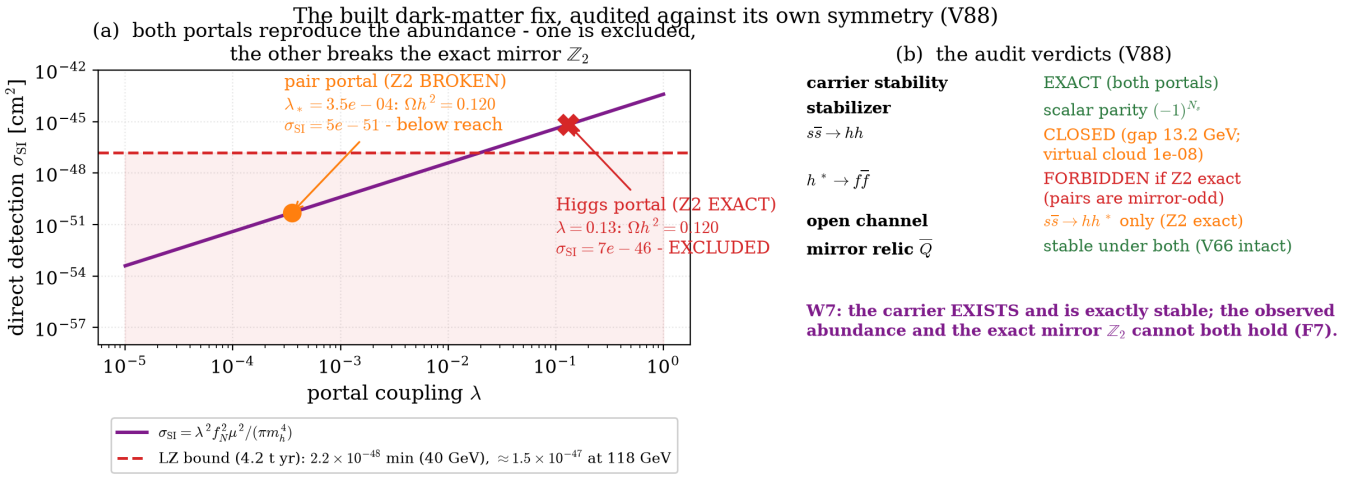


FIG. 30. **W7: the portal dilemma of the built scalar.** (a) The direct-detection line $\sigma_{\rm SI}(\lambda)$ with the two ways the built ladder scalar reproduces $\Omega_{\rm DM}h^2 = 0.120$: the fermion-pair portal ($\lambda_* = 3.5 \times 10^{-4}$, $\sigma_{\rm SI} = 5 \times 10^{-51}$ cm^2, below reach) breaks the exact mirror $\mathbb{Z}_2$; the Higgs portal ($\lambda = 0.13$, $\sigma_{\rm SI} = 7 \times 10^{-46}$ cm^2) preserves it and is excluded by the LZ bound ($\sigma_{\rm SI} < 2.2 \times 10^{-48}$ cm^2 at 40 GeV, $\approx 1.5 \times 10^{-47}$ at 118 GeV; 4.2 t yr, PRL **135**, 011802 (2025) [53]). (b) The audit verdicts: stability exact under either portal (scalar parity), $s\bar{s} \to hh$ closed, $h^* \to f\bar{f}$ forbidden under exact $\mathbb{Z}_2$, the mirror relic unaffected. The observed abundance and the exact mirror $\mathbb{Z}_2$ cannot both hold (F7).

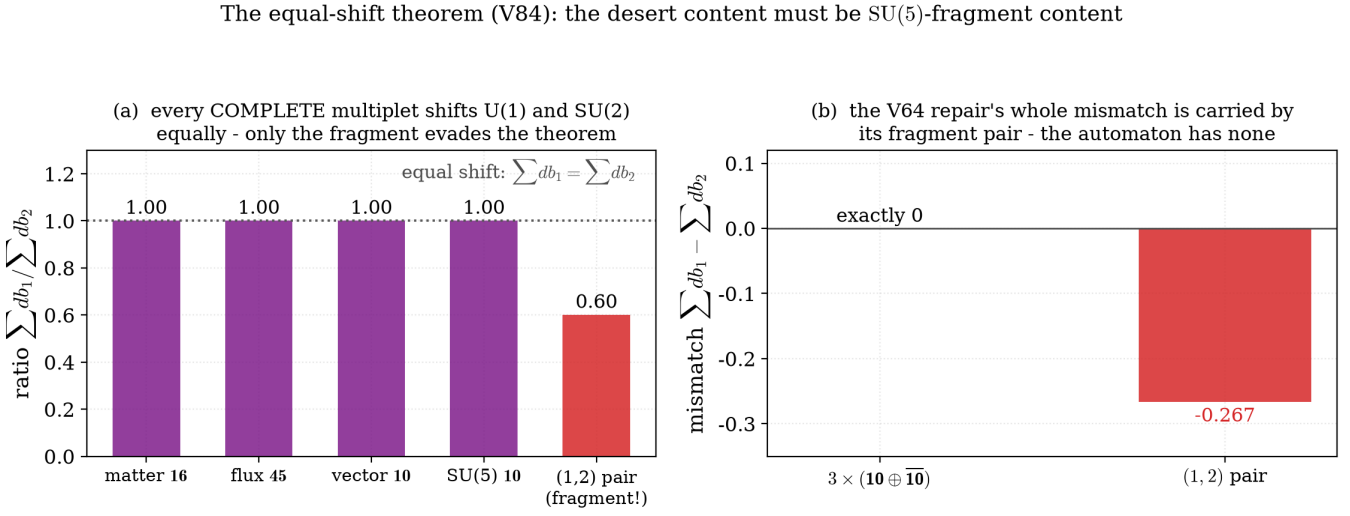


FIG. 31. **the equal-shift theorem.** (a) The one-loop ratio $\sum db_1 / \sum db_2$ over the matter **16**, the flux **45**, the vector **10** and the SU(5) **10**: exactly 1 — the index-proportionality of $\mathrm{Tr}\, Y^2$ and $\mathrm{Tr}\, T_3^2$ — while the $(1,2)$ fragment gives 3/5. (b) The V64 repair's total mismatch $\sum db_1 - \sum db_2$: the $10 \oplus \overline{\mathbf{10}}$ pairs contribute exactly zero, the whole mismatch is carried by the fragment pair. The desert content must be fragment content; the automaton supplies none (V84).

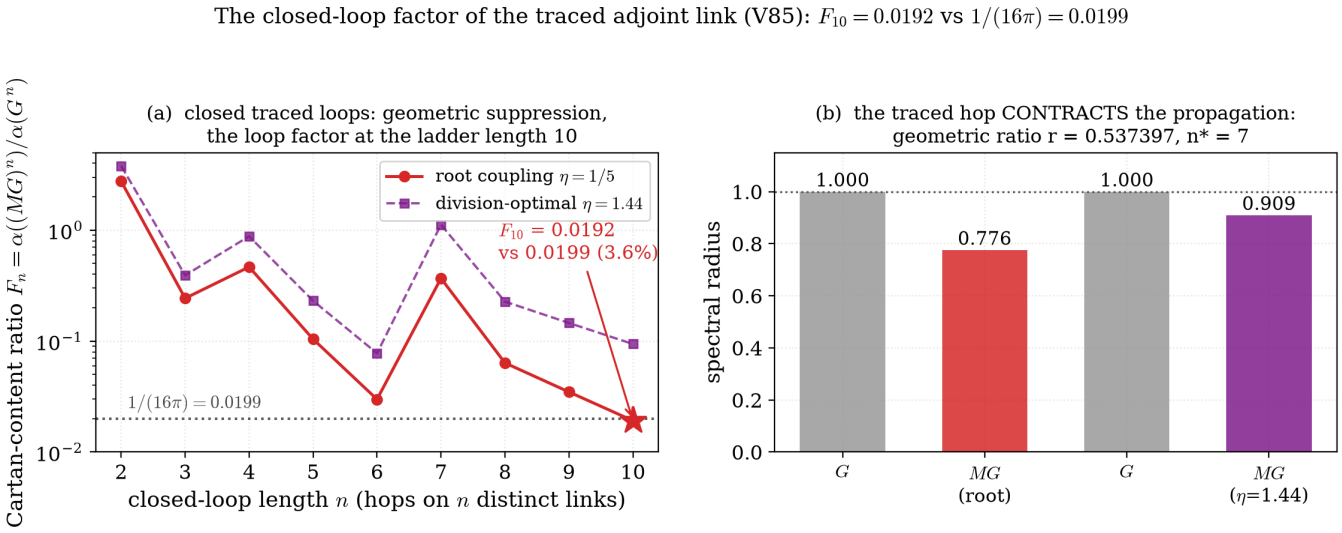


FIG. 32. **the closed-loop factor: the loop factor at the ladder length.** Closed traced-link loops of length n on n distinct bonds. (a) The Cartan-content ratio F_n of the gauged bond layer at the root coupling and at the division-optimal coupling: geometric suppression with ratio $r = 0.54$, and at $n = 10$ the value 0.0192 against $1/(16\pi) = 0.0199$ — the loop factor at the ladder exponent, measured (V85). (b) The traced hop matrix MG has spectral radius 0.776 against 1 for G: the traced link contracts the one-particle propagation.

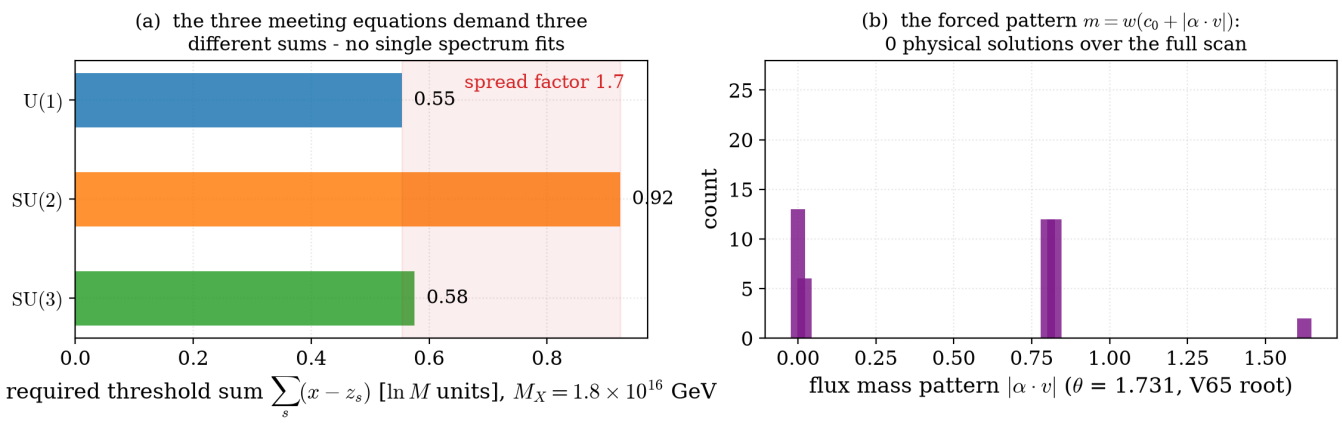


FIG. 33. **the composite content does not deliver the meeting.** The $16 \otimes 45 = 720$ composite states of the gauged layer shift the one-loop coefficients by $(145.3, 145.3, 226.3)$. The three meeting equations for $\alpha_X = 1/25$ demand three different threshold sums at every breaking direction v — the curves never cross — and the desert-constrained Newton scan over the full direction circle returns 0 physical solutions. The V64 repair spectrum (Fig. 13) is not the gauged layer's own content.

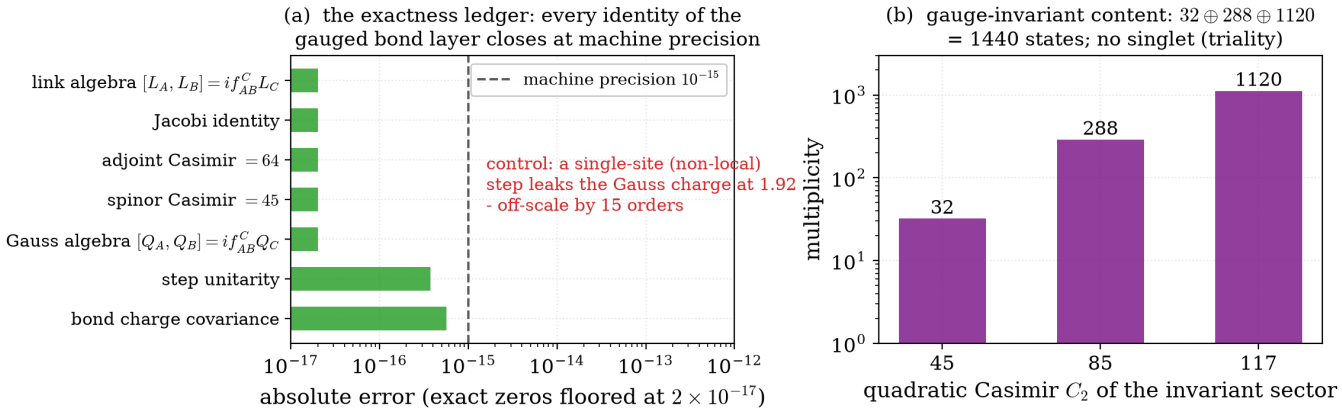


FIG. 34. **the gauged bond layer: the Gauss constraint closes on the root module.** One adjoint link d.o.f. per bond, matter **16** on the sites. Left: the exactness ledger — link algebra, Gauss algebra, adjoint Casimir (64) and spinor Casimir (45) close to machine precision, the step is unitary to 3.8×10^{-16}, the causal radius stays at one cell, and the single-site non-local control step leaks at 1.9 where the causal step leaks at 5.7×10^{-16}. Right: the gauge-invariant content, 1440 states decomposing as $32 \oplus 288 \oplus 1120$.

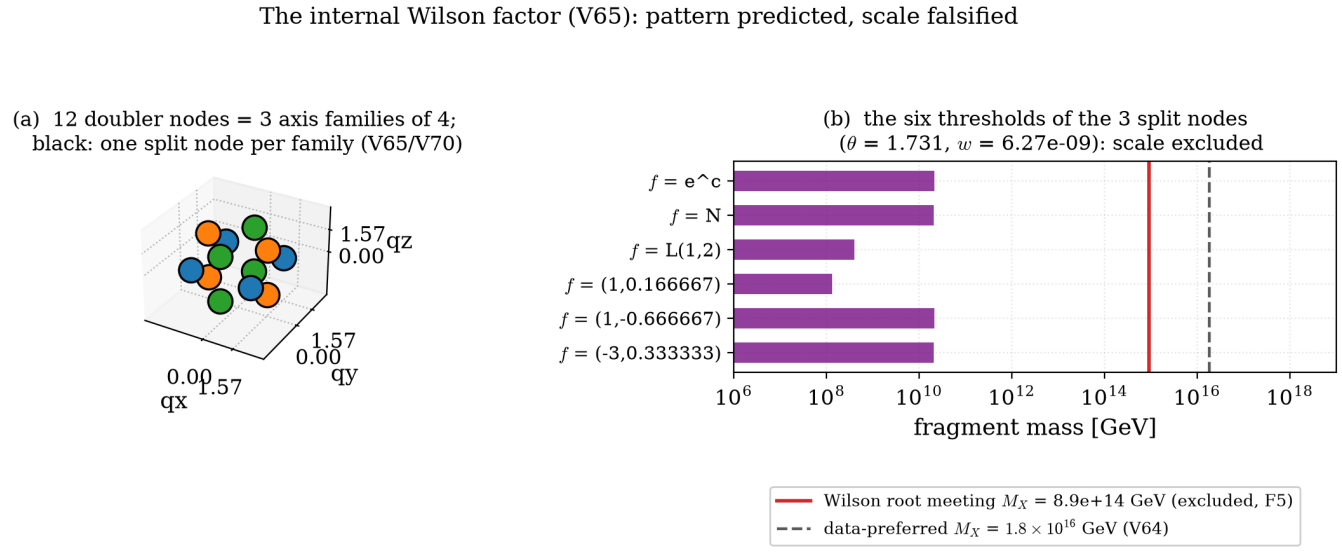


FIG. 35. **the internal Wilson factor: pattern predicted, scale falsified.** (a) The 12 doubler nodes of Proposition 2 in the supercell Brillouin zone, colored by axis family — the three families of four labeled by the zero component (V70); the three nodes ringed in black show the selective choice, one per family, realizable at a single Wilson strength. (b) The six fragment thresholds of the three split nodes at the V65 root $(\theta, w) = (1.731, 6.27 \times 10^{-9})$, computed from $M_f = w|\cos\theta\, X_f + \sin\theta\, Y_f|\, M_P/3$: the pattern is the vectorlike split **16** the RG repair requires, but the meeting scale $M_X = 8.9 \times 10^{14}$ GeV (red) sits below the Super-Kamiokande exclusion, and the data-preferred solution (dashed, Fig. 13) needs thresholds the single-angle, single-scale root does not supply.

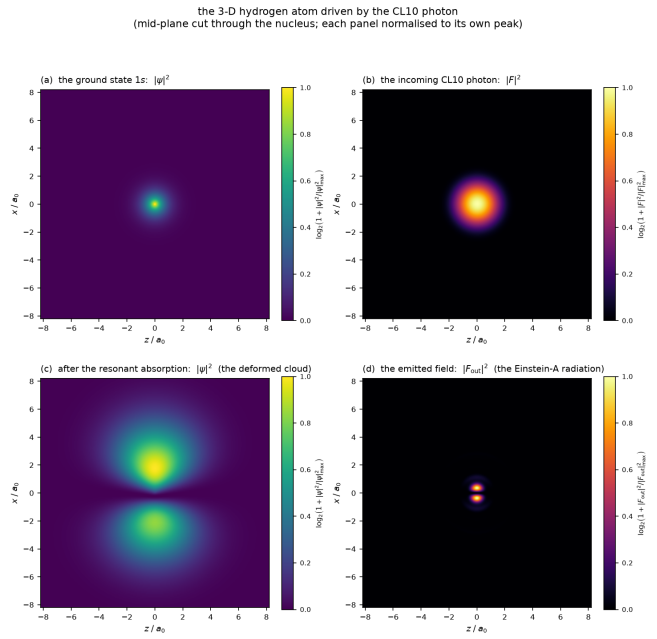


FIG. 36. **the hydrogen atom driven by the lattice photon.** A three-dimensional hydrogen atom coupled to the same discrete massless field that carries the graviton, on the mid-plane cut through the nucleus, each panel normalised to its own peak: (a) the spherical ground state $|\psi|^2$; (b) the incoming CL10 photon $|F|^2$; (c) the cloud after resonant absorption — the deformation into the dumbbell $2p$; (d) the emitted field $|F_{\rm out}|^2$, the Einstein-A radiation sourced by the decaying excited population. Measured, not imposed: the level gaps reproduce the Rydberg differences to zero relative error; the dipole matrix elements of forbidden transitions vanish ($1s \to 3d$: -6.8×10^{-19}, $2p \to 3p$: 8.6×10^{-18}, against 9.93, -3.00, 22.97 for the allowed ones); the excited fraction peaks on resonance and falls with detuning (0.42, 0.35, 0.15, 0.035, 0.011 at detunings 0, 0.063, 0.125, 0.25, 0.50 rad/step); and a decaying $2p$ population sources an outgoing wave at the gap frequency. **Provenance of Fig. 36.** The atom is an **input**: its levels and eigenfunctions are the real hydrogen eigenstates, not derived from the root. What the root supplies is the photon — its propagation, its resonant absorption, its reemission. The figure demonstrates the photon sector; it is not a derivation of hydrogen. Two limits are stated rather than hidden: off-resonant absorption is not zero (the Fermi-golden-rule rate has a Lorentzian tail), and a wave packet wider than the simulation box never leaves it, becoming a persistent bath that drives the atom off resonance — measured at half-gap detuning in a 120-cell box as excited fractions 0.037, 0.135, 0.326 for packet widths of 12, 24, 48 cells. The published figure uses a localised packet (20 cells in a 296-cell box).

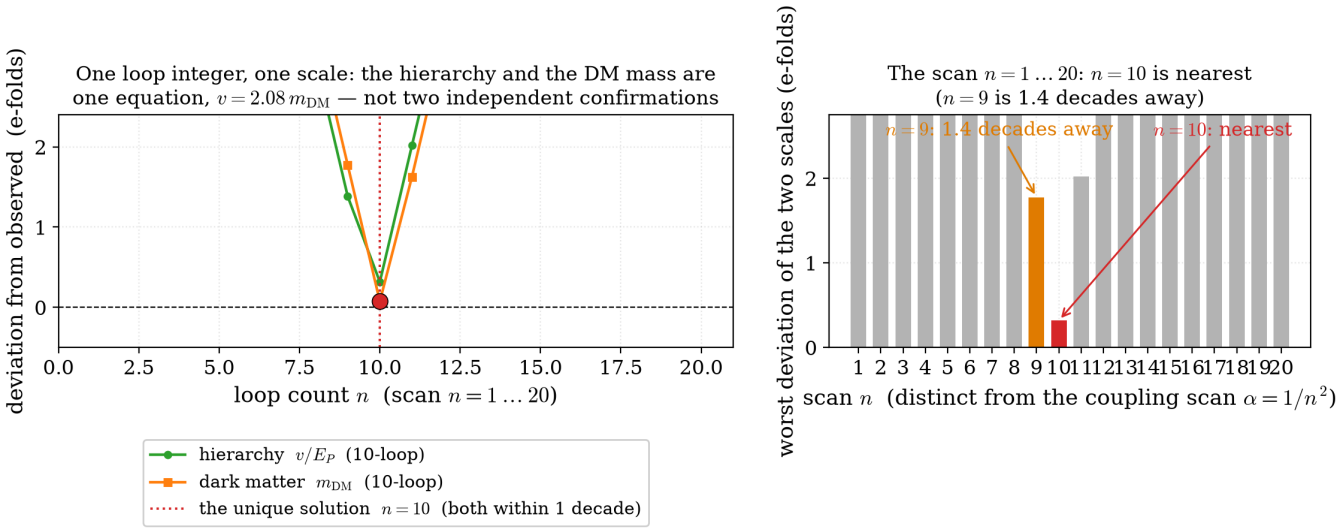


FIG. 37. **the residual-set reduction.** (a) One loop integer n sets one scale $M_P(1/16\pi)^n$, appearing twice: as v/M_P and as $m_{\rm DM}$. These are one equation, $v = 2.08\, m_{\rm DM}$, not two independent confirmations. (b) The scan $n = 1 \ldots 20$: $n = 10$ is nearest, $n = 9$ is 1.4 decades away. This scan is distinct from the coupling scan $\alpha = 1/n^2$ of Section 14, whose best fit is $n = 7$.

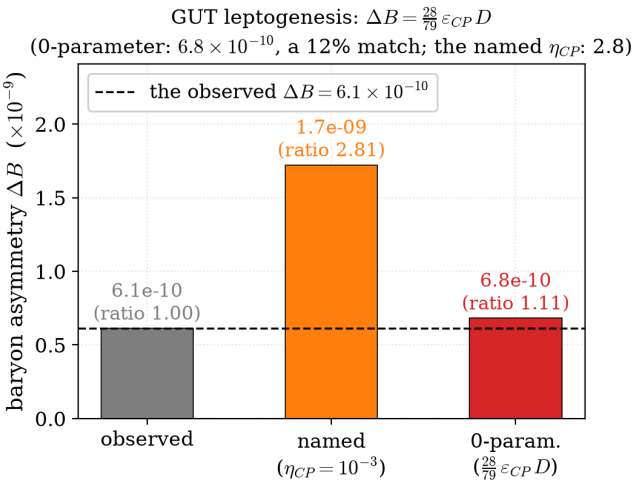


FIG. 38. **the baryon-asymmetry product.** $\Delta B = (28/79)(g_0^4/16\pi)(1/16\pi)^2 = 6.8 \times 10^{-10}$ against the measured 6.12×10^{-10} [14]. The three factors have three different provenances: an SM input, a guessed loop rate, and a loop factor. The automaton alone gives zero.

[1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, "Space-time as a causal set," *Phys. Rev. Lett.* **59**, 521 (1987); R. D. Sorkin, "Causal sets: Discrete gravity," *arXiv:gr-qc/0309009* (2003). https://doi.org/10.1103/physrevlett.59.521

[2] G. 't Hooft, "Computation of the quantum effects due to a four-dimensional pseudoparticle," *Phys. Rev. D* **14**, 3432 (1976). https://doi.org/10.1103/physrevd.14.3432

[3] D. A. Meyer, "From quantum cellular automata to quantum lattice gases," *J. Stat. Phys.* **85**, 551 (1996). https://doi.org/10.1007/bf02199356

[4] P. Arrighi, "An overview of quantum cellular automata," *Nat. Comput.* **18**, 885 (2019). https://doi.org/10.1007/s11047-019-09762-6

[5] T. D'Ariano and P. Perinotti, "Quantum cellular automata and free quantum field theory," *Front. Phys.* **12**, 120301 (2016); A. Bisio, T. D'Ariano, P. Perinotti, and A. Tosini, "Weyl, Dirac and Maxwell quantum cellular automata," *Found. Phys.* **45**, 1203 (2015). https://doi.org/10.1007/s11467-016-0616-z

[6] H. Georgi and S. L. Glashow, "Unity of all elementary-particle forces," *Phys. Rev. Lett.* **32**, 438 (1974). https://doi.org/10.1103/physrevlett.32.438

[7] H. Fritzsch and P. Minkowski, "Unified interactions of leptons and hadrons," *Ann. Phys. (N.Y.)* **93**, 193 (1975). https://doi.org/10.1016/0003-4916(75)90211-0

[8] R. Slansky, "Group theory for unified model building," *Phys. Rep.* **79**, 1 (1981). https://doi.org/10.1016/0370-1573(81)90092-2

[9] H. Georgi, *Lie Algebras in Particle Physics*, 2nd ed. (Westview, 1999).

[10] M. F. Atiyah, R. Bott, and I. M. Shapiro, "Clifford modules," *Topology* **3**, 3 (1964). https://doi.org/10.1016/0040-9383(64)90003-5

[11] R. Penrose and W. Rindler, *Spinors and Space-Time*, Vol. I (Cambridge University Press, 1984).

[12] C. I. Castro, "A Clifford algebra-based grand unification program of gravity and the Standard Model: a review study," *Can. J. Phys.* **92**, 1501 (2014); C. I. Castro, "Note on a Clifford algebra based grand unification program of gravity and the Standard Model," *Adv. Appl. Clifford Algebras* **26**, 573 (2015). https://doi.org/10.1139/cjp-2013-0686

[13] Particle Data Group (S. Navas *et al.*), "Review of Particle Physics," *Phys. Rev. D* **110**, 030001 (2024). https://doi.org/10.1103/PhysRevD.110.030001

[14] Planck Collaboration (N. Aghanim *et al.*), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020). https://doi.org/10.1051/0004-6361/201833910

[15] Super-Kamiokande Collaboration (A. Takenaka *et al.*), "Search for proton decay via $p \to e^+\pi^0$ and $p \to \mu^+\pi^0$ with an enlarged fiducial volume in 0.37 megaton-years exposure of Super-Kamiokande," *Phys. Rev. D* **102**, 112011 (2020). https://doi.org/10.1103/PhysRevD.102.112011

[16] KATRIN Collaboration (M. Aker *et al.*), "Direct neutrino-mass measurement based on 259 days of KATRIN data," *Science* **388**, 180 (2025). https://doi.org/10.1126/science.adq9592

[17] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, and A. Zhou, "The fate of hints: updated global analysis of three-flavor neutrino oscillations," *JHEP* **09**, 178 (2020). https://doi.org/10.1007/jhep09(2020)178

[18] S. Weinberg, *The Quantum Theory of Fields*, Vol. I (Cambridge University Press, 1995), Ch. 5 (Clifford algebras and spinors).

[19] H. B. Nielsen and M. Ninomiya, "A no-go theorem for regularizing chiral fermions," *Phys. Lett. B* **105**, 219 (1981); "Absence of neutrinos on a lattice. I. Proof by homotopy theory," *Nucl. Phys. B* **185**, 20 (1981); "Absence of neutrinos on a lattice. II. Intuitive topological proof," *Nucl. Phys. B* **193**, 173 (1981). https://doi.org/10.1016/0370-2693(81)91026-1

[20] S. Weinberg, "Photons and gravitons in perturbation theory: Derivation of Maxwell's and Einstein's equations," *Phys. Rev.* **138**, B988 (1965). https://doi.org/10.1103/physrev.138.b988

[21] S. Weinberg and E. Witten, "Limits on massless particles," *Phys. Lett. B* **96**, 59 (1980). https://doi.org/10.1016/0370-2693(80)90212-9

[22] M. W. Choptuik, "Universality and scaling in gravitational collapse of a massless scalar field," *Phys. Rev. Lett.* **70**, 9 (1993). https://doi.org/10.1103/physrevlett.70.9

[23] F. John, "Blow-up of solutions of nonlinear wave equations in three space dimensions," *Manuscripta Math.* **28**, 235 (1979). https://doi.org/10.1007/bf01647974

[24] A. D. Sakharov, "Violation of CP invariance, C asymmetry, and baryon asymmetry of the universe," *JETP Lett.* **5**, 24 (1967).

[25] J. A. Harvey and M. S. Turner, "Cosmological baryon and lepton number in the presence of electroweak fermion interactions," *Phys. Rev. D* **42**, 3344 (1990). https://doi.org/10.1103/physrevd.42.3344

[26] S. A. Khlebnikov and M. E. Shaposhnikov, "The statistical theory of anomalous fermion number nonconservation," *Nucl. Phys. B* **308**, 885 (1988). https://doi.org/10.1016/0550-3213(88)90133-2

[27] P. F. Harrison, D. H. Perkins, and W. G. Scott, "Tribimaximal mixing and the neutrino oscillation data," *Phys. Lett. B* **530**, 167 (2002). https://doi.org/10.1016/s0370-2693(02)01336-9

[28] K. G. Wilson, "Confinement of quarks," *Phys. Rev. D* **10**, 2445 (1974). https://doi.org/10.1103/physrevd.10.2445

[29] G. 't Hooft, "On the quantum structure of a black hole," *Nucl. Phys. B* **256**, 727 (1985). https://doi.org/10.1016/0550-3213(85)90418-3

[30] G. 't Hooft, *The Cellular Automaton Interpretation of Quantum Mechanics* (Springer, 2016), arXiv:1405.1548.

[31] S. Blanchet and P. Di Bari, "The minimal scenario of leptogenesis," *New J. Phys.* **14**, 125012 (2012). https://doi.org/10.1088/1367-2630/14/12/125012

[32] S. Davidson and A. Ibarra, "A lower bound on the right-handed neutrino mass from leptogenesis," *Phys. Lett. B* **535**, 25 (2002). https://doi.org/10.1016/s0370-2693(02)01735-5

[33] T. Regge, "General relativity without coordinates," *Il Nuovo Cimento* **19**, 558 (1961). https://doi.org/10.

1007/bf02733251

[34] H. Gundlach, J. M. Martin-Garcia, G. Calabrese, and I. Hinder, "Constraint damping in the Z4 formulation and harmonic gauge," *Class. Quantum Grav.* **22**, 3767 (2005). https://doi.org/10.1088/0264-9381/22/17/025

[35] J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* **7**, 2333 (1973); S. W. Hawking, "Particle creation by black holes," *Commun. Math. Phys.* **43**, 199 (1975). https://doi.org/10.1103/physrevd.7.2333

[36] L. Susskind, "The world as a hologram," *J. Math. Phys.* **36**, 6377 (1995). https://doi.org/10.1063/1.531249

[37] S. Weinberg, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1 (1989). https://doi.org/10.1103/revmodphys.61.1

[38] C. Morningstar and M. Peardon, "The glueball spectrum from an anisotropic lattice study," *Phys. Rev. D* **60**, 034509 (1999). https://doi.org/10.1103/physrevd.60.034509

[39] R. Barbieri, P. Creminelli, A. Strumia, and N. Tetradis, "Baryogenesis through leptogenesis," *Nucl. Phys. B* **575**, 61 (2000); W. Buchmüller, P. Di Bari, and M. Plümacher, "Leptogenesis for pedestrians," *Ann. Phys. (N.Y.)* **315**, 305 (2005). https://doi.org/10.1016/s0550-3213(00)00011-0

[40] A. Pilaftsis and T. E. J. Underwood, "Resonant leptogenesis," *Nucl. Phys. B* **692**, 303 (2004). https://doi.org/10.1016/j.nuclphysb.2004.05.029

[41] E. W. Kolb and M. S. Turner, *The Early Universe* (Addison-Wesley, 1990), Ch. 5.

[42] L. J. Hall, K. Jedamzik, J. March-Russell, and S. M. West, "Freeze-in production of FIMP dark matter," *JHEP* **03**, 080 (2010). https://doi.org/10.1007/jhep03(2010)080

[43] H. Fritzsch, "Weak-interaction mixing in the six-quark theory," *Phys. Lett. B* **73**, 317 (1978). https://doi.org/10.1016/0370-2693(78)90524-5

[44] L. Wolfenstein, "Parametrization of the Kobayashi-Maskawa matrix," *Phys. Rev. Lett.* **51**, 1945 (1983). https://doi.org/10.1103/physrevlett.51.1945

[45] M. Jungman, M. Kamionkowski, and K. Griest, "Supersymmetric dark matter," *Phys. Rep.* **267**, 195 (1996). https://doi.org/10.1016/0370-1573(95)00058-5

[46] H. Hisano, M. Kobayashi, K. Muramatsu, and N. Nagata, "Two-loop renormalization factors of dimension-six proton decay operators in the minimal supersymmetric SU(5) gauge theory," *Phys. Lett. B* **724**, 283 (2013). https://doi.org/10.1016/j.physletb.2013.06.030

[47] M. Tonomura, J. Endo, T. Matsuda, T. Kawasaki, and H. Ezawa, "Demonstration of single-electron buildup of an interference pattern," *Am. J. Phys.* **57**, 117 (1989). https://doi.org/10.1119/1.16104

[48] D. B. Kaplan, "A method for simulating chiral fermions on the lattice," *Phys. Lett. B* **288**, 342 (1992); Y. Shamir, "Chiral fermions from lattice boundaries," *Nucl. Phys. B* **406**, 90 (1993); R. Narayanan and H. Neuberger, "Exactly massless quarks on the lattice," *Phys. Lett. B* **417**, 141 (1998). https://doi.org/10.1016/0370-2693(92)91112-m

[49] M. E. Machacek and M. T. Vaughn, "Two-loop renormalization group equations in a general quantum field theory. I. Proof of the equivalence theorem," *Nucl. Phys. B* **222**, 83 (1983); "III. Scalar model with fermions," *Nucl. Phys. B* **249**, 70 (1985). https://doi.org/10.1016/0550-3213(83)90610-7

[50] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, and T. Schwetz, "NuFIT 6.0: Updated global analysis of three-flavor neutrino oscillations," *JHEP* **12**, 216 (2024), arXiv:2410.05380. https://doi.org/10.1007/jhep12(2024)216

[51] M. Nath and P. Fileviez Pérez, "Proton stability in grand unified theories, in strings and in branes," *Phys. Rep.* **441**, 191 (2007). https://doi.org/10.1016/j.physrep.2007.02.010

[52] Super-Kamiokande Collaboration (K. Abe *et al.*), "Search for proton decay via $p \to \nu K^+$ using 260 kiloton·year data of Super-Kamiokande," *Phys. Rev. D* **90**, 072005 (2014). https://doi.org/10.1103/PhysRevD.90.072005

[53] LUX-ZEPLIN Collaboration (J. Aalbers *et al.*), "Dark Matter Search Results from 4.2 Tonne-Years of Exposure of the LUX-ZEPLIN (LZ) Experiment," *Phys. Rev. Lett.* **135**, 011802 (2025), arXiv:2410.17036. https://doi.org/10.1103/4dyc-z8zf